Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df(data):
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()



    # Removing of the columns where there is more than 15% of nan
    #df_features = pd.DataFrame(features_all)
    cols_to_check = target_col + features_X
```

```python
    nan_ratio = df[cols_to_check].isna().mean()
    valid_features = nan_ratio[nan_ratio <= 0.15].index
    valid_features_list = list(valid_features)

    #df = df[valid_features]
    # Removing of last np.nan
    mask = np.isfinite(df[valid_features]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(valid_features_list + #target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    new_feats_X = [x for x in features_X if x in valid_features_list]
    new_feats_add = [x for x in feats_eng if x in valid_features_list]

    data["df"]=df
    data["features_X"] = new_feats_X
    data["feats_added"] = new_feats_add

    column_removed = [x for x in cols_to_check if x not in valid_features_list]
    print(f'Column removed because np.isnan > 15%: {column_removed}')

    return data
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        file_path = "csv_decomposed/innova/df_signal_decomposition_NH3_junesept.csv"
        #file_path = "csv/innova/switched/df_signal_decomposition_10_50_200.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv_decomposed/innova/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
```

```
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv_decomposed/innova/features_decomposition_NH3_list_junesept.json", "r")
        #with open("csv/innova/switched/features_decomposition_list_10_50_200.json", "r")as f
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
```

```
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Column removed because np.isnan > 15%: ['r_NH3_fft_4h_fft_f2_cph', 'r_NH3_fft_4h_fft_period2_h', 'r_NH3_fft_4h_fft_power2', 'r_NH3_fft_4h_fft_f3_cph', 'r_NH3_fft_4h_fft_period3_h', 'r_NH3_fft_4h_fft_power3', 'r_temperature_fft_4h_fft_f2_cph', 'r_temperature_fft_4h_fft_period2_h', 'r_temperature_fft_4h_fft_power2', 'r_temperature_fft_4h_fft_f3_cph', 'r_temperature_fft_4h_fft_period3_h', 'r_temperature_fft_4h_fft_power3', 'r_umidity_fft_4h_fft_f2_cph', 'r_umidity_fft_4h_fft_period2_h', 'r_umidity_fft_4h_fft_power2', 'r_umidity_fft_4h_fft_f3_cph', 'r_umidity_fft_4h_fft_period3_h', 'r_umidity_fft_4h_fft_power3']

1.2 Metadata

Nombre de lignes df	8012
Nombre de lignes df clean	8012
Nombre de features	14
% de NaN	0.0
Taille dataset entraînement	5608
Taille dataset test	2403
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_4h_fft_f1_cph, r_NH3_fft_4h_fft_period1_h, r_NH3_fft_4h_fft_power1, r_temperature_fft_4h_fft_f1_cph, r_temperature_fft_4h_fft_period1_h, r_temperature_fft_4h_fft_power1, r_umidity_fft_4h_fft_f1_cph, r_umidity_fft_4h_fft_period1_h, r_umidity_fft_4h_fft_power1

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

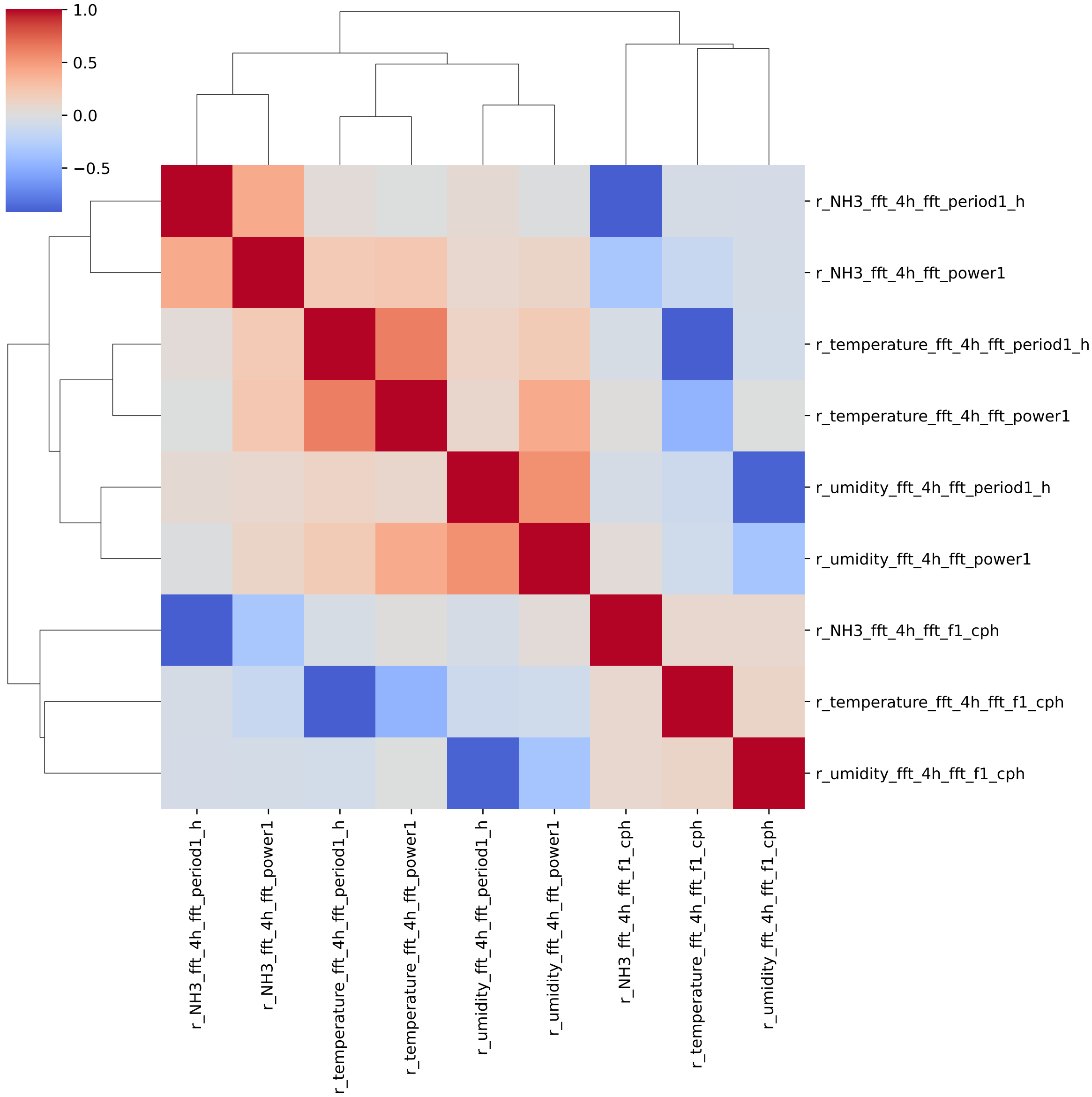
The following section has been done with a threshold choosen of : 0.8

cluster 1: r, *, N, H, 3 ;* ***cluster 2:*** *r,* , t, e, m, p, e, r, a, t, u, r, e ; **cluster 3:** r, *, u, m, i, d, i, t, y ;* ***cluster 4:*** *r,* , C, O, 2 ; **cluster 5:** r, _, T, H, I ; **cluster 6:** r_NH3_fft_4h_fft_period1_h, r_NH3_fft_4h_fft_f1_cph ; **cluster 7:** r_NH3_fft_4h_fft_power1 ; **cluster 8:** r_temperature_fft_4h_fft_period1_h, r_temperature_fft_4h_fft_f1_cph ; **cluster 9:** r_temperature_fft_4h_fft_power1 ; **cluster 10:** r_umidity_fft_4h_fft_f1_cph, r_umidity_fft_4h_fft_period1_h ; **cluster 11:** r_umidity_fft_4h_fft_power1 ;

2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Choosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_4h_fft_period1_h, r_NH3_fft_4h_fft_power1, r_temperature_fft_4h_fft_period1_h, r_temperature_fft_4h_fft_power1, r_umidity_fft_4h_fft_f1_cph, r_umidity_fft_4h_fft_power1



3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_fft_4h_fft_period1_h, r_NH3_fft_4h_fft_power1, r_temperature_fft_4h_fft_period1_h, r_temperature_fft_4h_fft_power1, r_umidity_fft_4h_fft_f1_cph, r_umidity_fft_4h_fft_power1

Len(Features_X): 11

Config Gridsearch:{'show': True, 'search_type': 'optuna', 'model_name': None, 'model': None, 'param_grid': None, 'best_param': None, 'cv_func': ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), 'save_models': True, 'sumup-table': True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

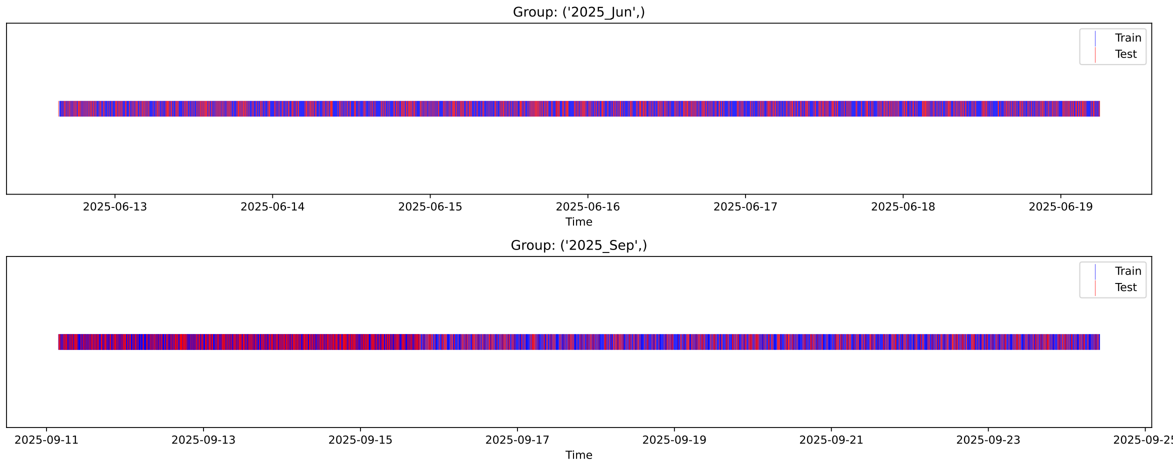
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_train vs X_test

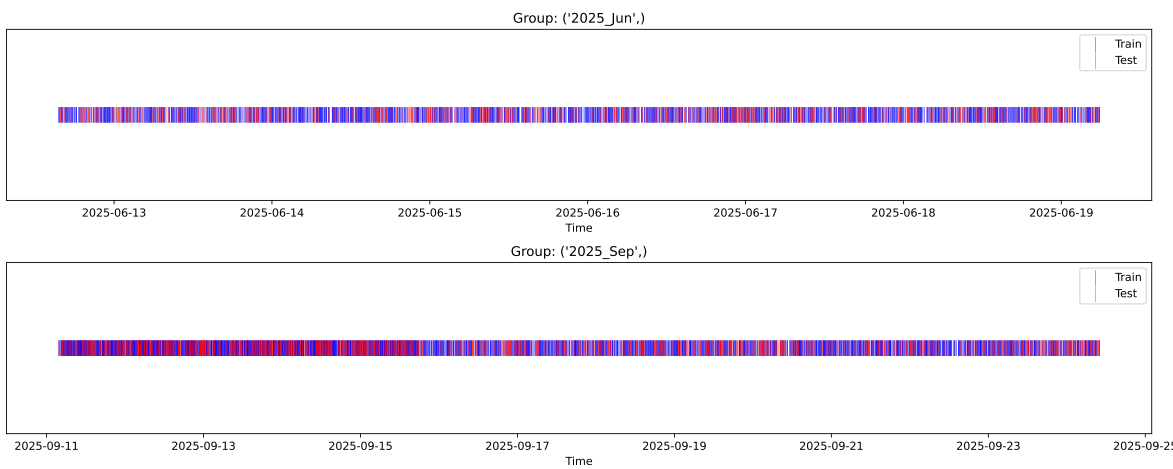
External Split: Train: 5608 samples Test: 2404 samples



3.3.2 Internal CV Folds

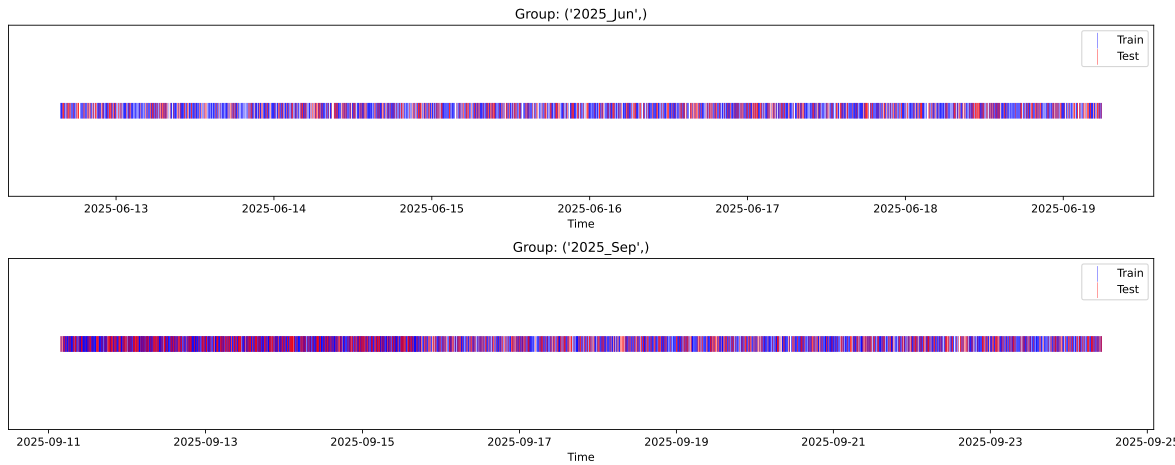
Fold 1/5

Train: 3925 samples Test: 1683 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3925 samples Test: 1683 samples Excluded: 0 samples Window exclusion: 0h

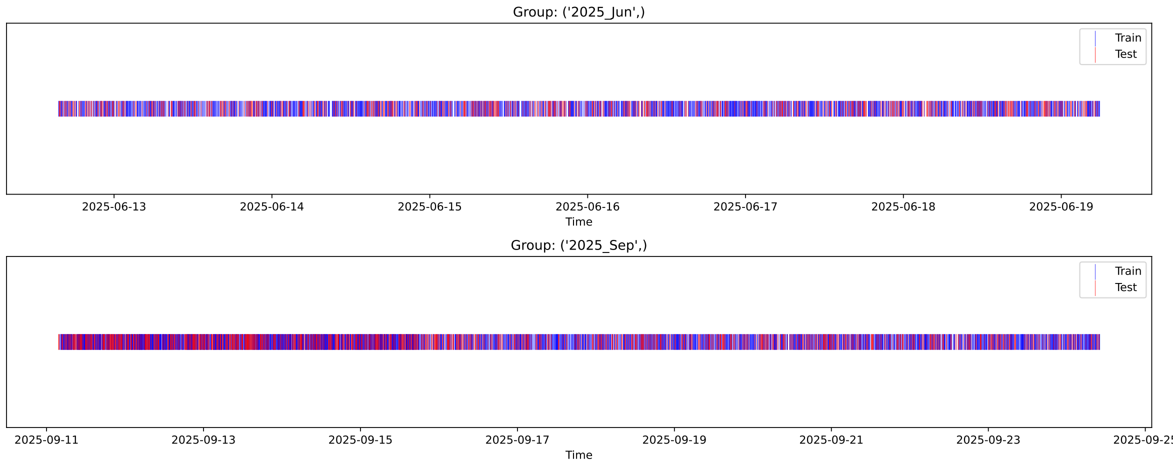


Fold 3/5

Train: 3925 samples Test: 1683 samples Excluded: 0 samples Window exclusion: 0h

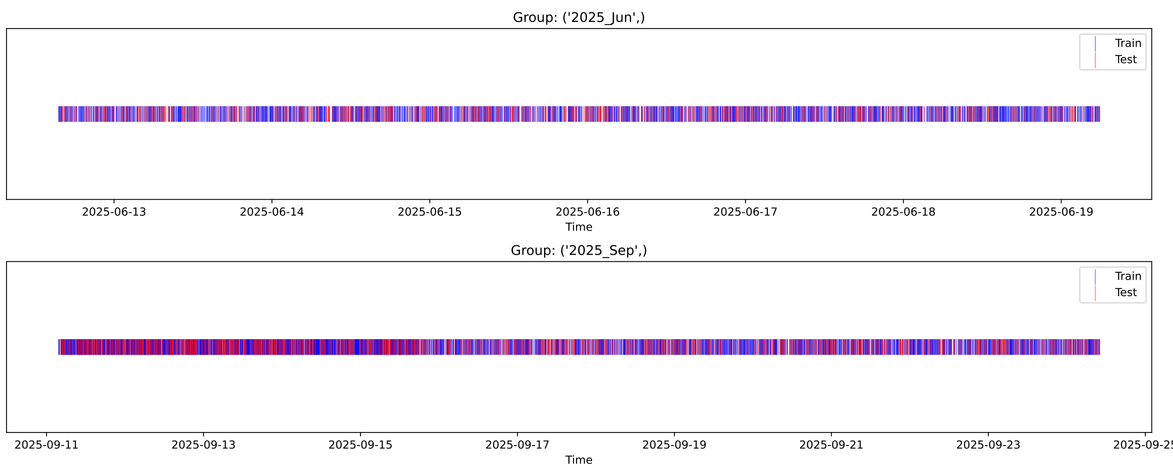
Fold 4/5

Train: 3925 samples Test: 1683 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3925 samples Test: 1683 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.346, std=0.795, min=0.959, max=8.682 y_test: mean=2.366, std=0.806, min=1.041, max=13.845

Fold 0 — y_test: mean=2.33, std=0.80, y_train: mean=2.35, std=0.79

Fold 1 — y_test: mean=2.35, std=0.79, y_train: mean=2.35, std=0.80

Fold 2 — y_test: mean=2.33, std=0.79, y_train: mean=2.35, std=0.80

Fold 3 — y_test: mean=2.33, std=0.77, y_train: mean=2.35, std=0.81

Fold 4 — y_test: mean=2.33, std=0.80, y_train: mean=2.35, std=0.79

train scores CV: [0.42345667 0.41338491 0.41069222 0.41001976 0.41998443]

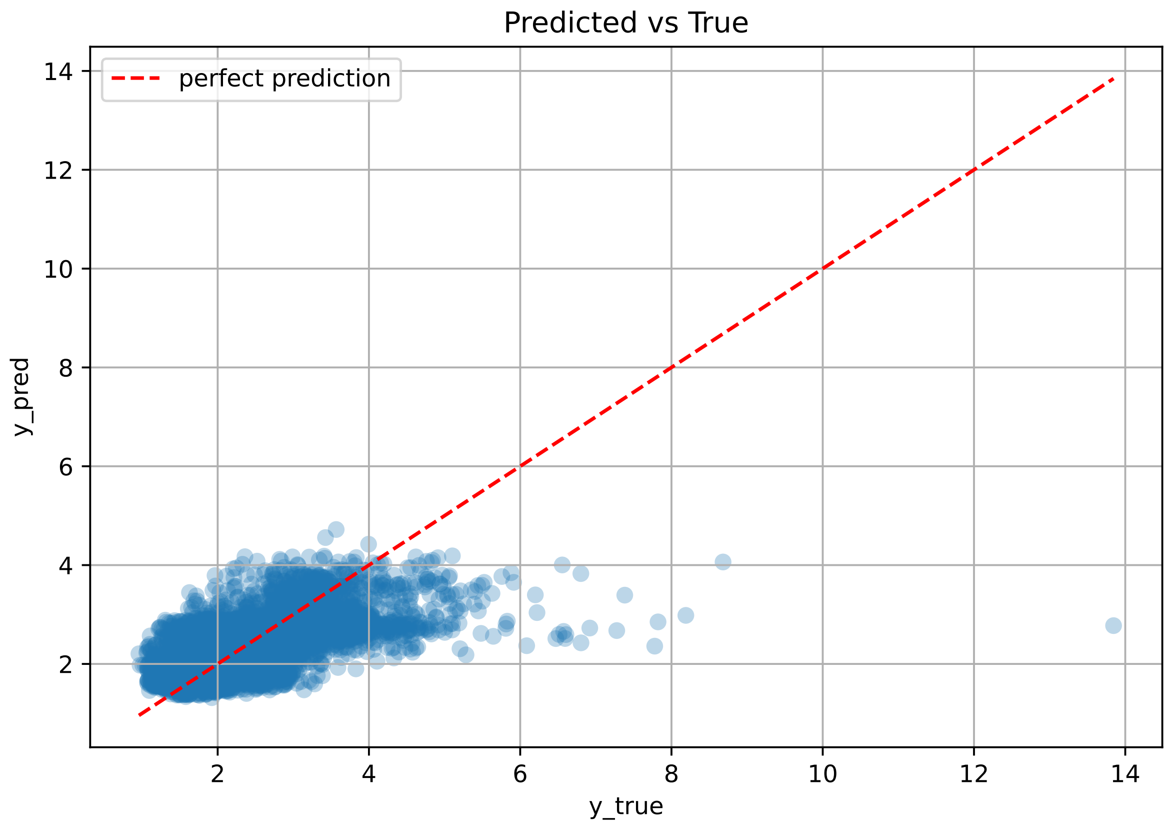
test scores CV: [0.39182174 0.41507948 0.42212897 0.42391461 0.39959774]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4155	0.4105	0.0127	0.3998	0.3999	-0.0106	0.9743	1.0122

Coefficients:

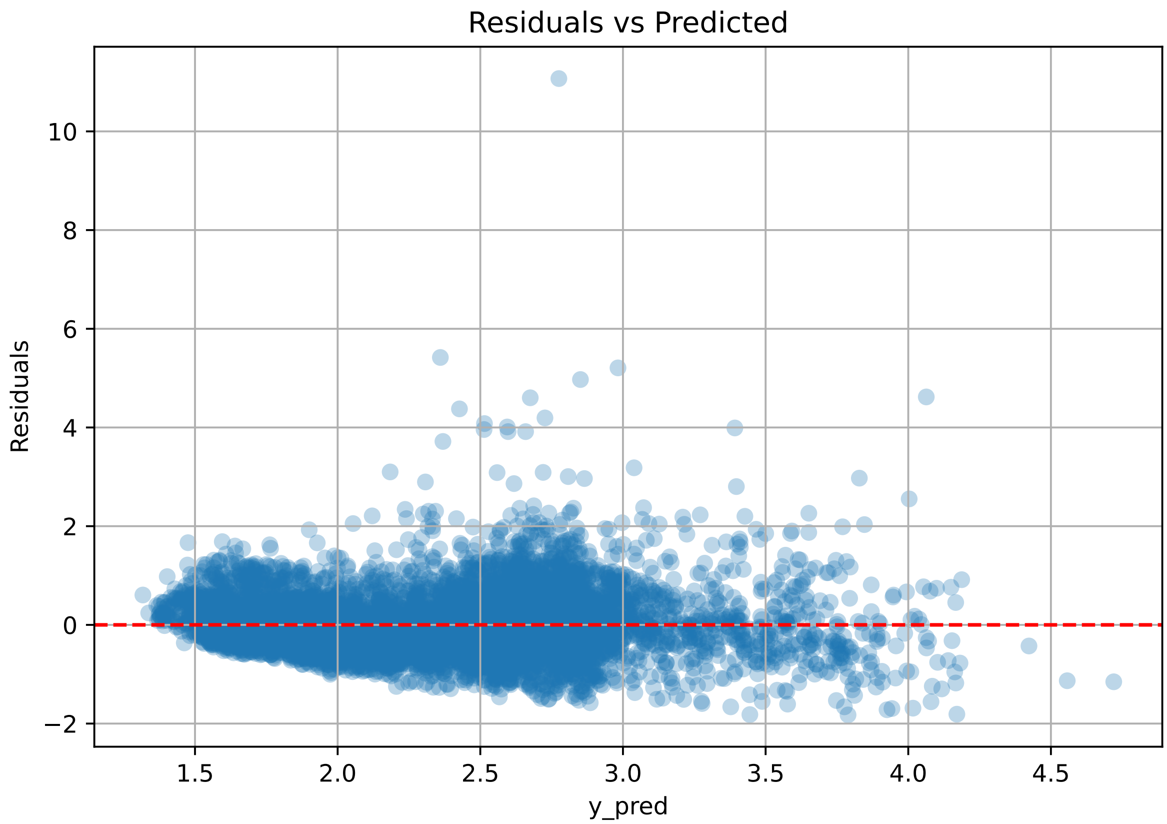
coef_r__CO2 = -0.0392, coef_r__NH3 = -0.0536, coef_r__NH3__fft__4h__fft__period1__h = 0.0056, coef_r__NH3__fft__4h__fft__power1 = 0.0057, coef_r__THI = 0.6714, coef_r__temperature = -0.8321, coef_r__temperature__fft__4h__fft__period1__h = -0.0579, coef_r__temperature__fft__4h__fft__power1 = 0.1786, coef_r__umidity = -0.6161, coef_r__umidity__fft__4h__fft__f1__cph = 0.0102, coef_r__umidity__fft__4h__fft__power1 = 0.0762, intercept = 2.3463,

3.4.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

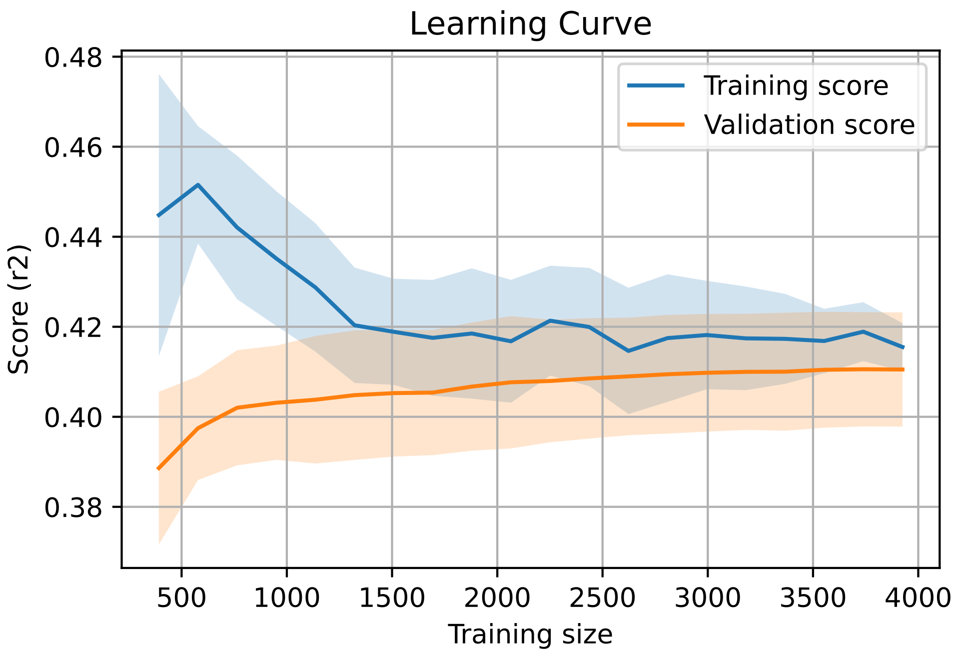
3.4.3 Scatter residuals



3.4.4 Learning curve

Len(group) : 8012, **Len(training)** : 5609,

Len(training_real) : 3927, **Len(test_of_training)** : 1682, **Len(test)** : 2403

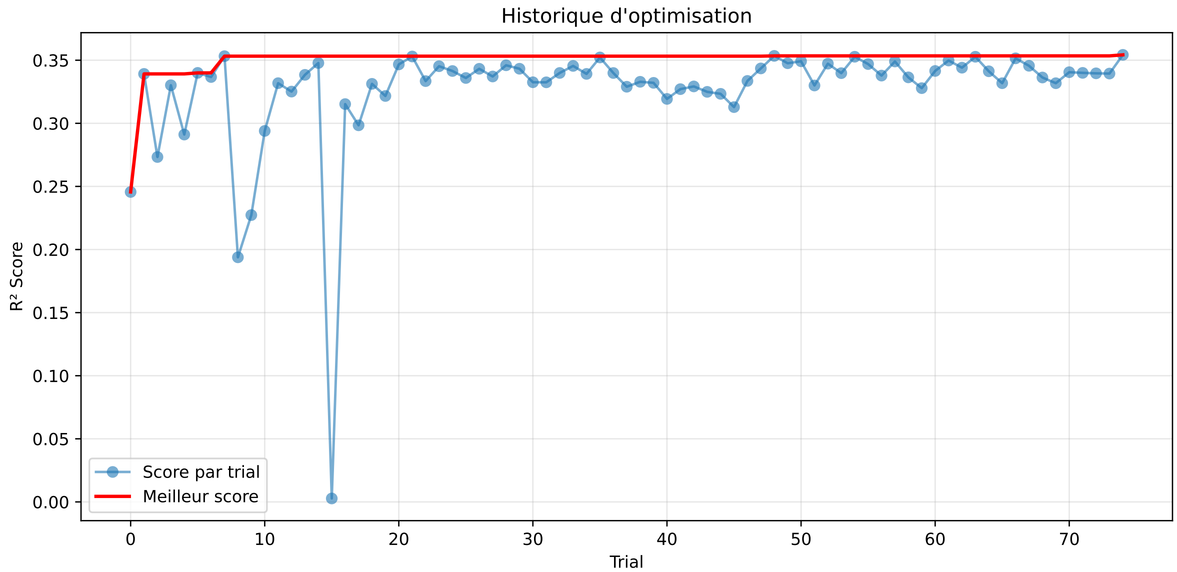


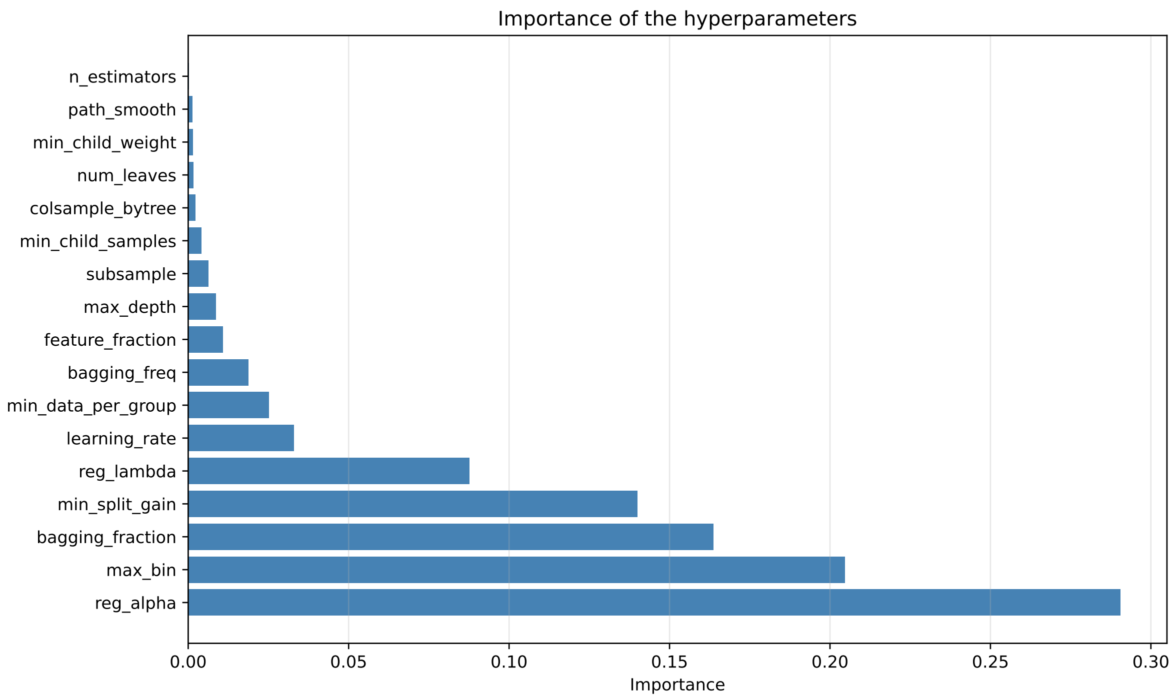
3.5 Model : LGBM

3.5.1 Gridsearch

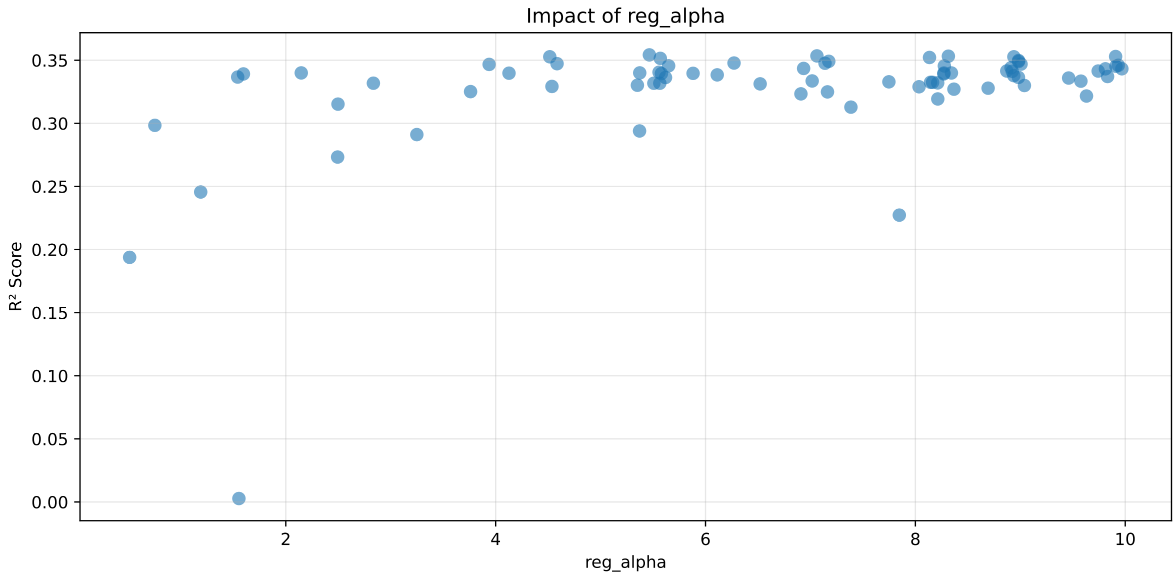
Distribution y_train vs y_test: y_train: mean=2.346, std=0.795, min=0.959, max=8.682 y_test: mean=2.366, std=0.806, min=1.041, max=13.845

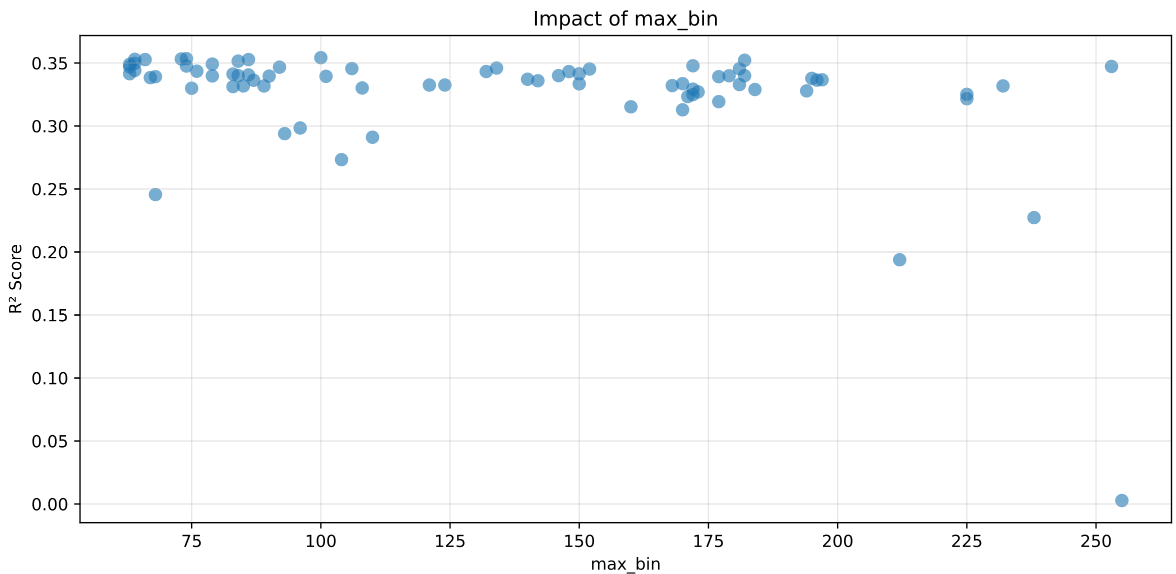
==================== TOP Importance FEATURES ==================== feature importance r_CO2 282 r_umidity 185 r_NH3 164 r_THI 142 r_temperature 138 r_temperature_fft_4h_fft_power1 104 r_temperature_fft_4h_fft_period1_h 83 r_NH3_fft_4h_fft_power1 79 r_umidity_fft_4h_fft_power1 60 r_umidity_fft_4h_fft_f1_cph 31 r_NH3_fft_4h_fft_period1_h 8

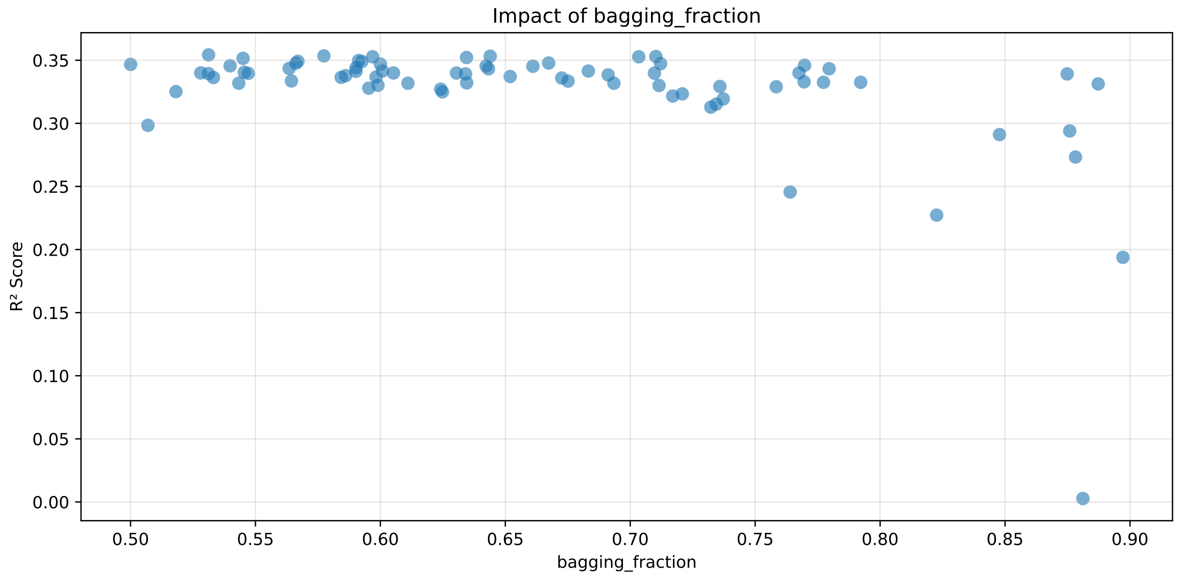


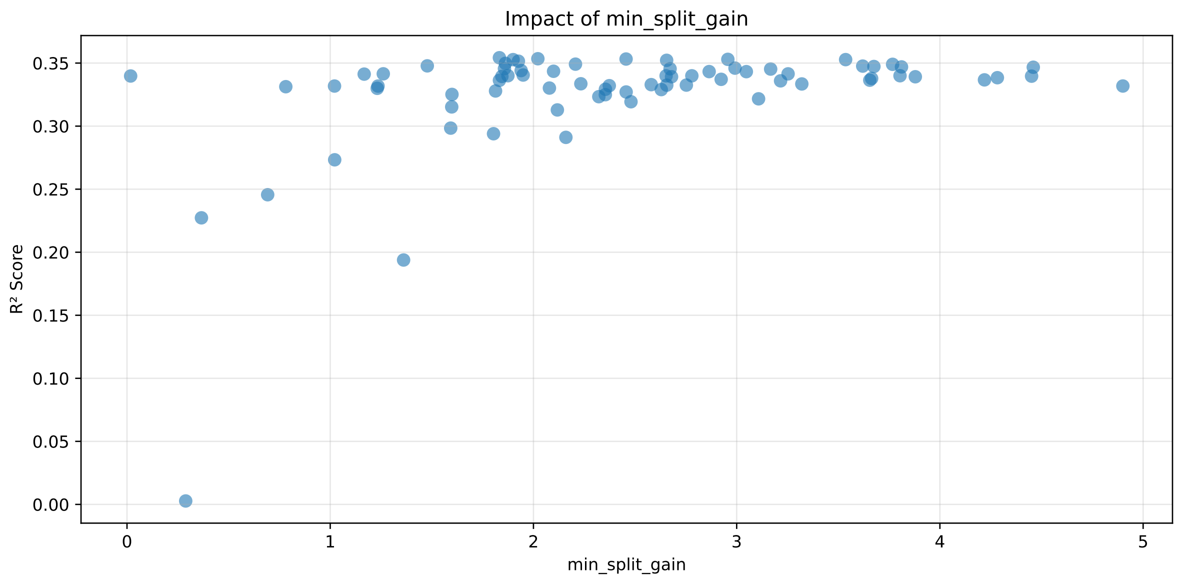


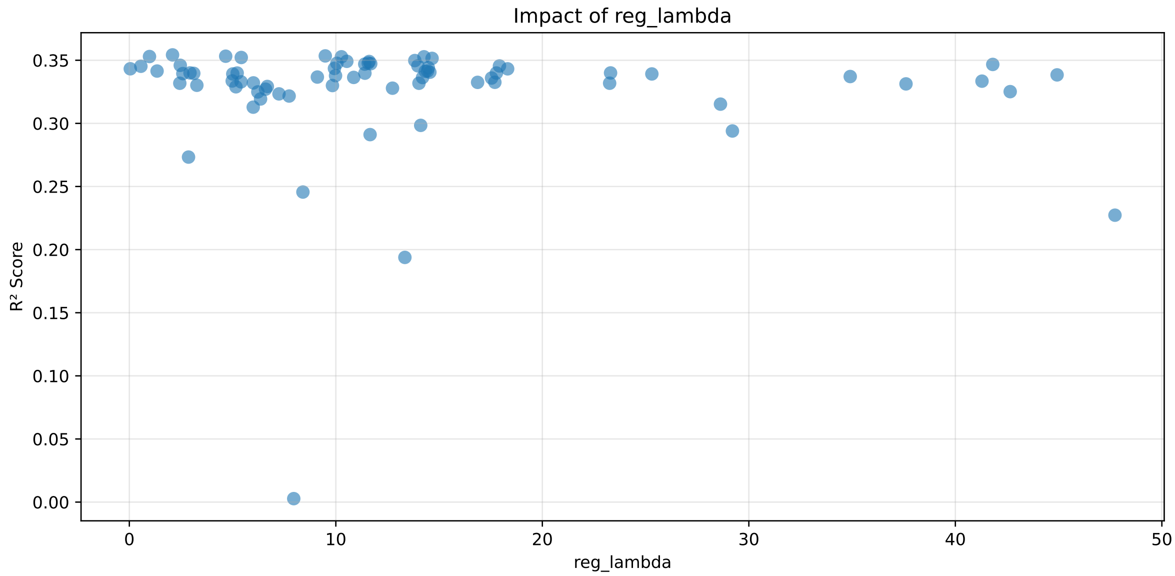
Paramètres avec >5% d'importance : ['reg_alpha', 'max_bin', 'bagging_fraction', 'min_split_gain', 'reg_lambda']


Impact of reg_alpha
R² Score
reg_alpha
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
2
4
6
8
10


Impact of max_bin
R² Score
max_bin
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
75
100
125
150
175
200
225
250

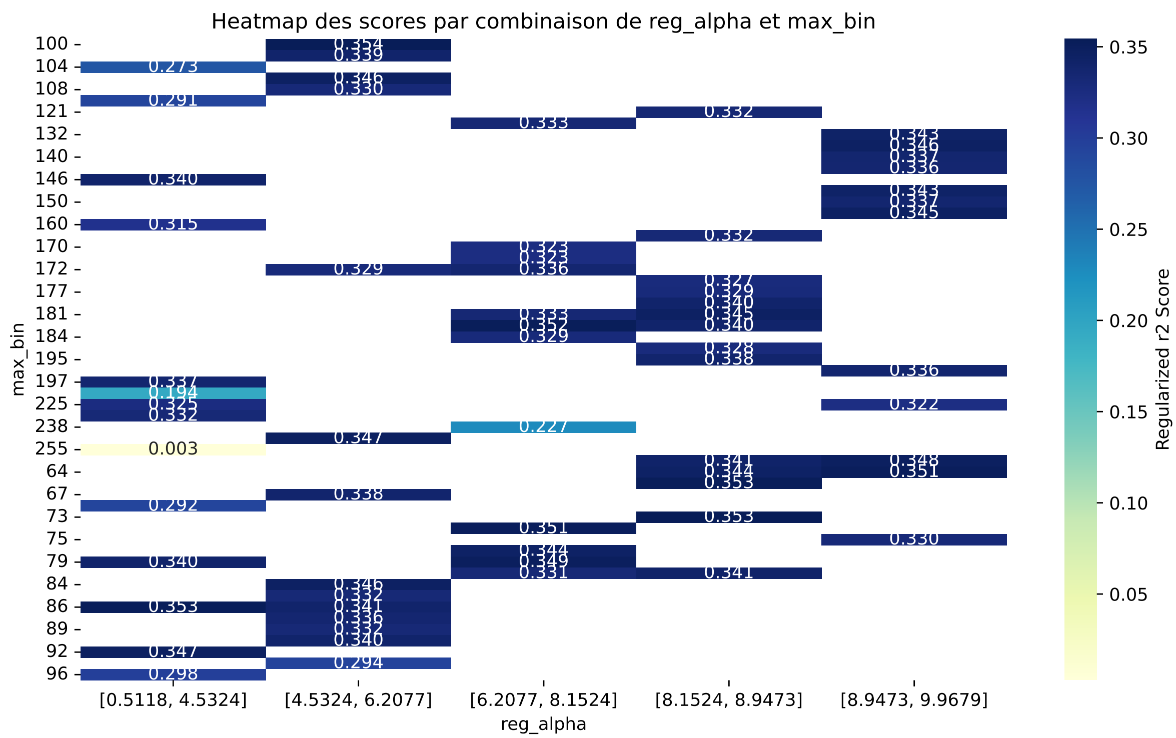

Impact of bagging_fraction
R² Score
bagging_fraction
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90


Impact of min_split_gain
R² Score
min_split_gain
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
1
2
3
4
5



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5575	0.5236	0.512	0.0119	0.3542	0.5379	0.0143	1.0272	1.0647
2	0.5418	0.5104	0.5	0.0151	0.3534	0.5268	0.0164	1.032	1.0615
3	0.5277	0.4986	0.4885	0.0109	0.3532	0.5074	0.0087	1.0175	1.0583
4	0.5155	0.4884	0.4781	0.0133	0.353	0.499	0.0106	1.0216	1.0555
5	0.5258	0.497	0.4862	0.0114	0.3527	0.5005	0.0036	1.0072	1.058
6	0.5393	0.5082	0.4999	0.0152	0.3527	0.5185	0.0103	1.0204	1.0612
7	0.5239	0.4953	0.4848	0.014	0.3522	0.5048	0.0095	1.0193	1.0578
8	0.5449	0.5126	0.5022	0.013	0.3515	0.522	0.0093	1.0182	1.0629
9	0.5266	0.4972	0.4876	0.0143	0.3498	0.5104	0.0133	1.0267	1.0593
10	0.5333	0.5026	0.492	0.0152	0.3492	0.5179	0.0152	1.0303	1.0611
---	---	---	---	---	---	---	---	---	---
75	0.7598	0.6336	0.6223	0.0072	0.0027	0.6461	0.0125	1.0198	1.1991

Hyperparameters:

Rank 1: {’reg__n_estimators’: 200, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.02830508793785416, ’reg__num_leaves’: 28, ’reg__subsample’: 0.5853489834195856, ’reg__colsample_bytree’: 0.8558494214655722, ’reg__min_child_samples’: 40, ’reg__reg_alpha’: 5.46484466893167, ’reg__reg_lambda’: 2.0925827463677815, ’reg__min_split_gain’: 1.832463583411663, ’reg__path_smooth’: 2.155848928179081, ’reg__min_child_weight’: 9.100411684645273, ’reg__feature_fraction’: 0.5469288413059802, ’reg__bagging_fraction’: 0.5312371826799387, ’reg__bagging_freq’: 2, ’reg__max_bin’: 100, ’reg__min_data_per_group’: 131}

Rank 2: {’reg__n_estimators’: 600, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.013694455222741906, ’reg__num_leaves’: 24, ’reg__subsample’: 0.549695216827611, ’reg__colsample_bytree’: 0.8975548780171638, ’reg__min_child_samples’: 49, ’reg__reg_alpha’: 7.061763220638129, ’reg__reg_lambda’: 9.489166359688808, ’reg__min_split_gain’: 2.021736373392107, ’reg__path_smooth’: 2.7752518895612988, ’reg__min_child_weight’: 0.010888130825273967, ’reg__feature_fraction’: 0.6689999074577963, ’reg__bagging_fraction’: 0.5773753651792504, ’reg__bagging_freq’: 5, ’reg__max_bin’: 74, ’reg__min_data_per_group’: 158}

Rank 3: {’reg__n_estimators’: 400, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.012156471148929754, ’reg__num_leaves’: 45, ’reg__subsample’: 0.5692139194962347, ’reg__colsample_bytree’: 0.5606471560518608, ’reg__min_child_samples’: 18, ’reg__reg_alpha’: 8.314469998655557, ’reg__reg_lambda’: 4.666276652282458, ’reg__min_split_gain’: 2.455753424284853, ’reg__path_smooth’: 4.6025839957639665, ’reg__min_child_weight’: 2.5365769430202048, ’reg__feature_fraction’: 0.683520949469173, ’reg__bagging_fraction’: 0.6439501310739414, ’reg__bagging_freq’: 5, ’reg__max_bin’: 73, ’reg__min_data_per_group’: 141}

Rank 4: {’reg__n_estimators’: 400, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.019744450154811768, ’reg__num_leaves’: 16, ’reg__subsample’: 0.61455021939739, ’reg__colsample_bytree’: 0.7875837069120541, ’reg__min_child_samples’: 26, ’reg__reg_alpha’: 9.908204758738297, ’reg__reg_lambda’: 0.9765075430593981, ’reg__min_split_gain’: 2.9565420877163486, ’reg__path_smooth’: 1.5692920024640054, ’reg__min_child_weight’: 0.44077155470081947, ’reg__feature_fraction’: 0.5448514114970999, ’reg__bagging_fraction’: 0.710257749633685, ’reg__bagging_freq’: 4, ’reg__max_bin’: 64, ’reg__min_data_per_group’: 125}

Rank 5: {’reg__n_estimators’: 300, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.021026305414017895, ’reg__num_leaves’: 42, ’reg__subsample’: 0.5425133388847636, ’reg__colsample_bytree’: 0.6504255399746337, ’reg__min_child_samples’: 44, ’reg__reg_alpha’: 4.515824068026284, ’reg__reg_lambda’: 10.270976812149579, ’reg__min_split_gain’: 3.536079605492688, ’reg__path_smooth’: 2.194685517877023, ’reg__min_child_weight’: 7.336022847550067, ’reg__feature_fraction’: 0.6076838033986501, ’reg__bagging_fraction’: 0.7034043464466385, ’reg__bagging_freq’: 2, ’reg__max_bin’: 86, ’reg__min_data_per_group’: 157}

Rank 6: {’reg__n_estimators’: 300, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.021794790104117796, ’reg__num_leaves’: 37, ’reg__subsample’: 0.5876052649595652, ’reg__colsample_bytree’: 0.8729940282020211, ’reg__min_child_samples’: 43, ’reg__reg_alpha’: 8.938507477722807, ’reg__reg_lambda’: 14.267332967280439, ’reg__min_split_gain’: 1.899802218126092, ’reg__path_smooth’: 3.040832563795225, ’reg__min_child_weight’: 0.10567927173800436, ’reg__feature_fraction’: 0.6166759301674227, ’reg__bagging_fraction’: 0.5968899923428918, ’reg__bagging_freq’: 2, ’reg__max_bin’: 66, ’reg__min_data_per_group’: 178}

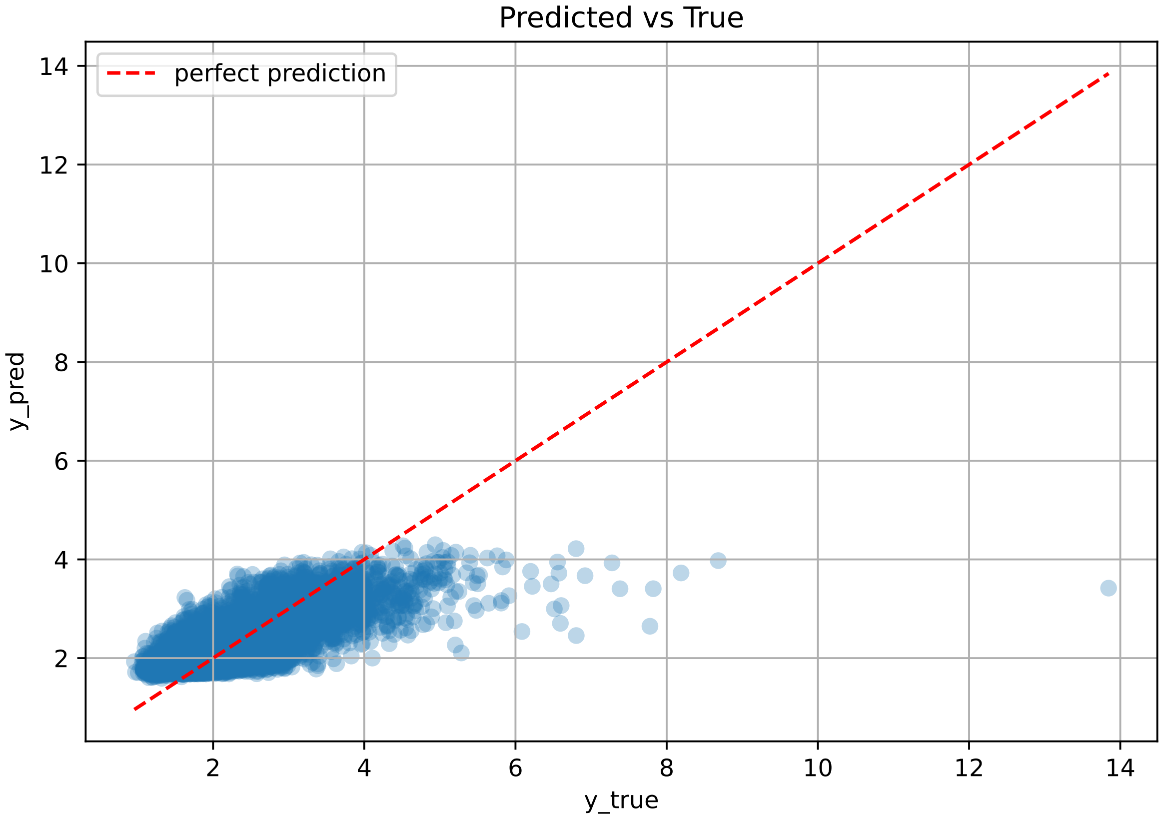
Rank 7: {’reg__n_estimators’: 400, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.024438575867492953, ’reg__num_leaves’: 16, ’reg__subsample’: 0.650316890394232, ’reg__colsample_bytree’: 0.8026105333374797, ’reg__min_child_samples’: 33, ’reg__reg_alpha’: 8.134628193663715, ’reg__reg_lambda’: 5.420154759992375, ’reg__min_split_gain’: 2.6554137270616707, ’reg__path_smooth’: 1.2761960203670295, ’reg__min_child_weight’: 1.3076850472992023, ’reg__feature_fraction’: 0.5739333997998532, ’reg__bagging_fraction’: 0.6344784199131764, ’reg__bagging_freq’: 4, ’reg__max_bin’: 182, ’reg__min_data_per_group’: 144}

Rank 8: {’reg__n_estimators’: 200, ’reg__max_depth’: 7, ’reg__learning_rate’: 0.02076961535955711, ’reg__num_leaves’: 38, ’reg__subsample’: 0.5267742621438185, ’reg__colsample_bytree’: 0.8756626440739224, ’reg__min_child_samples’: 42, ’reg__reg_alpha’: 5.569126562809492, ’reg__reg_lambda’: 14.660206512430246, ’reg__min_split_gain’: 1.9259640990818196, ’reg__path_smooth’: 2.7733133820631393, ’reg__min_child_weight’: 9.337166088758137, ’reg__feature_fraction’: 0.6045232375815162, ’reg__bagging_fraction’: 0.5450478248794509, ’reg__bagging_freq’: 2, ’reg__max_bin’: 84, ’reg__min_data_per_group’: 199}

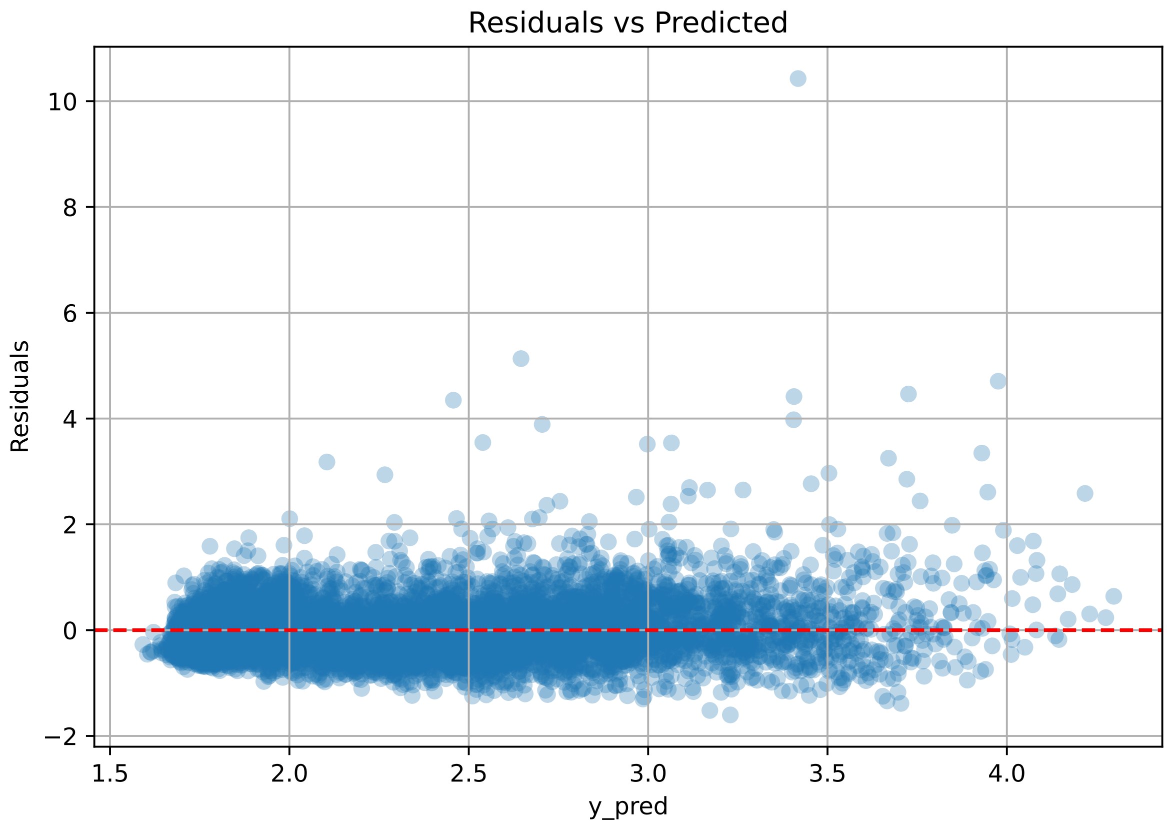
Rank 9: {'reg__n_estimators': 300, 'reg__max_depth': 6, 'reg__learning_rate': 0.012222585498853878, 'reg__num_leaves': 29, 'reg__subsample': 0.5369353025209055, 'reg__colsample_bytree': 0.8756610377245209, 'reg__min_child_samples': 43, 'reg__reg_alpha': 8.983499149257408, 'reg__reg_lambda': 13.82350107847163, 'reg__min_split_gain': 1.8620262354718422, 'reg__path_smooth': 2.7469490970904467, 'reg__min_child_weight': 9.427348862723047, 'reg__feature_fraction': 0.6139357787321014, 'reg__bagging_fraction': 0.591297384129214, 'reg__bagging_freq': 5, 'reg__max_bin': 64, 'reg__min_data_per_group': 200}

Rank 10: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.013642115162982181, 'reg__num_leaves': 23, 'reg__subsample': 0.5677105094067201, 'reg__colsample_bytree': 0.8991854638330564, 'reg__min_child_samples': 46, 'reg__reg_alpha': 7.174632388625024, 'reg__reg_lambda': 10.531756295316333, 'reg__min_split_gain': 2.2067950569109547, 'reg__path_smooth': 0.2817314151413961, 'reg__min_child_weight': 0.18837870884867366, 'reg__feature_fraction': 0.6690245602809884, 'reg__bagging_fraction': 0.5669875012342883, 'reg__bagging_freq': 5, 'reg__max_bin': 79, 'reg__min_data_per_group': 159}

3.5.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

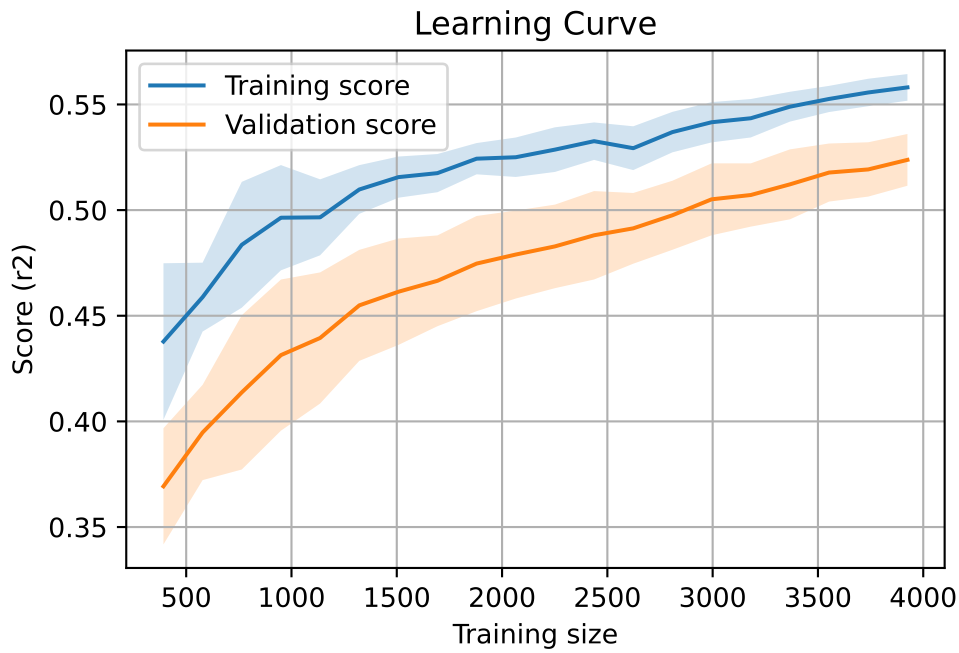
3.5.3 Scatter residuals



3.5.4 Learning curve

Len(group) : 8012, **Len(training)** : 5609,

Len(training_real) : 3927, **Len(test_of_training)** : 1682, **Len(test)** : 2403

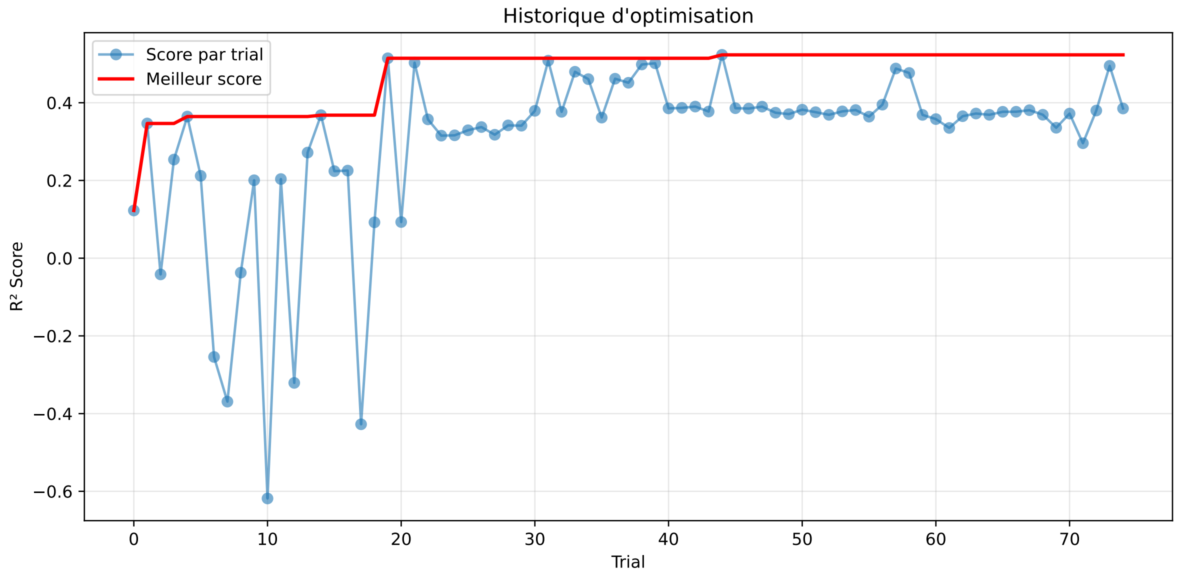


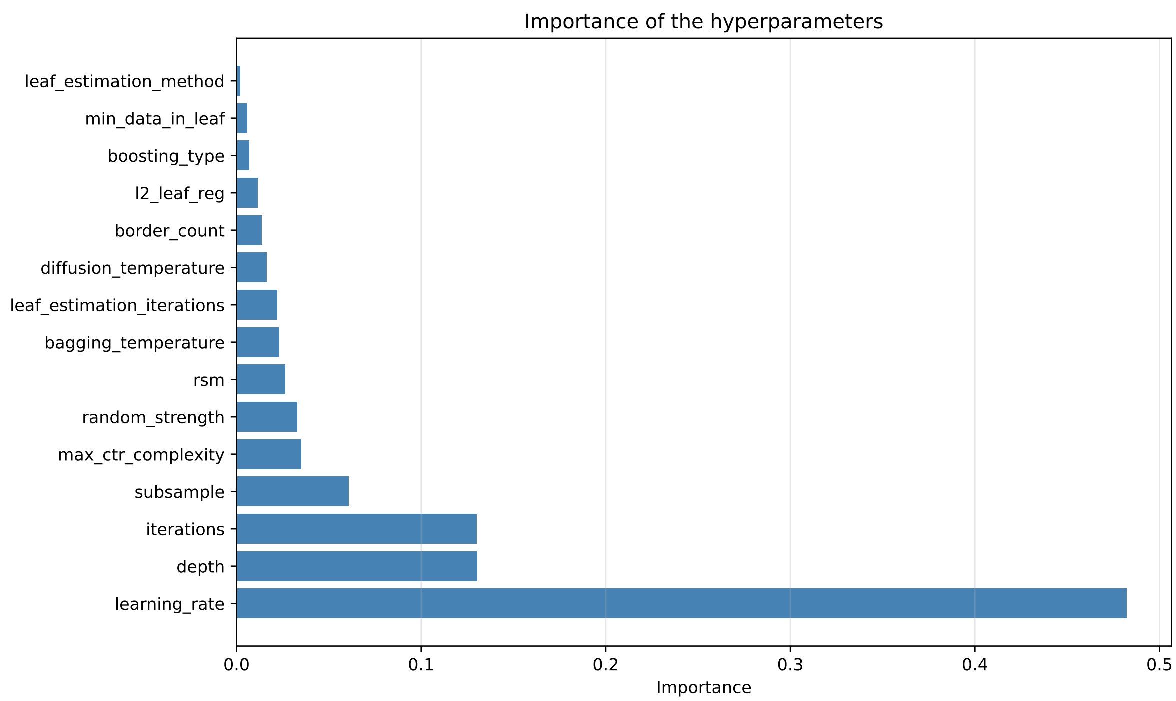
3.6 Model : CatBoost

3.6.1 Gridsearch

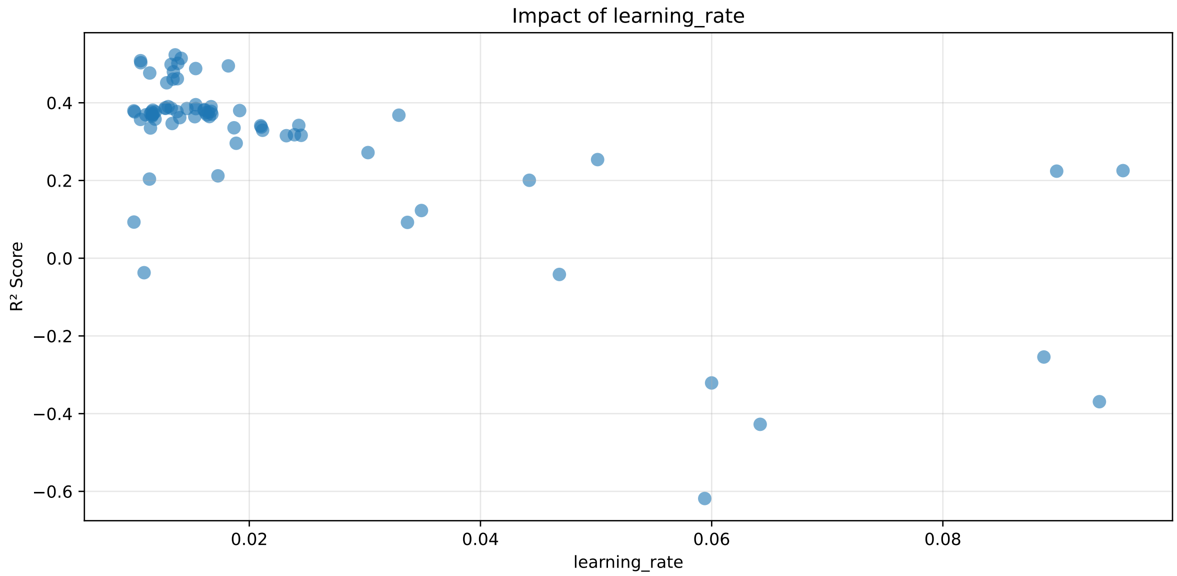
Distribution y_train vs y_test: y_train: mean=2.346, std=0.795, min=0.959, max=8.682 y_test: mean=2.366, std=0.806, min=1.041, max=13.845

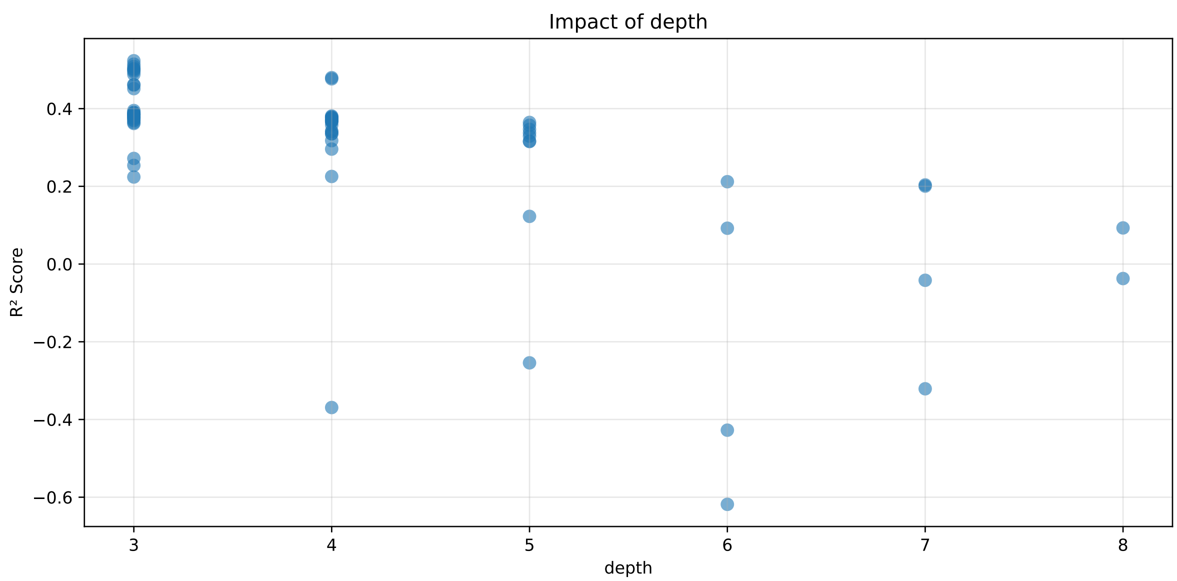
==================== TOP Importance FEATURES ==================== feature importance r_umidity 40.393078 r_temperature_fft_4h_fft_power1 11.380640 r_temperature 10.710744 r_NH3 10.632743 r_CO2 8.483034 r_THI 6.327373 r_NH3_fft_4h_fft_power1 4.501414 r_umidity_fft_4h_fft_power1 3.977251 r_temperature_fft_4h_fft_period1_h 2.495226 r_umidity_fft_4h_fft_f1_cph 0.663853 r_NH3_fft_4h_fft_period1_h 0.434645

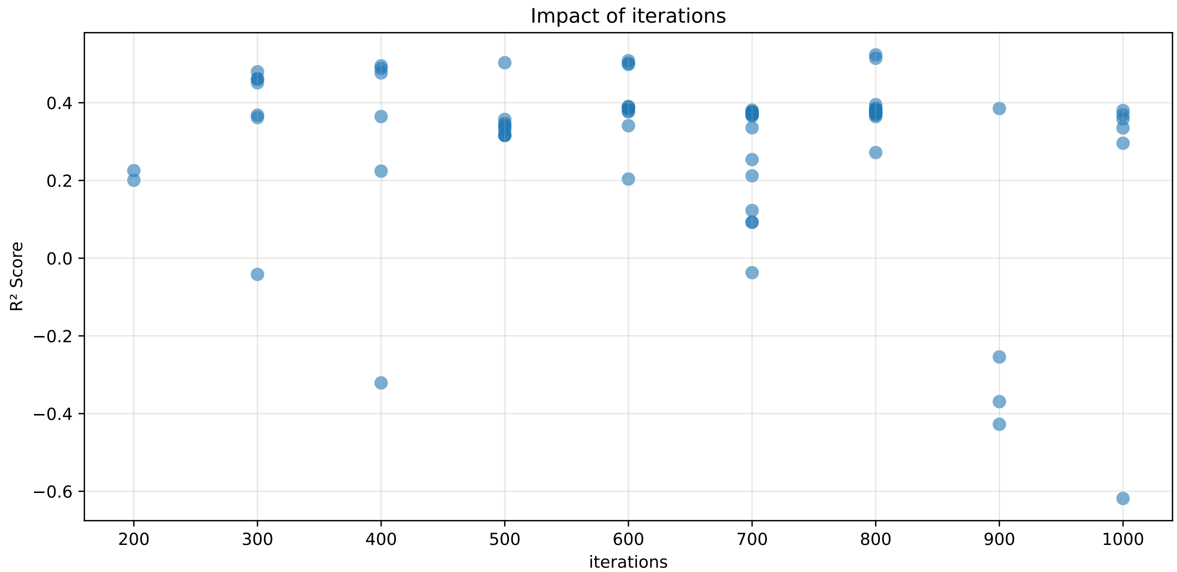


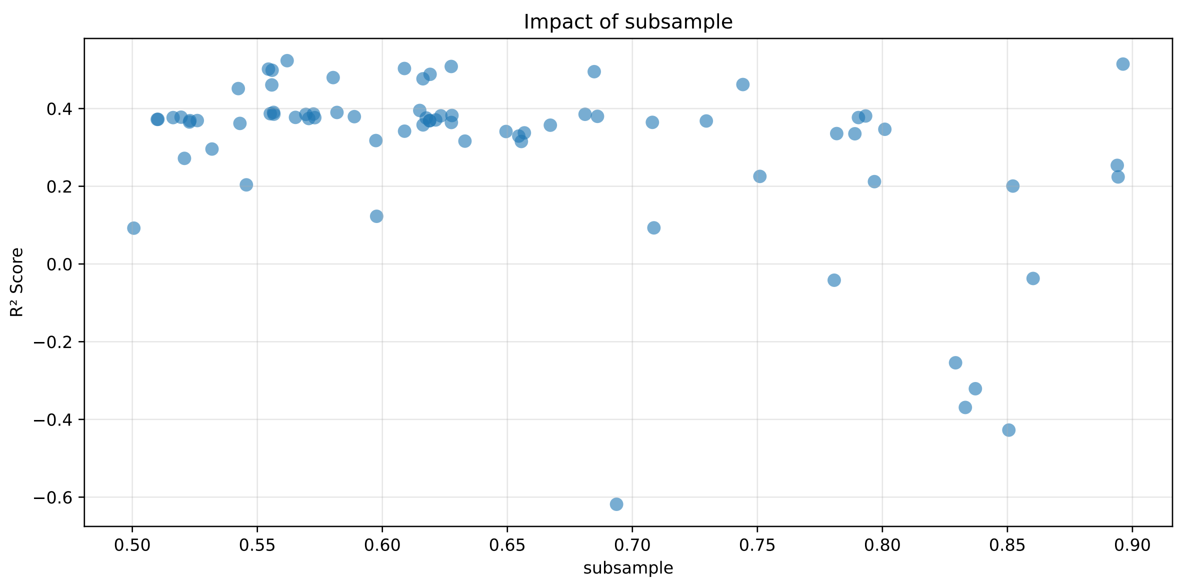


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'iterations', 'subsample']


Impact of learning_rate
R² Score
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.02
0.04
0.06
0.08
learning_rate

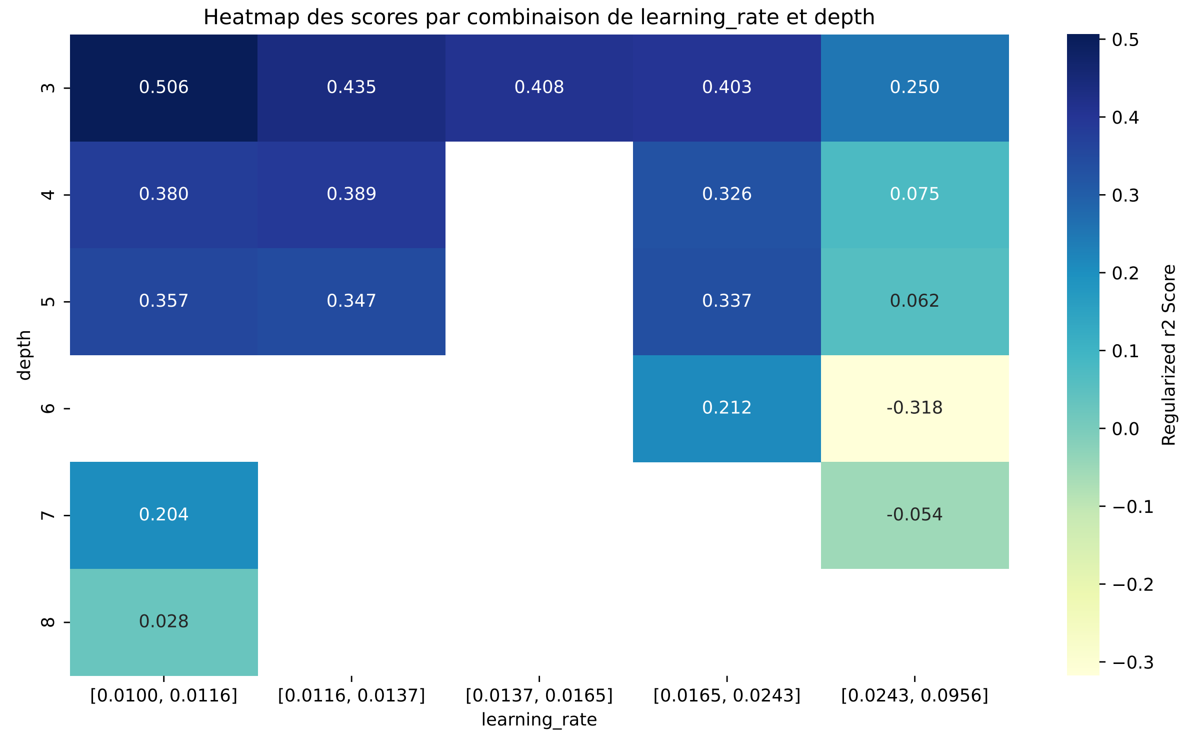

Impact of depth
R² Score
0.4
0.2
0.0
−0.2
−0.4
−0.6
3
4
5
6
7
8
depth






```
C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu
```



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5488	0.5232	0.509	0.0142	0.5232	0.5129	-0.0104	0.9802	1.0489
2	0.5379	0.5145	0.4999	0.0134	0.5145	0.5106	-0.0038	0.9925	1.0455
3	0.532	0.5083	0.4945	0.013	0.5083	0.498	-0.0103	0.9798	1.0466
4	0.5261	0.5031	0.4889	0.0131	0.5031	0.4949	-0.0082	0.9836	1.0457
5	0.5231	0.5015	0.4887	0.0135	0.5015	0.4953	-0.0062	0.9877	1.043
6	0.5197	0.4985	0.4862	0.015	0.4985	0.4906	-0.0079	0.9842	1.0424
7	0.5153	0.4949	0.4813	0.0153	0.4949	0.4862	-0.0088	0.9823	1.0412
8	0.508	0.4881	0.4769	0.0154	0.4881	0.4785	-0.0096	0.9804	1.0408
9	0.5013	0.4797	0.4696	0.016	0.4797	0.4749	-0.0048	0.99	1.0449
10	0.499	0.4766	0.4678	0.0158	0.4766	0.4708	-0.0058	0.9878	1.047
---	---	---	---	---	---	---	---	---	---
75	0.9266	0.6691	0.6525	0.0029	-0.6184	0.6681	-0.0011	0.9984	1.3848

Hyperparameters:

Rank 1: {'reg___iterations': 800, 'reg___depth': 3, 'reg___learning_rate': 0.013591031654497538, 'reg___l2_leaf_reg': 44.915271724589275, 'reg___bagging_temperature': 1.371080508624839, 'reg___random_strength': 4.979329395295099, 'reg___subsample': 0.5619252015937989, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.8499884335435609, 'reg___border_count': 74, 'reg___leaf_estimation_method': 'Newton',

’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4075.7350078503646}

Rank 2: {’reg__iterations’: 800, ’reg__depth’: 3, ’reg__learning_rate’: 0.01410938200100792, ’reg__l2_leaf_reg’: 38.746358657302636, ’reg__bagging_temperature’: 4.567443722310311, ’reg__random_strength’: 4.95775243504962, ’reg__subsample’: 0.8962367190415966, ’reg__min_data_in_leaf’: 24, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’: 0.7356080615900916, ’reg__border_count’: 45, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 1868.010464810934}

Rank 3: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.010591852609097206, ’reg__l2_leaf_reg’: 5.284607493328661, ’reg__bagging_temperature’: 1.78112536713494, ’reg__random_strength’: 2.1814604246610845, ’reg__subsample’: 0.6275573162479436, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 6, ’reg__rsm’: 0.8210480584480272, ’reg__border_count’: 74, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 9829.94464098498}

Rank 4: {’reg__iterations’: 500, ’reg__depth’: 3, ’reg__learning_rate’: 0.01061533096151017, ’reg__l2_leaf_reg’: 2.3122758216271784, ’reg__bagging_temperature’: 1.7992389451507393, ’reg__random_strength’: 1.559736290232412, ’reg__subsample’: 0.6088411534100494, ’reg__min_data_in_leaf’: 50, ’reg__leaf_estimation_iterations’: 1, ’reg__rsm’: 0.8557447890694618, ’reg__border_count’: 119, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 2, ’reg__diffusion_temperature’: 9898.119684216637}

Rank 5: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.013821039242850403, ’reg__l2_leaf_reg’: 10.400290652320423, ’reg__bagging_temperature’: 1.4416003693352963, ’reg__random_strength’: 4.978884433937079, ’reg__subsample’: 0.5544479691619021, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 2, ’reg__rsm’: 0.8169182561782957, ’reg__border_count’: 74, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 4202.573786228546}

Rank 6: {’reg__iterations’: 600, ’reg__depth’: 3, ’reg__learning_rate’: 0.01323255320863796, ’reg__l2_leaf_reg’: 8.57091744463375, ’reg__bagging_temperature’: 1.2488594025905841, ’reg__random_strength’: 4.926991149915073, ’reg__subsample’: 0.5559119307100187, ’reg__min_data_in_leaf’: 26, ’reg__leaf_estimation_iterations’: 2, ’reg__rsm’: 0.8139720772603135, ’reg__border_count’: 75, ’reg__leaf_estimation_method’: ‘Newton’, ’reg__boosting_type’: ‘Ordered’, ’reg__max_ctr_complexity’: 1, ’reg__diffusion_temperature’: 6391.500281514372}

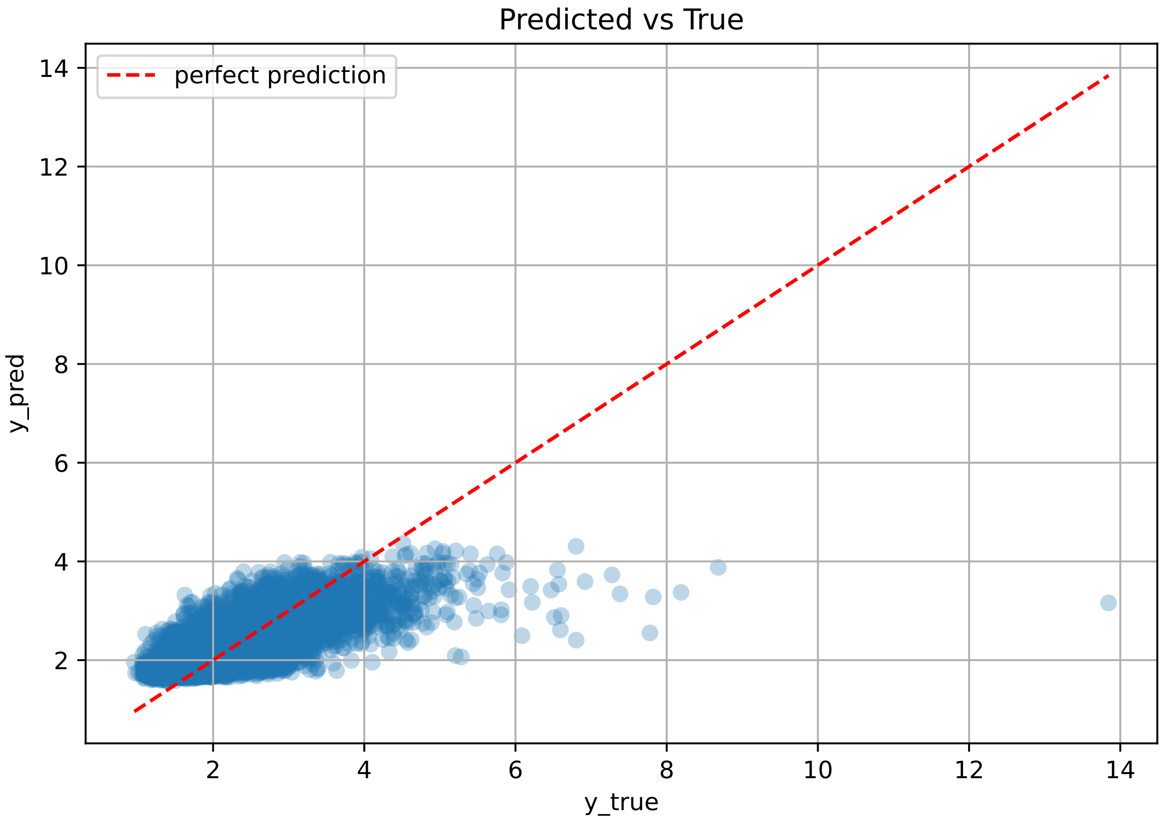
Rank 7: {’reg__iterations’: 400, ’reg__depth’: 3, ’reg__learning_rate’: 0.018190362898281506, ’reg__l2_leaf_reg’: 6.296674739122489, ’reg__bagging_temperature’: 0.5464838556207563, ’reg__random_strength’: 4.332332118818856, ’reg__subsample’: 0.6847483281339739, 'reg___min_data_in_leaf': 21, 'reg___leaf_estimation_iterations': 1, 'reg___rsm': 0.7301764392488712, 'reg___border_count': 128, 'reg___leaf_estimation_method': 'Gradient', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 2383.2609980498455}

Rank 8: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.01536317058773128, 'reg___l2_leaf_reg': 6.322268378160859, 'reg___bagging_temperature': 0.9097396043872946, 'reg___random_strength': 3.9167009964468367, 'reg___subsample': 0.6190944382331595, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.792270057111746, 'reg___border_count': 94, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 2050.810505324253}

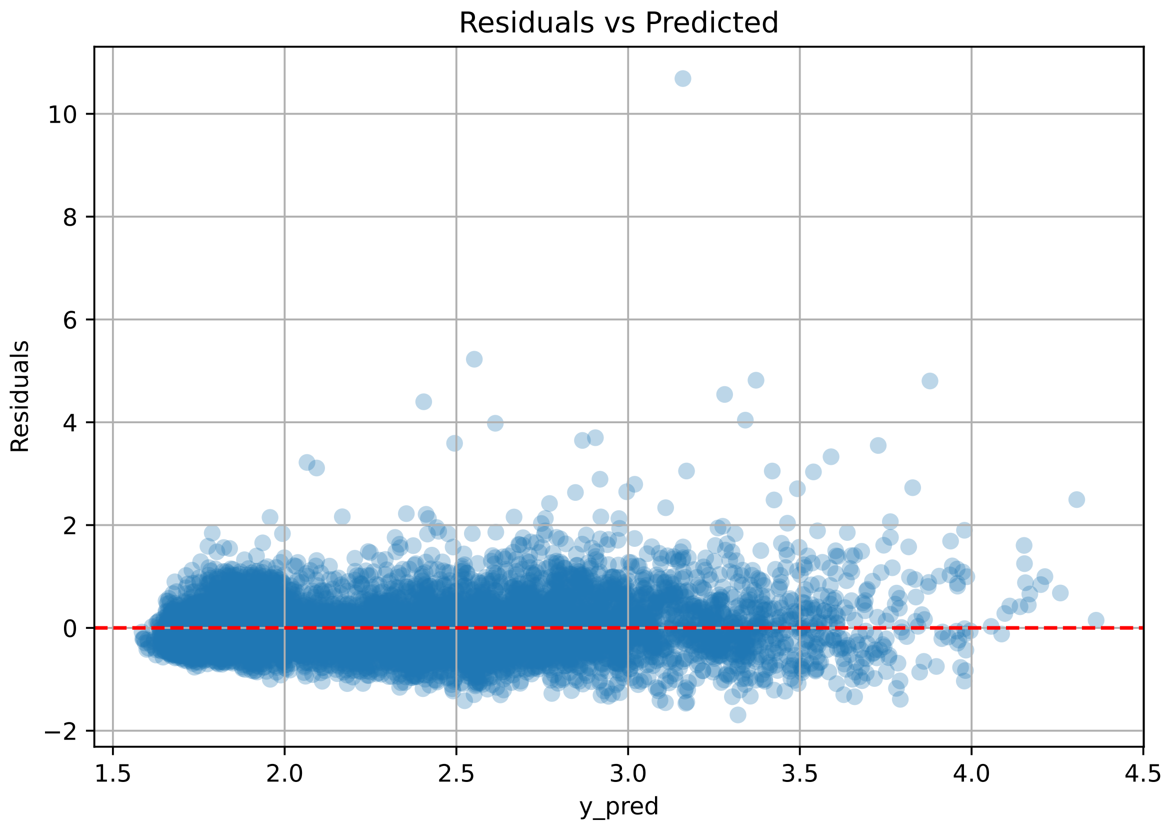
Rank 9: {'reg___iterations': 300, 'reg___depth': 4, 'reg___learning_rate': 0.013434553037867902, 'reg___l2_leaf_reg': 9.677982573449126, 'reg___bagging_temperature': 1.432751722340306, 'reg___random_strength': 4.9434237876739955, 'reg___subsample': 0.580335120620103, 'reg___min_data_in_leaf': 27, 'reg___leaf_estimation_iterations': 2, 'reg___rsm': 0.8414588960650655, 'reg___border_count': 75, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 1, 'reg___diffusion_temperature': 9896.230083413788}

Rank 10: {'reg___iterations': 400, 'reg___depth': 4, 'reg___learning_rate': 0.011388024544389693, 'reg___l2_leaf_reg': 13.897613876980882, 'reg___bagging_temperature': 4.818274187628269, 'reg___random_strength': 3.8866120088025125, 'reg___subsample': 0.6162557345987246, 'reg___min_data_in_leaf': 30, 'reg___leaf_estimation_iterations': 3, 'reg___rsm': 0.7919647536557833, 'reg___border_count': 94, 'reg___leaf_estimation_method': 'Newton', 'reg___boosting_type': 'Ordered', 'reg___max_ctr_complexity': 3, 'reg___diffusion_temperature': 111.31721494211479}

3.6.2 Scatter


Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

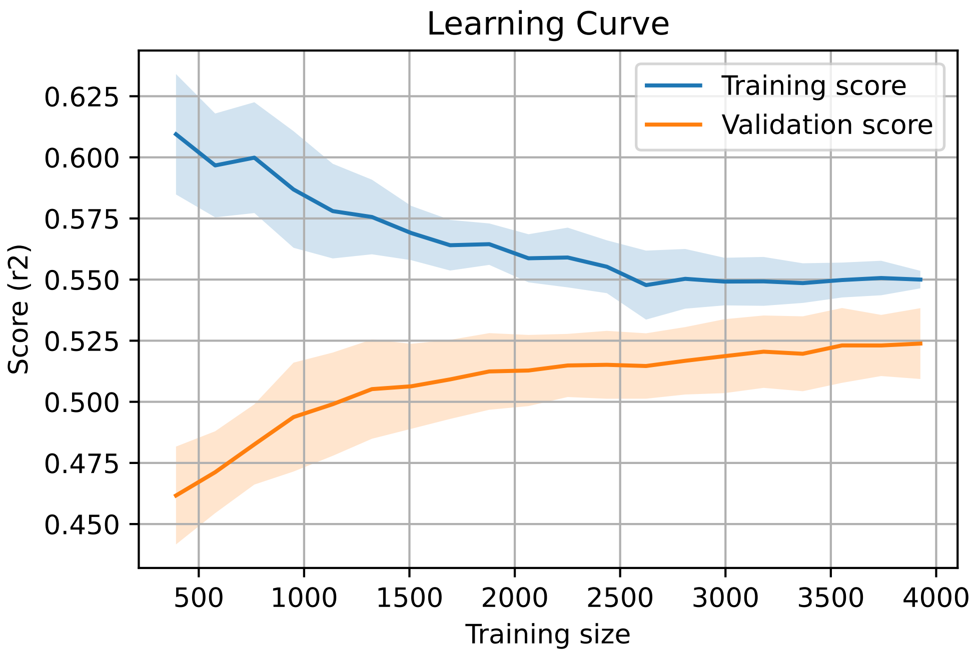
3.6.3 Scatter residuals



3.6.4 Learning curve

Len(group) : 8012, **Len(training)** : 5609,

Len(training_real) : 3927, **Len(test_of_training)** : 1682, **Len(test)** : 2403



3.7 Sumup-table

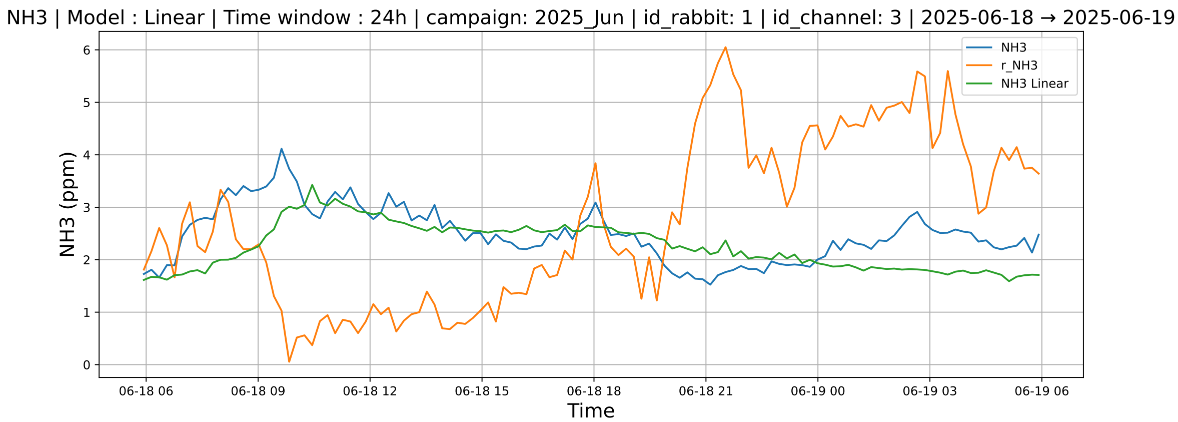
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.416	0.41	0.013	0.4	0.4	-0.011	0.974	1.012	nan
LGBM	0.558	0.524	0.012	0.512	0.538	0.014	1.027	1.065	0.3542
CatBoost	0.549	0.523	0.014	0.509	0.513	-0.01	0.98	1.049	0.5232

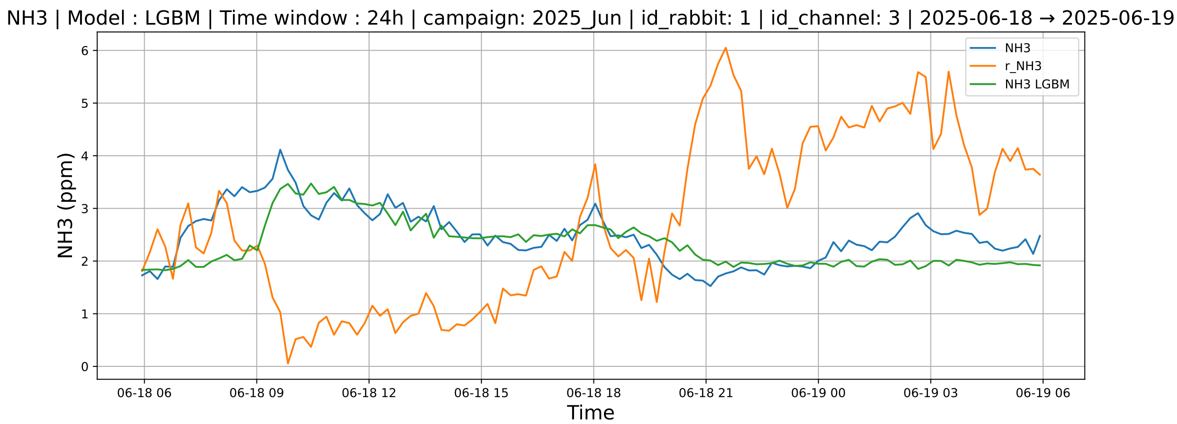
4 Plot

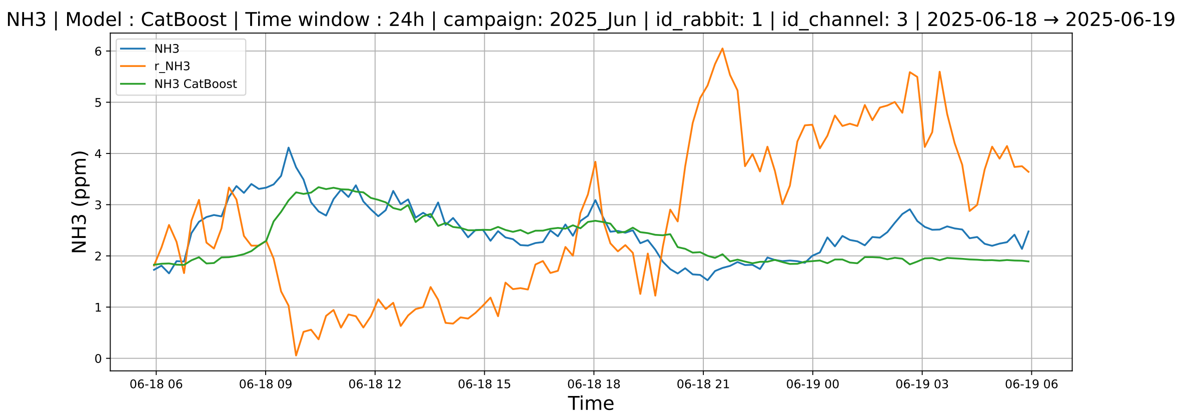
4.1 Standalone plots

4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

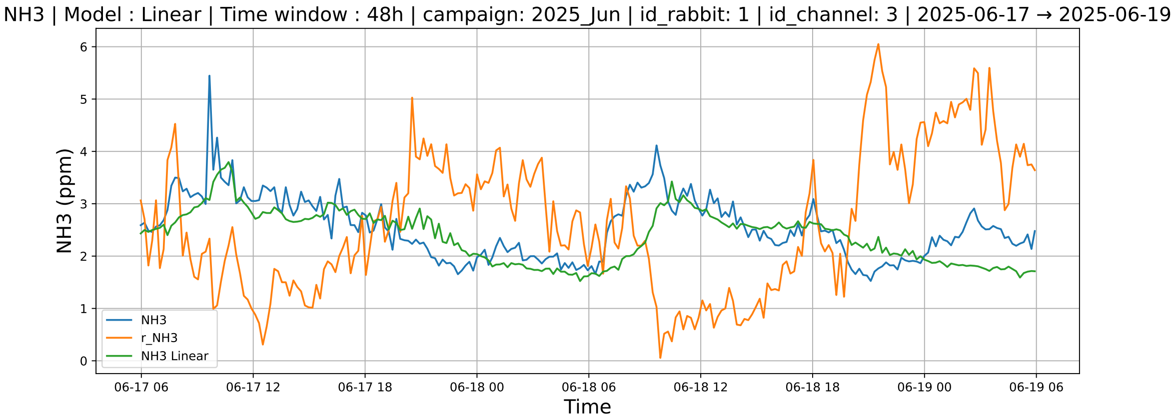
4.1.1.1 Time window : 24

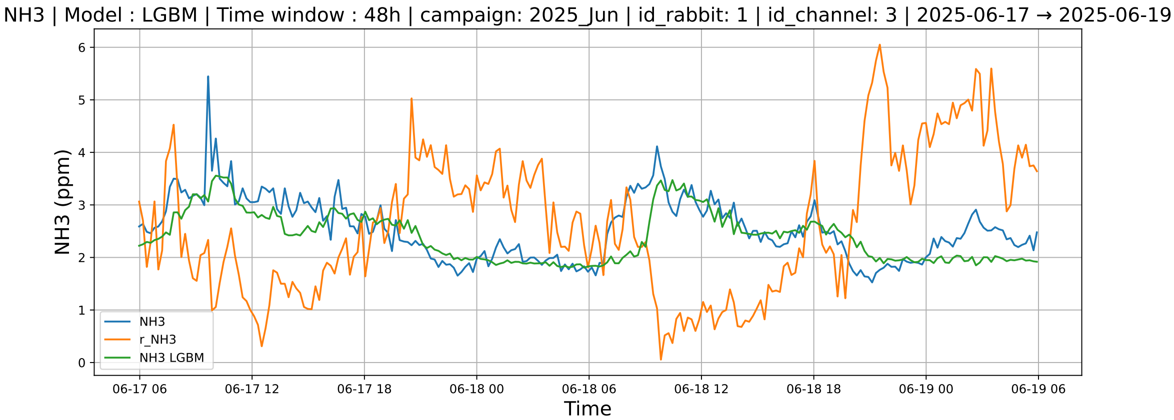


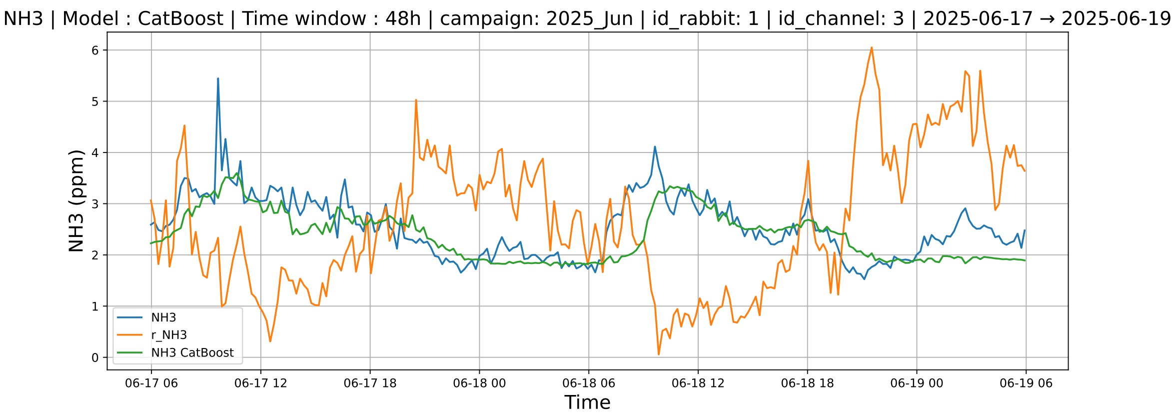




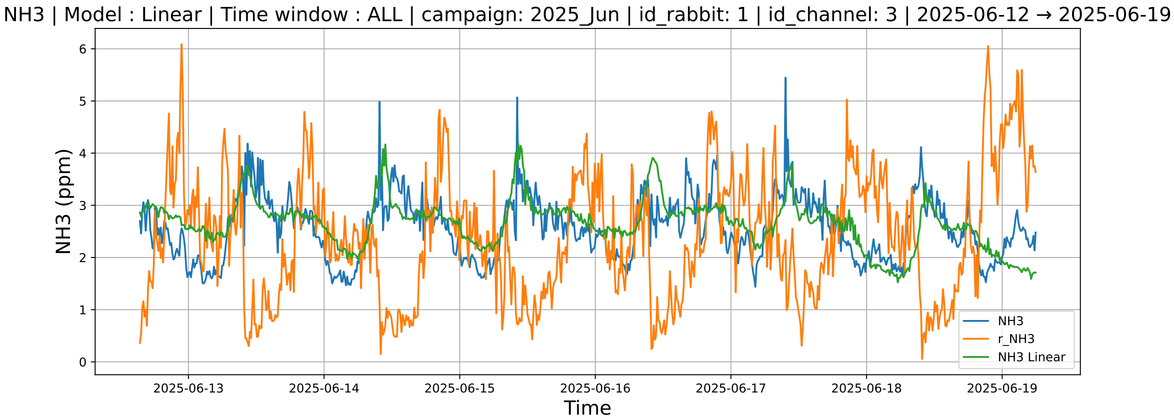
4.1.1.2 Time window : 48

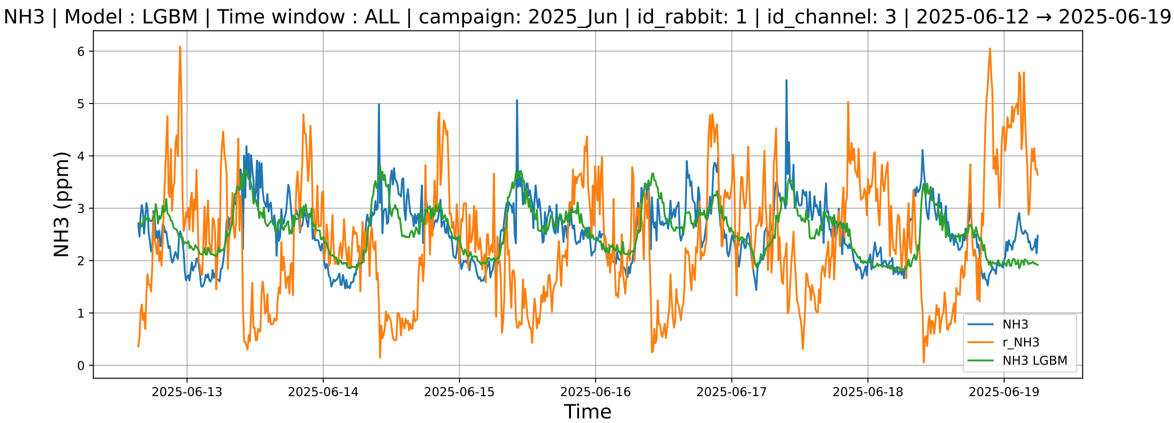


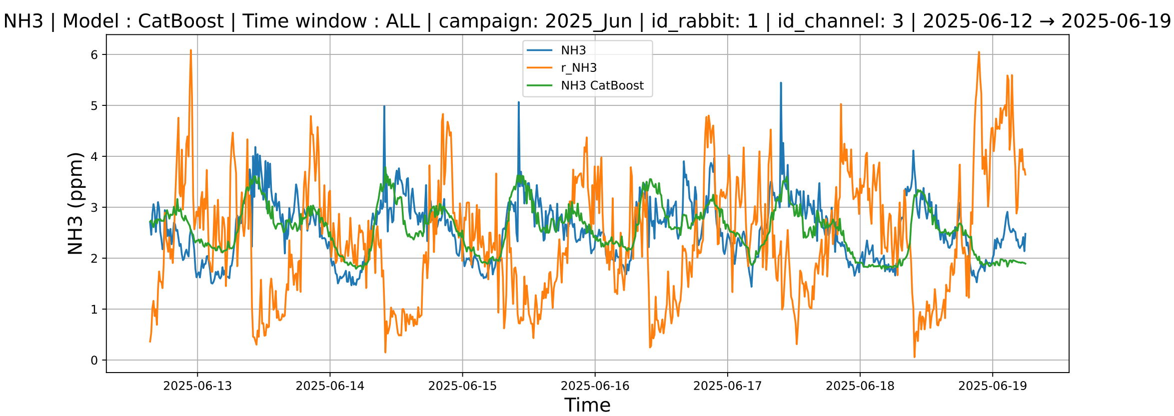




4.1.1.3 Time window : ALL

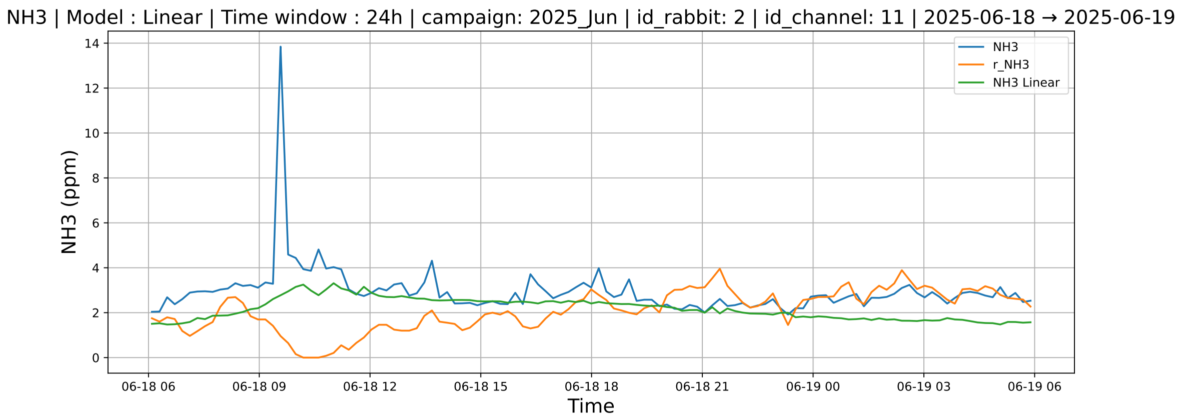


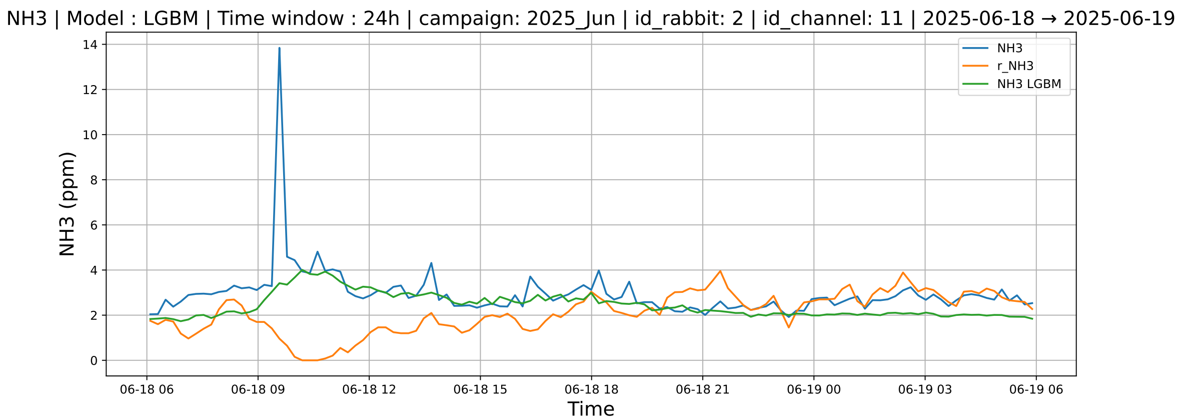


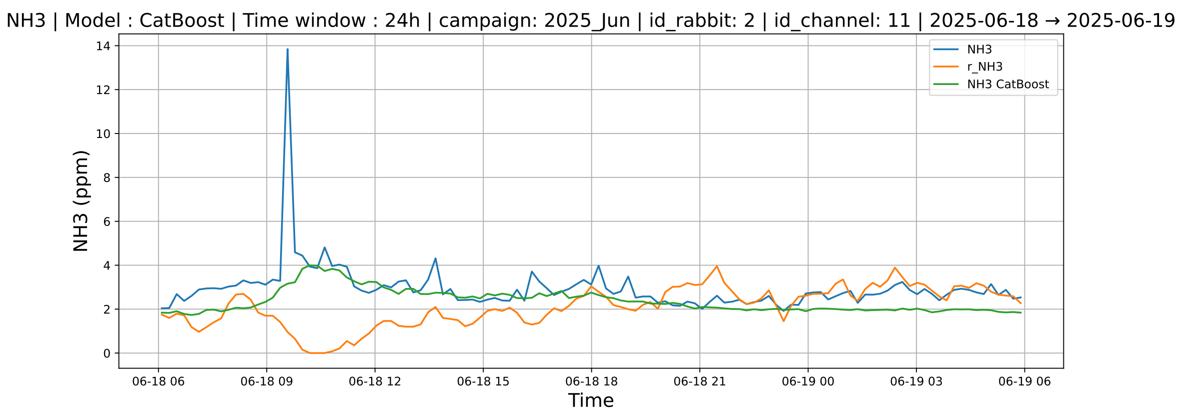


4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

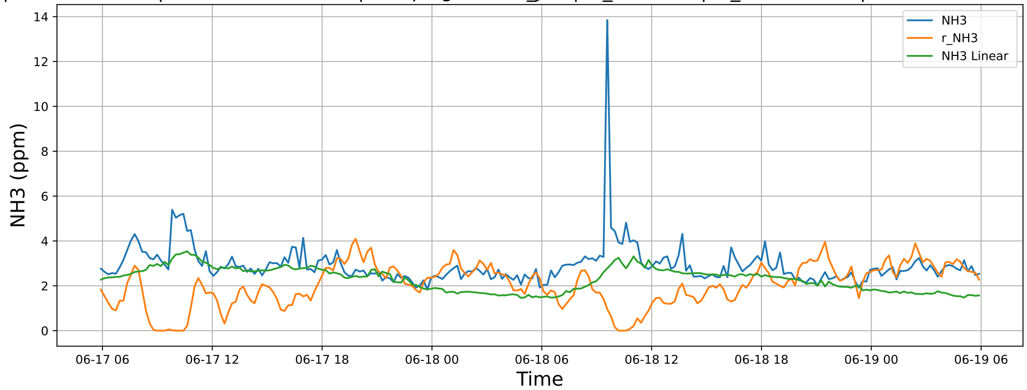




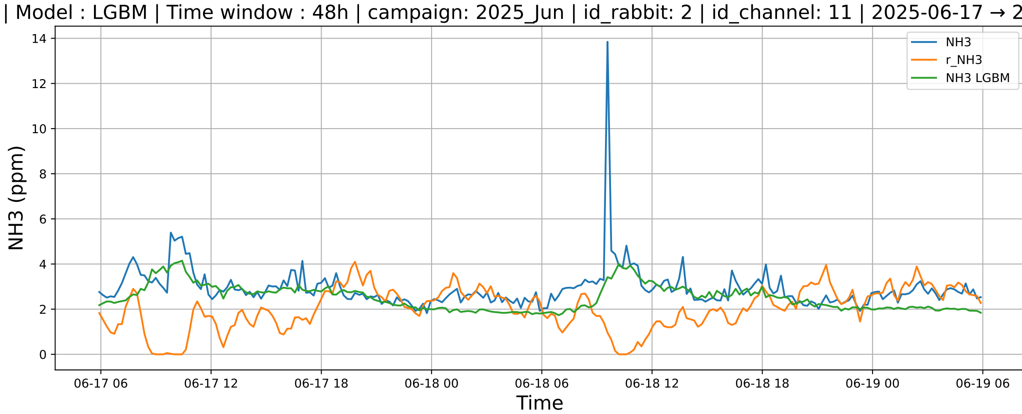


4.1.2.2 Time window : 48

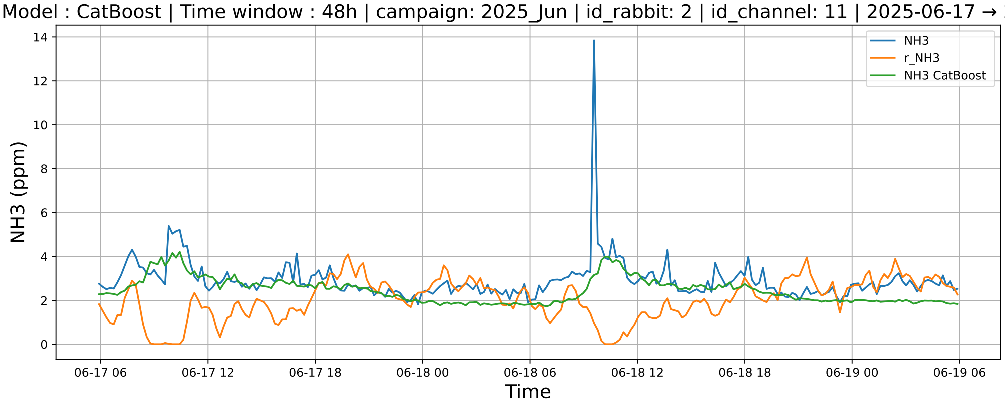
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

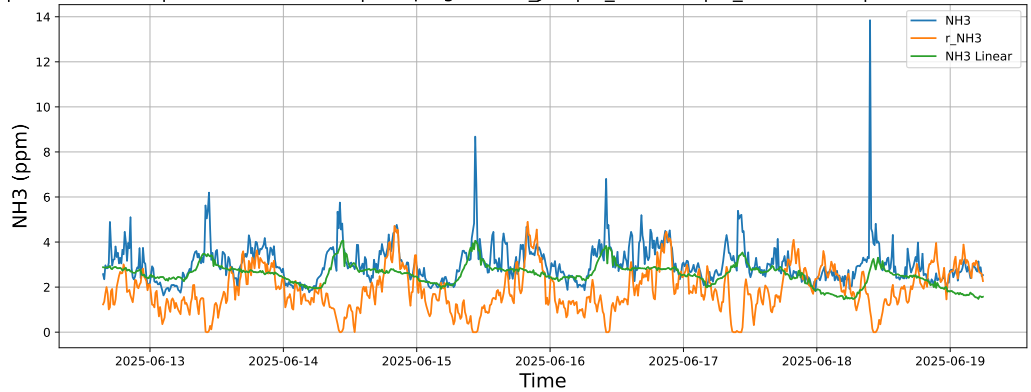


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

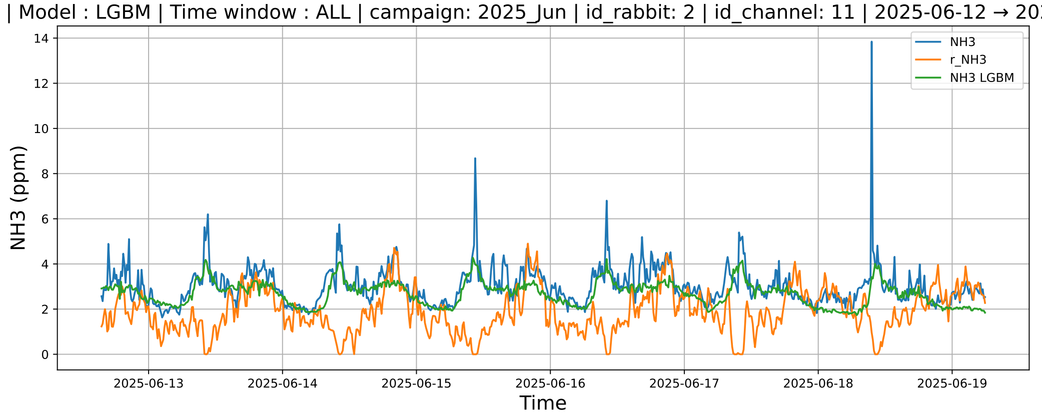


4.1.2.3 Time window : ALL

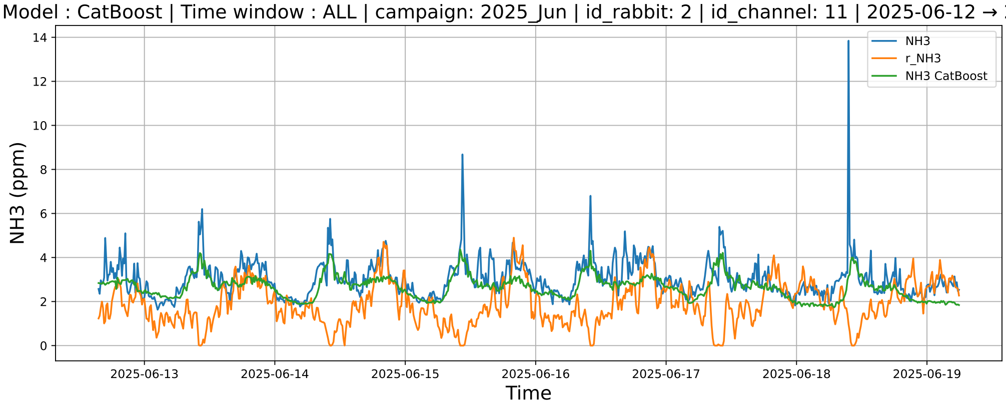
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



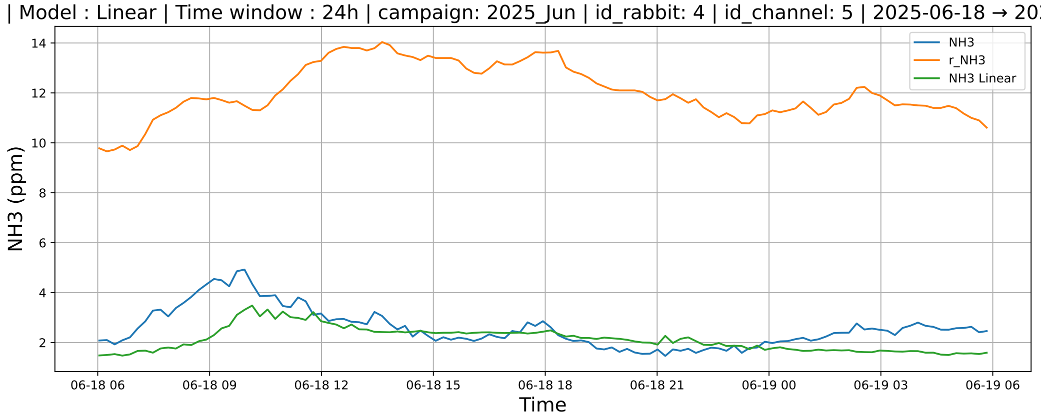
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



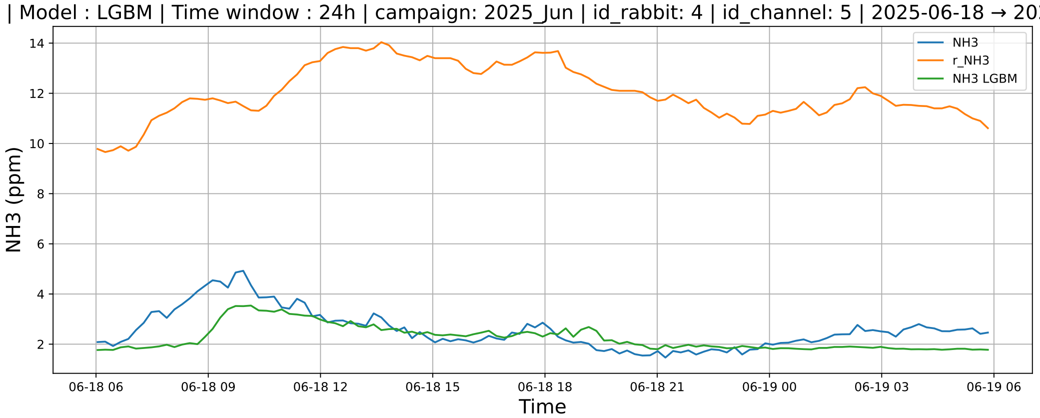
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

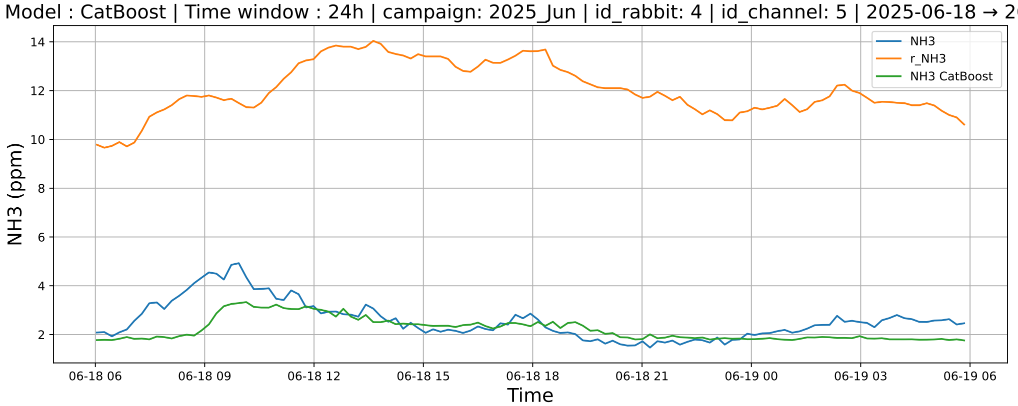
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



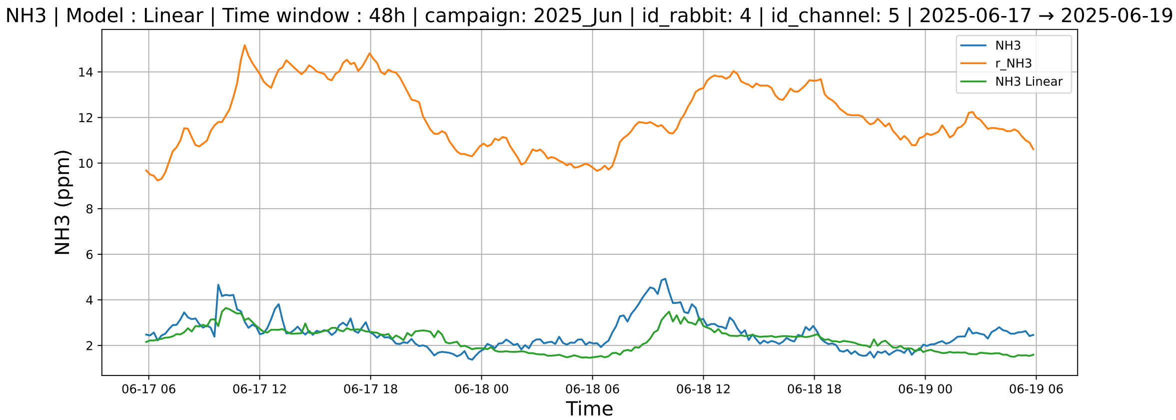
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

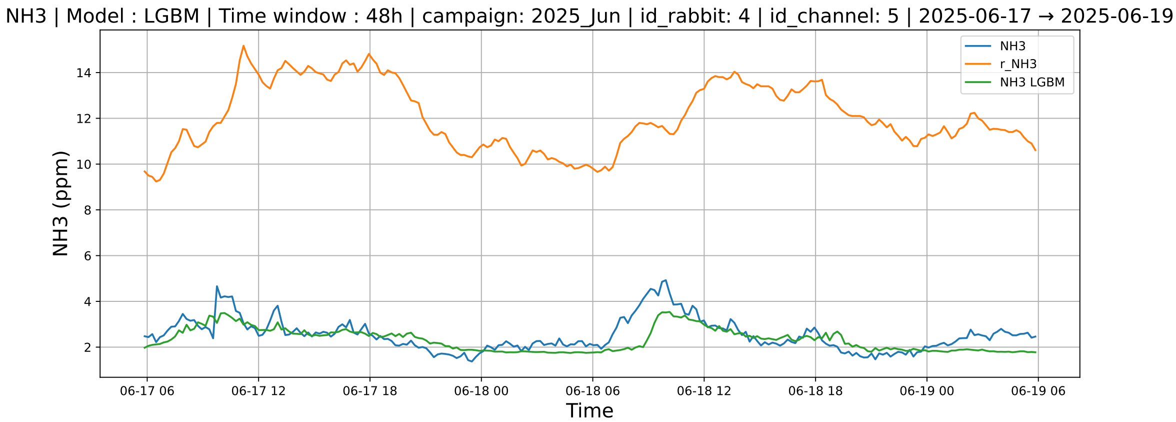


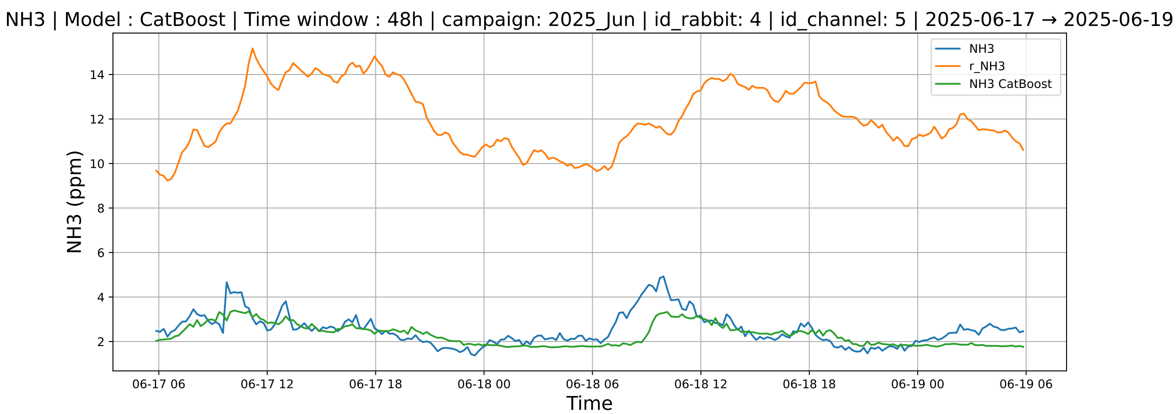
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



4.1.3.2 Time window : 48

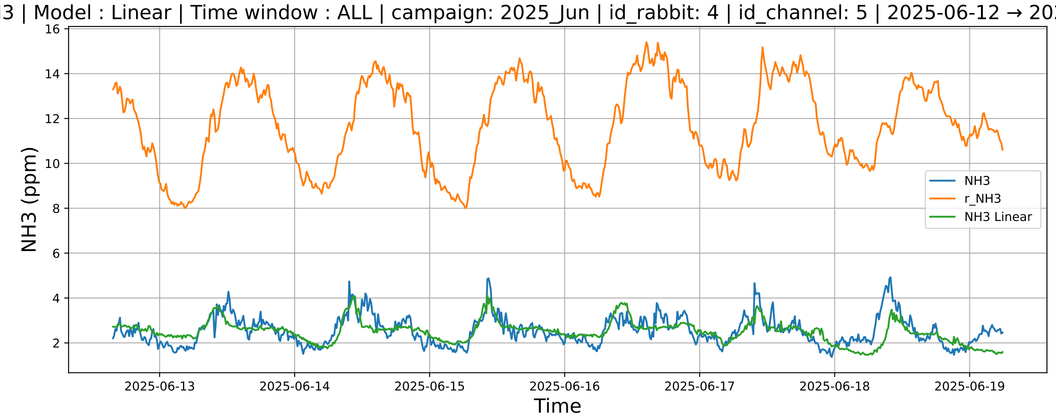




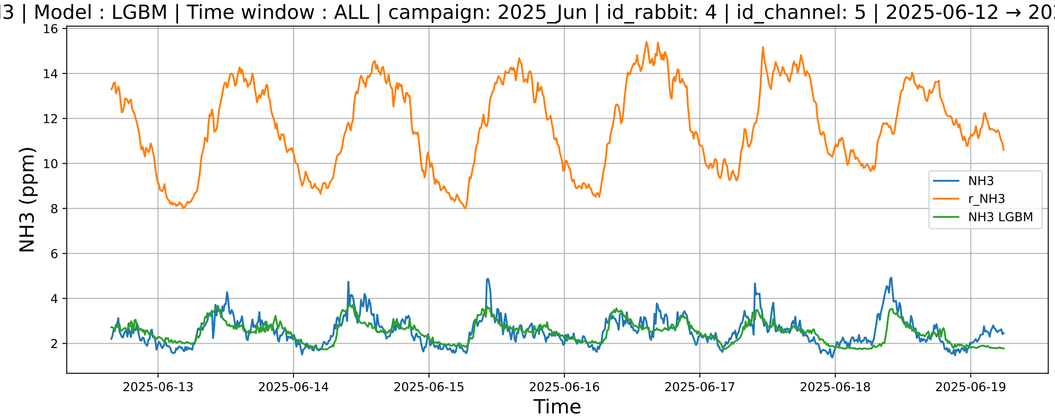


4.1.3.3 Time window : ALL

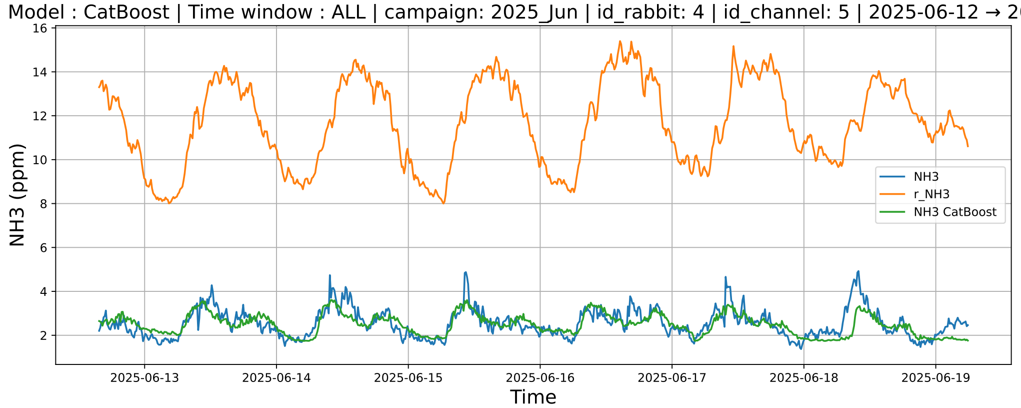
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

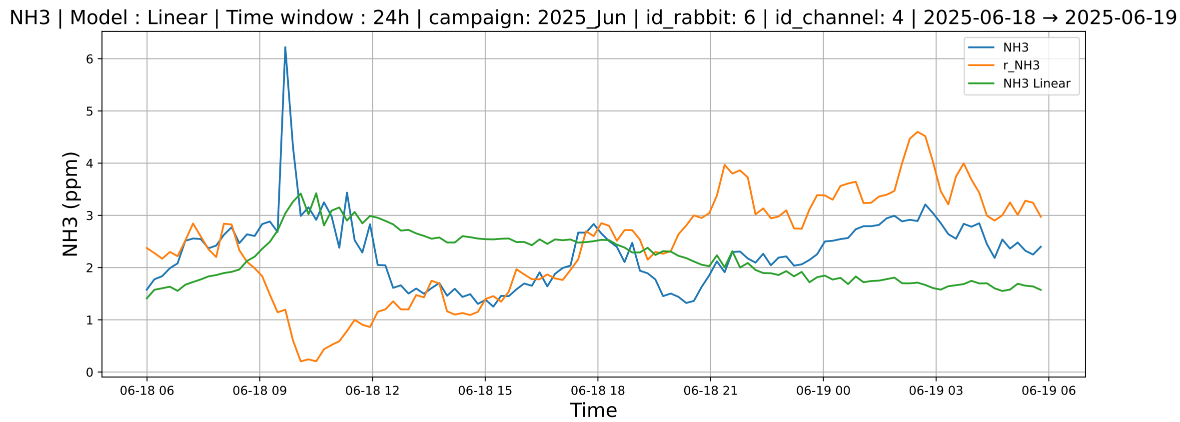


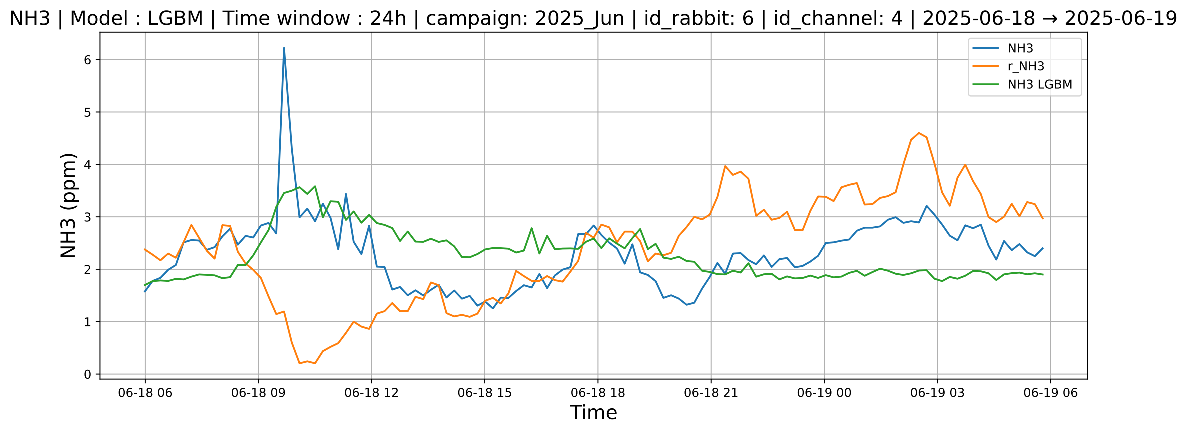
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

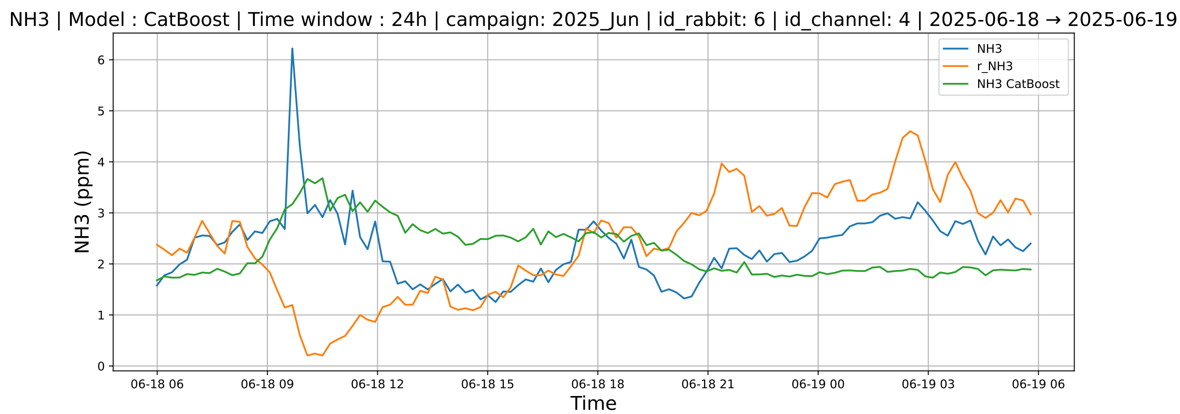


4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

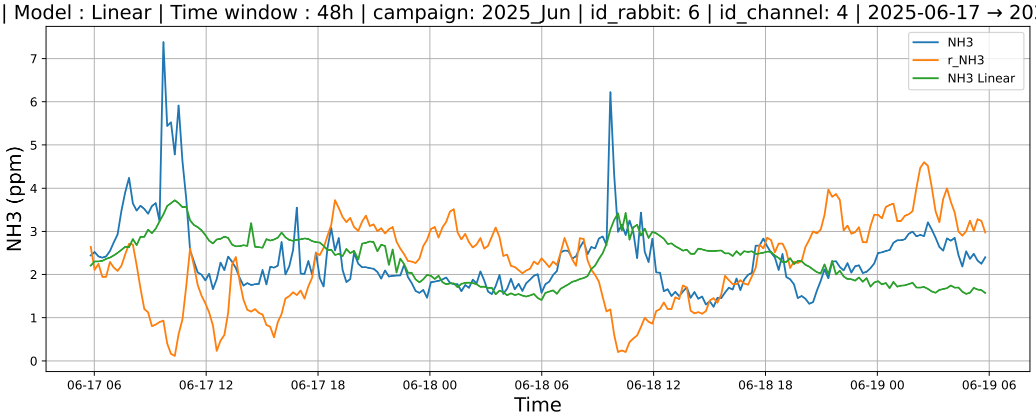




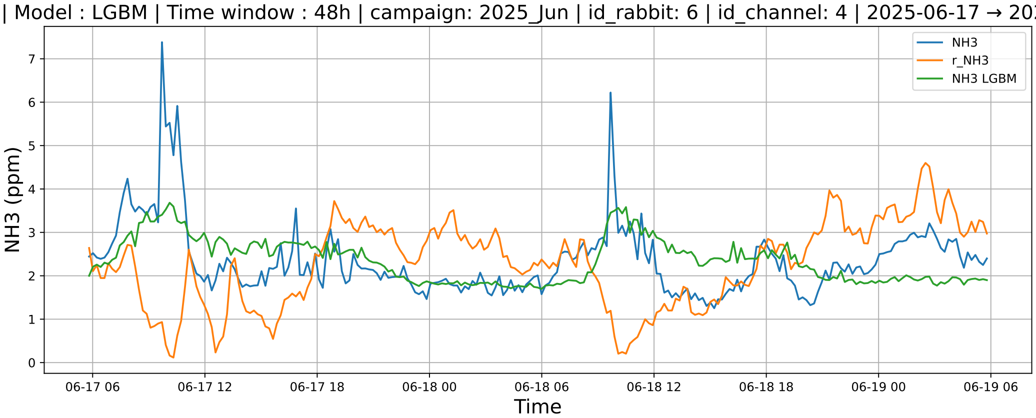


4.1.4.2 Time window : 48

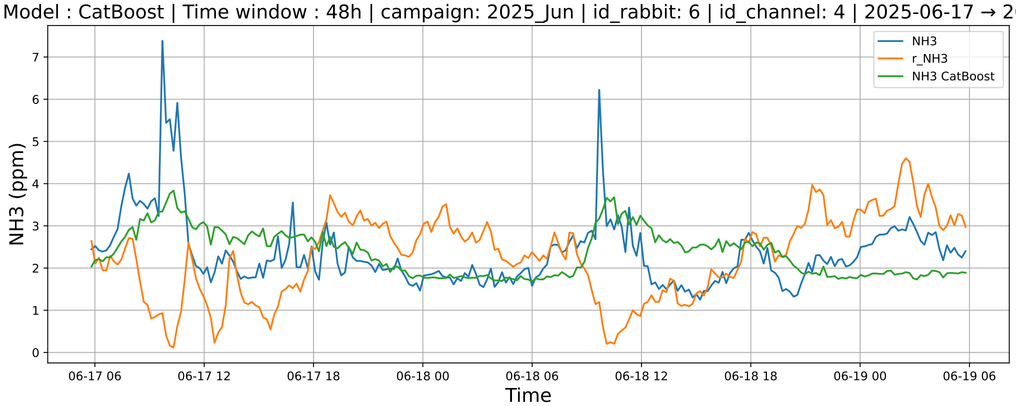
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

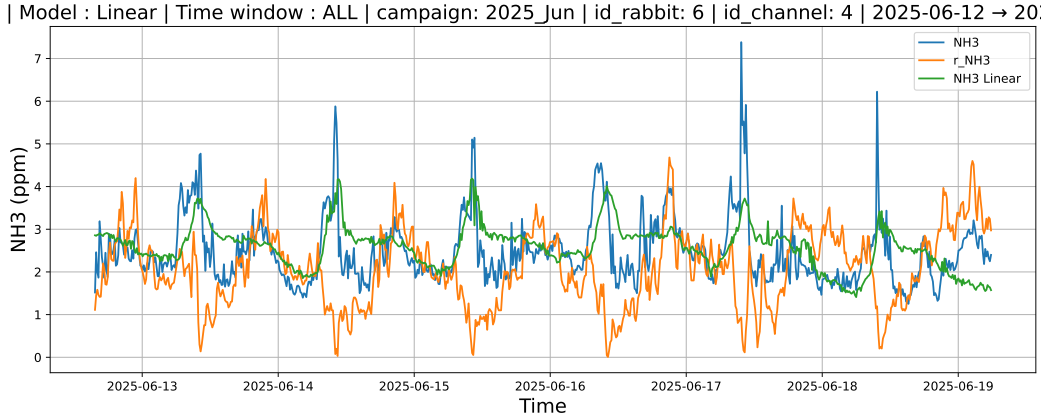


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

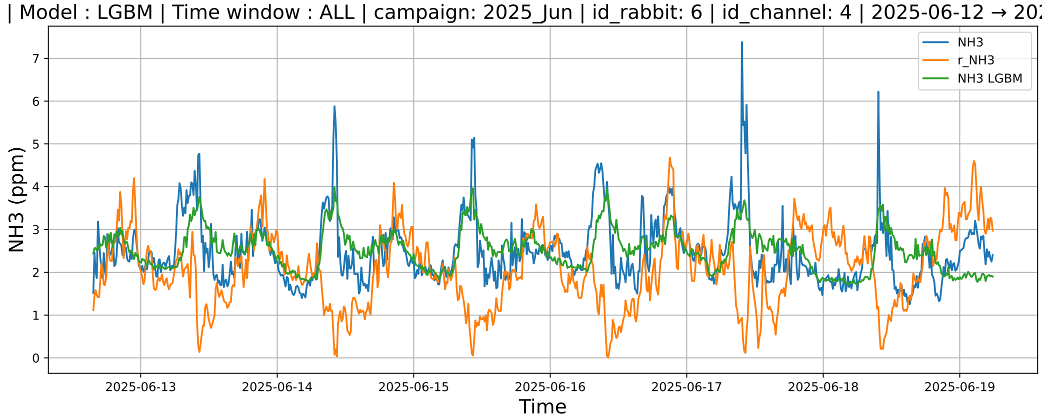


4.1.4.3 Time window : ALL

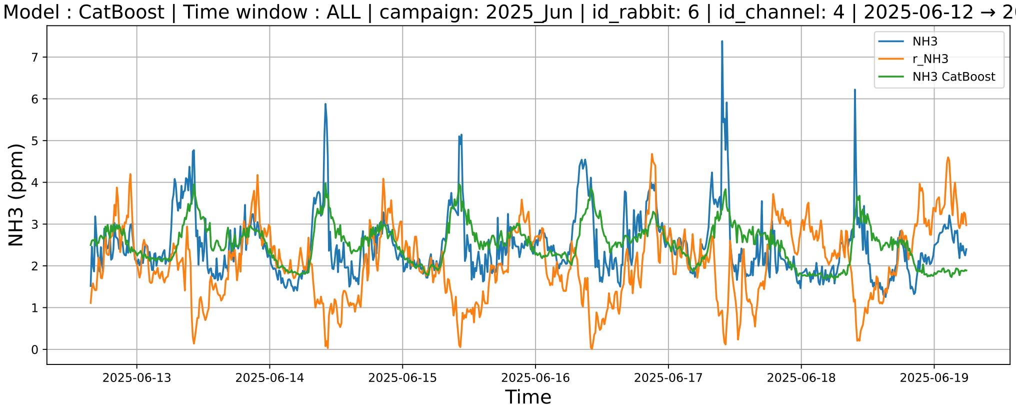
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

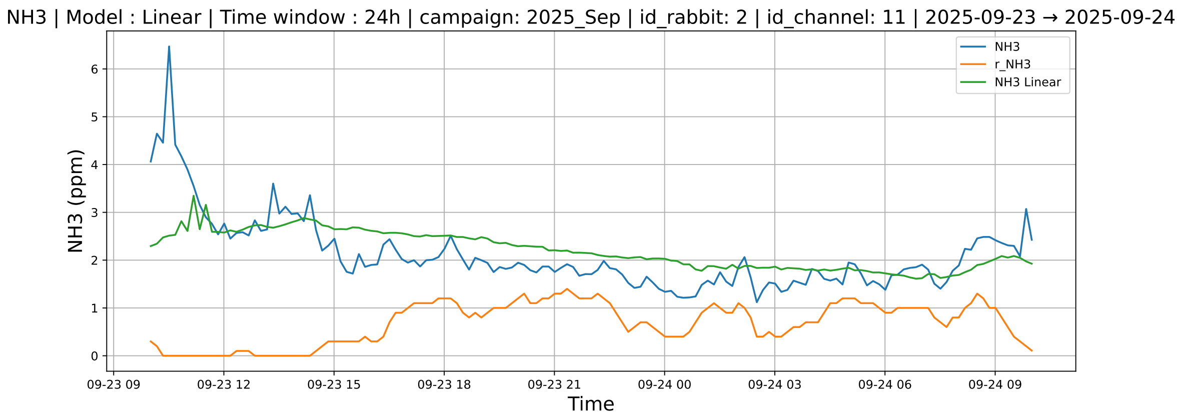


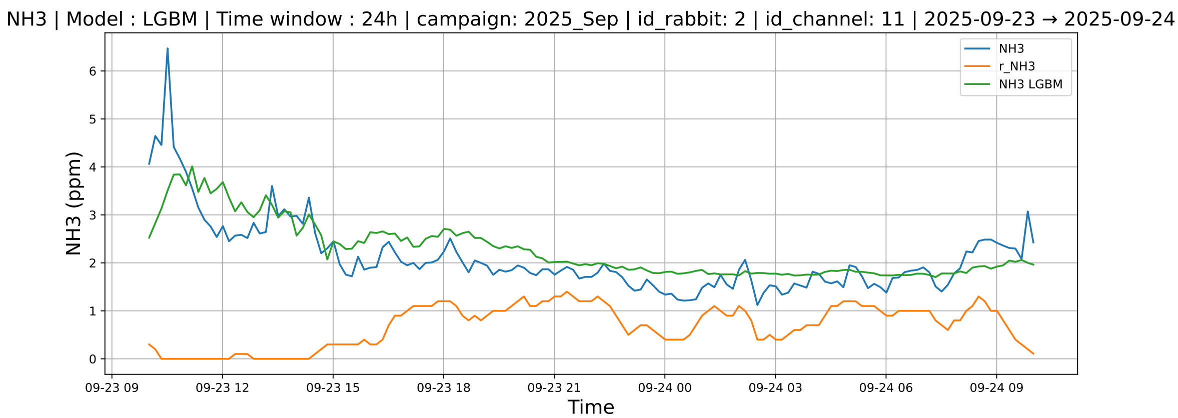
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

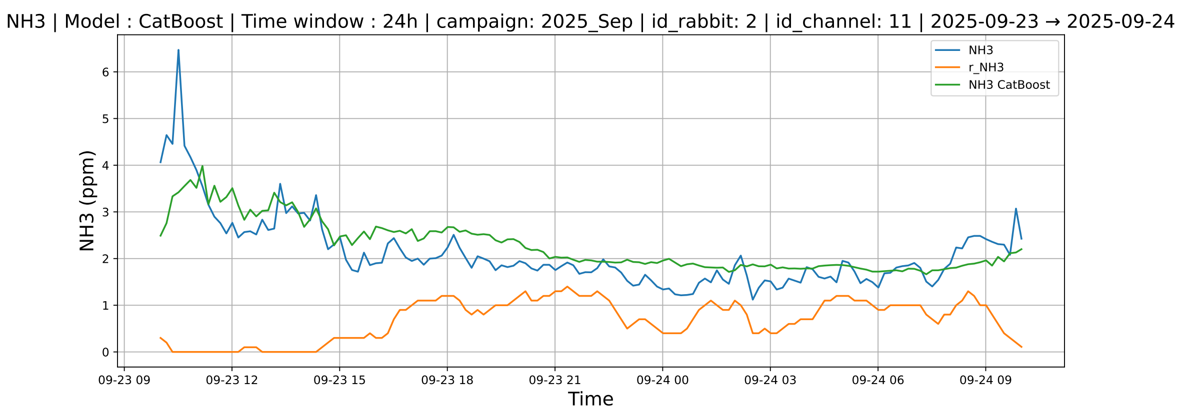


4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

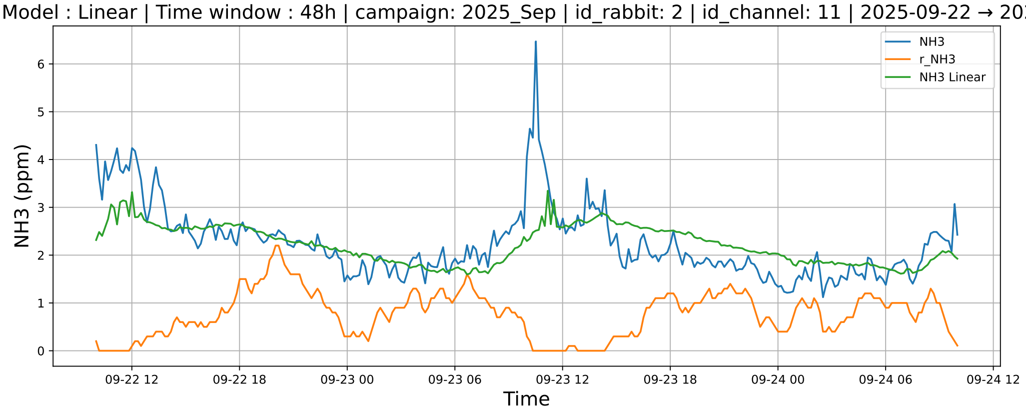




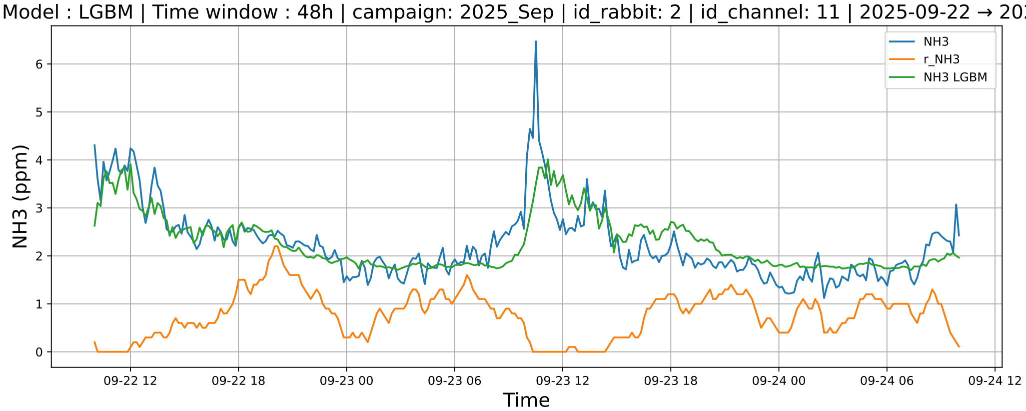


4.1.5.2 Time window : 48

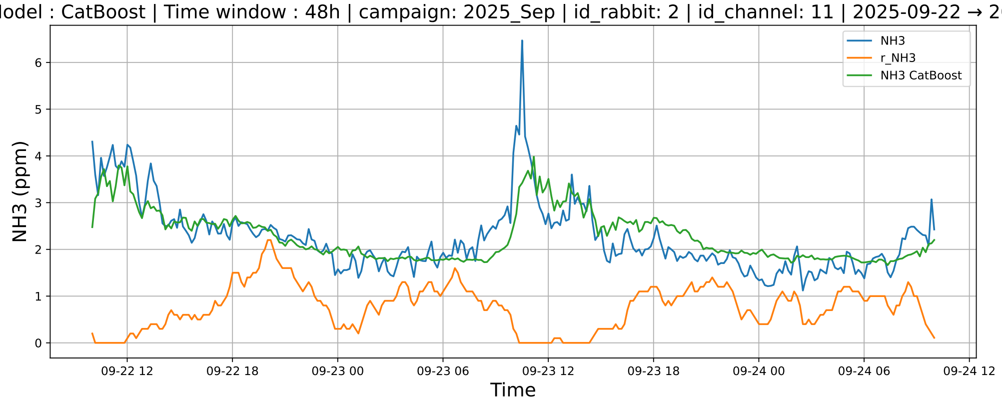
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



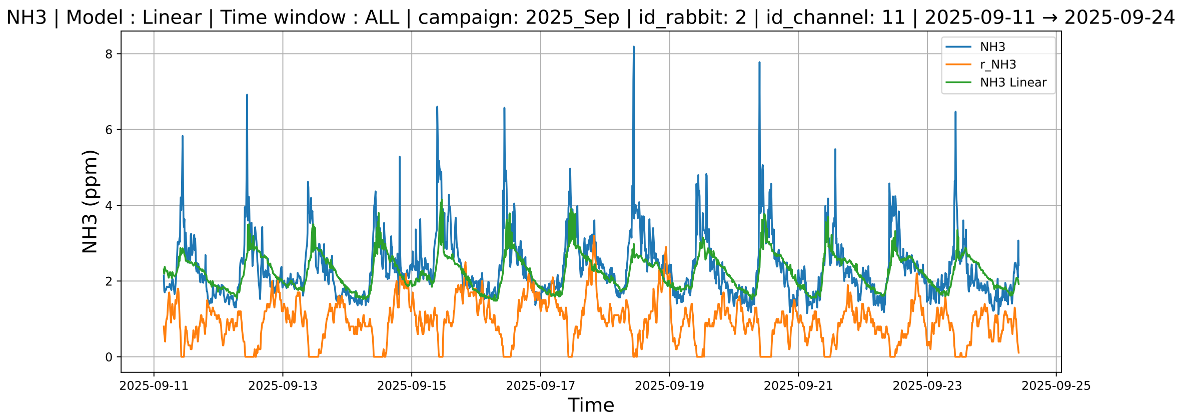
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

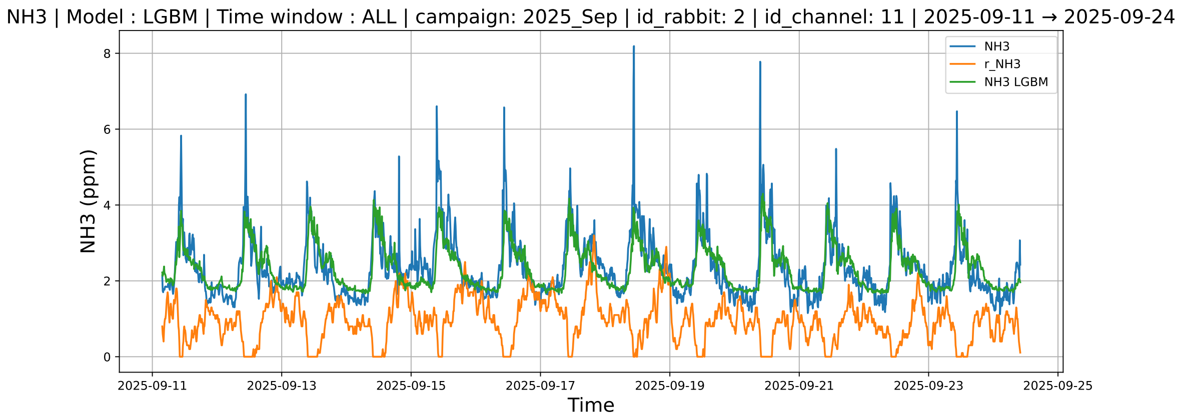


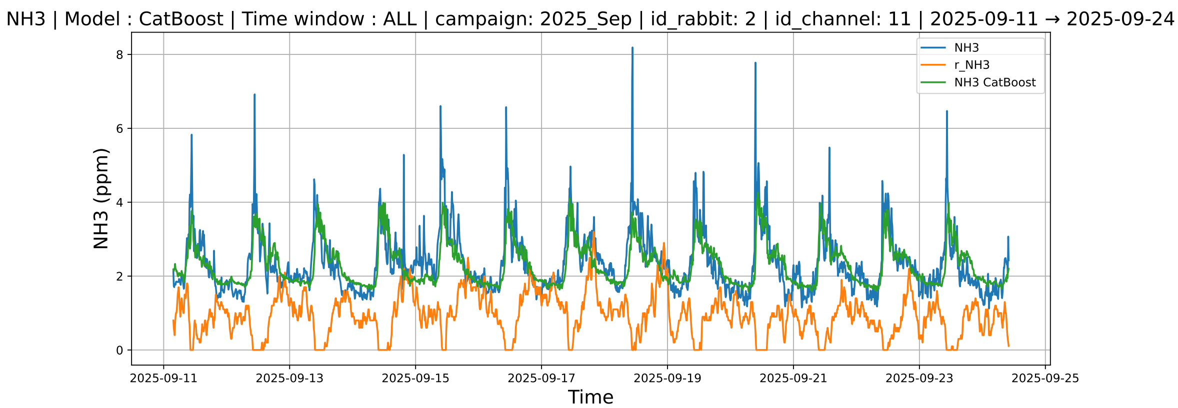
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.1.5.3 Time window : ALL

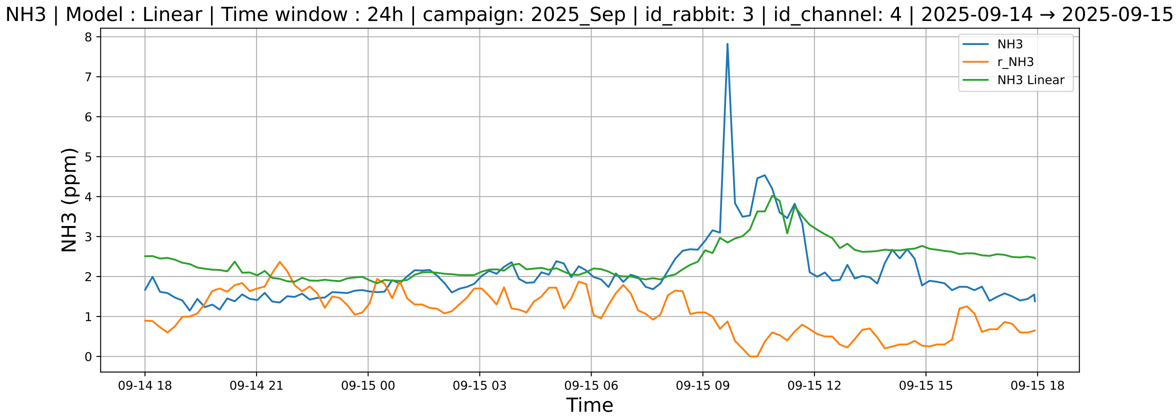


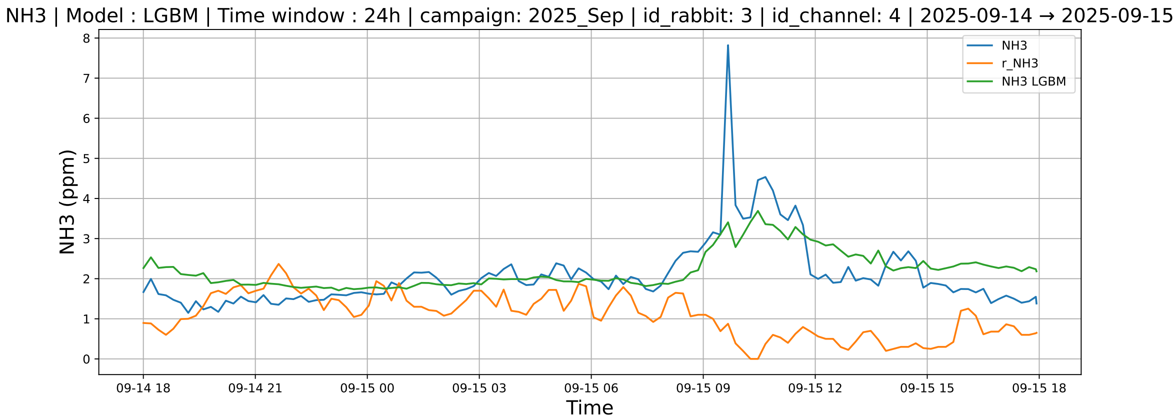


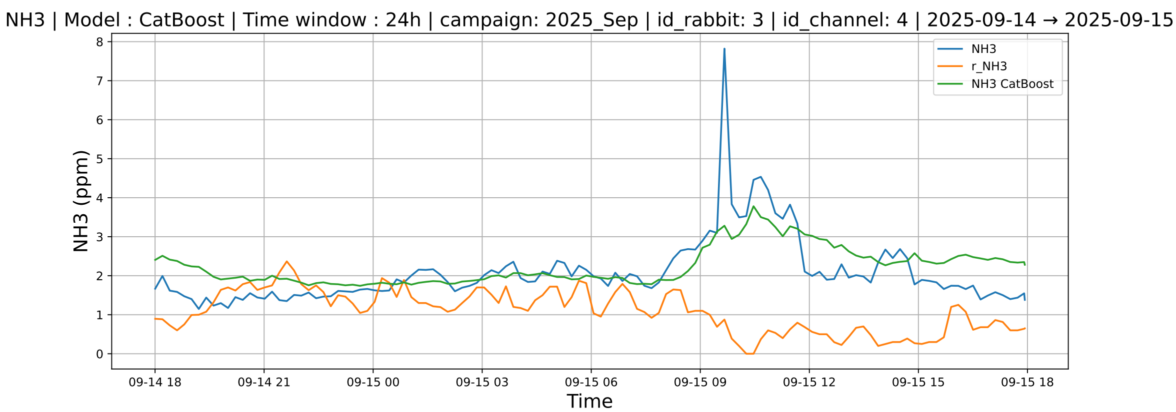


4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

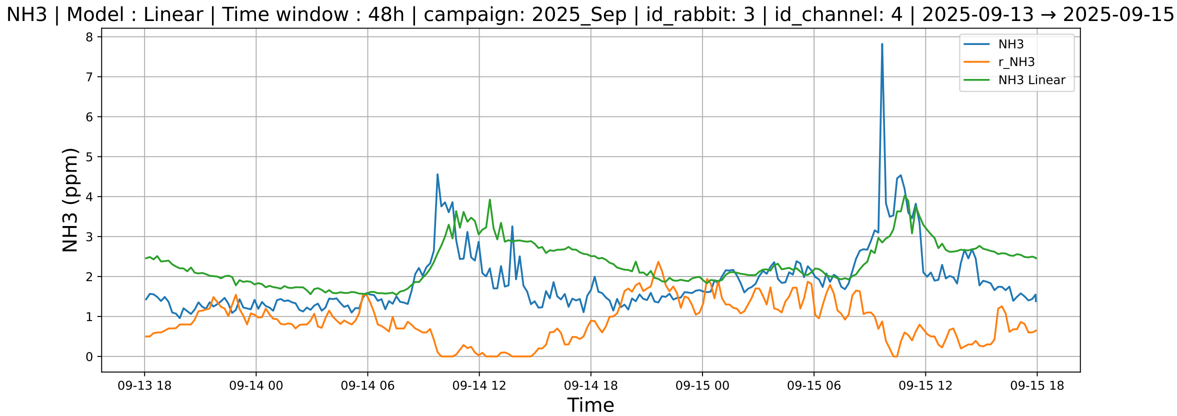
4.1.6.1 Time window : 24

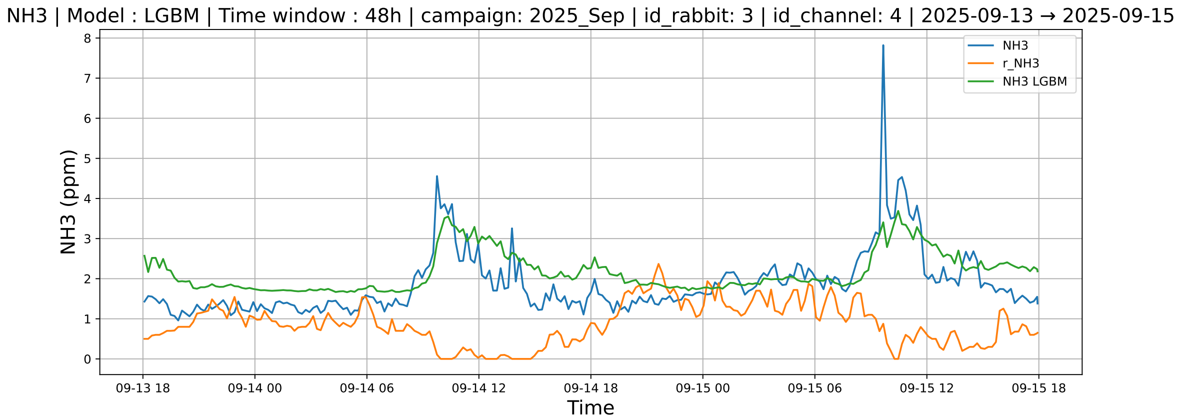


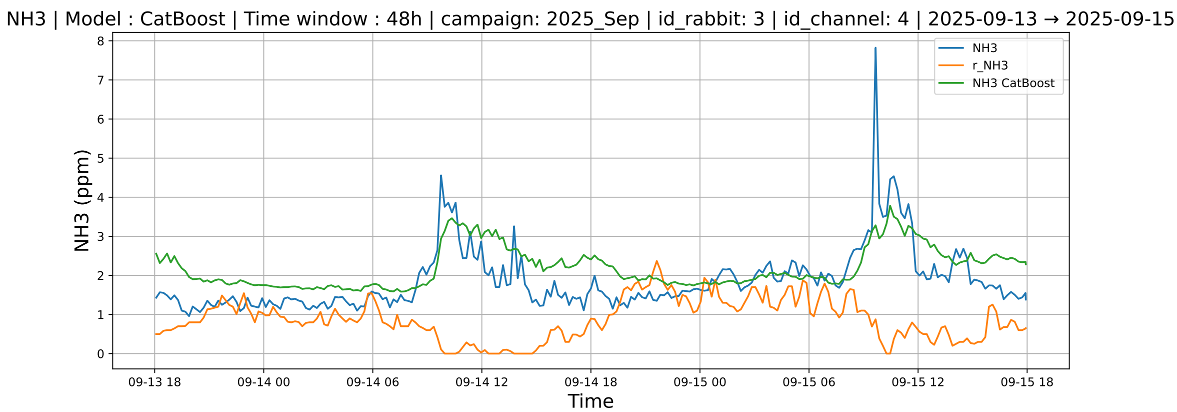




4.1.6.2 Time window : 48

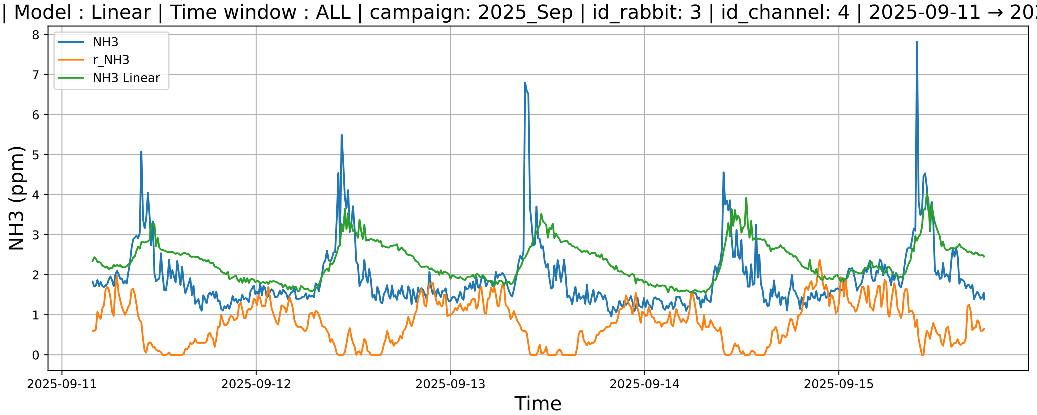




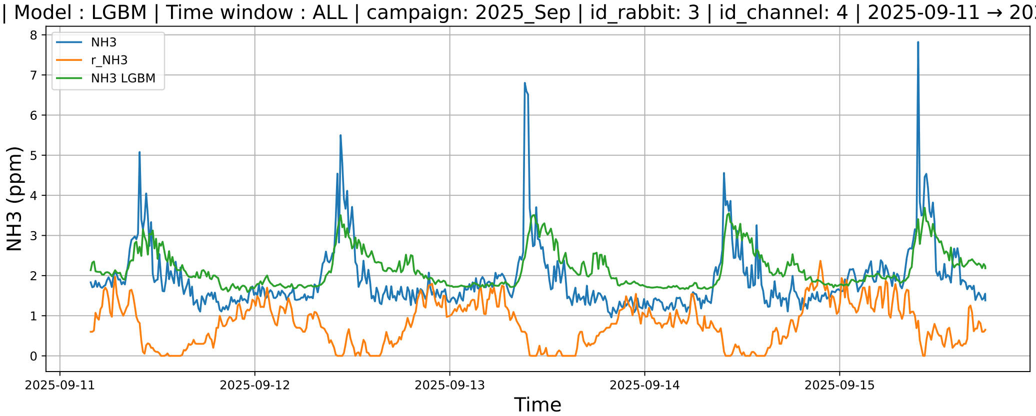


4.1.6.3 Time window : ALL

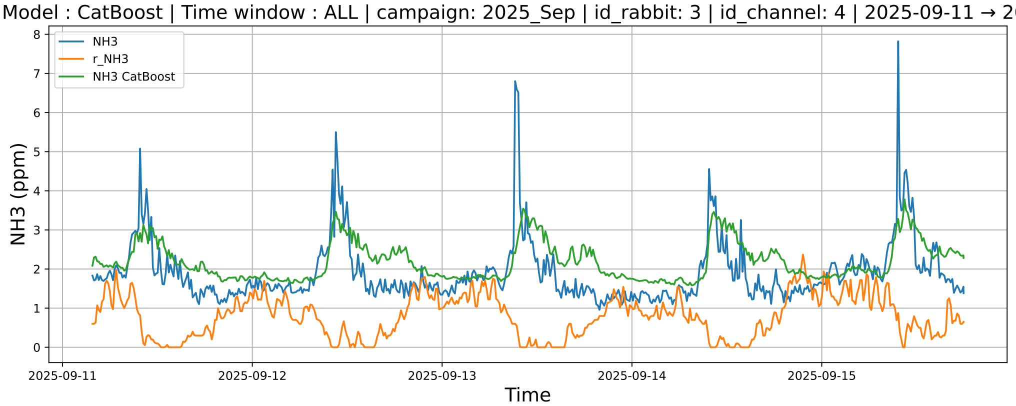
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

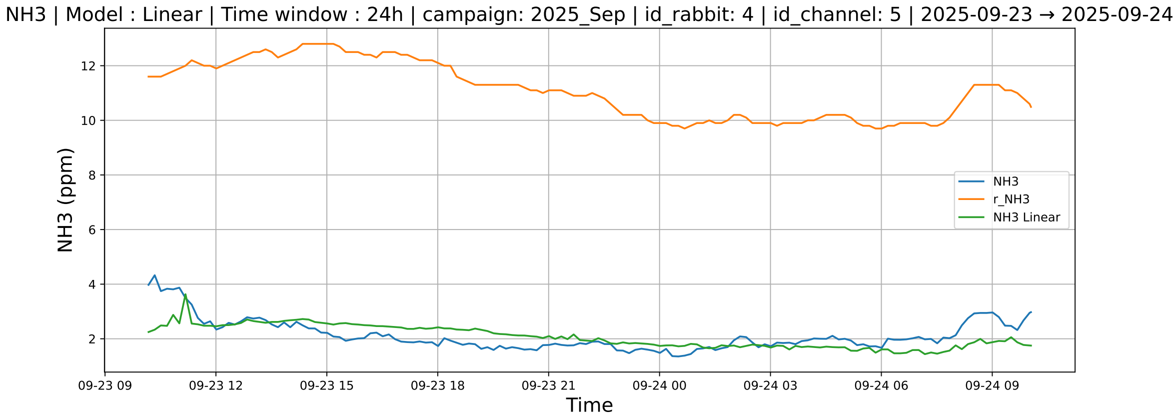


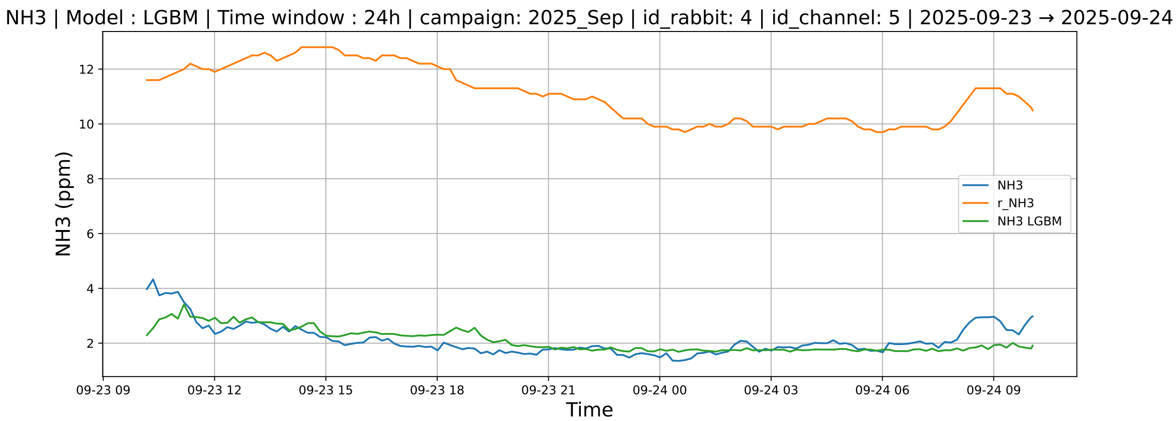
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15

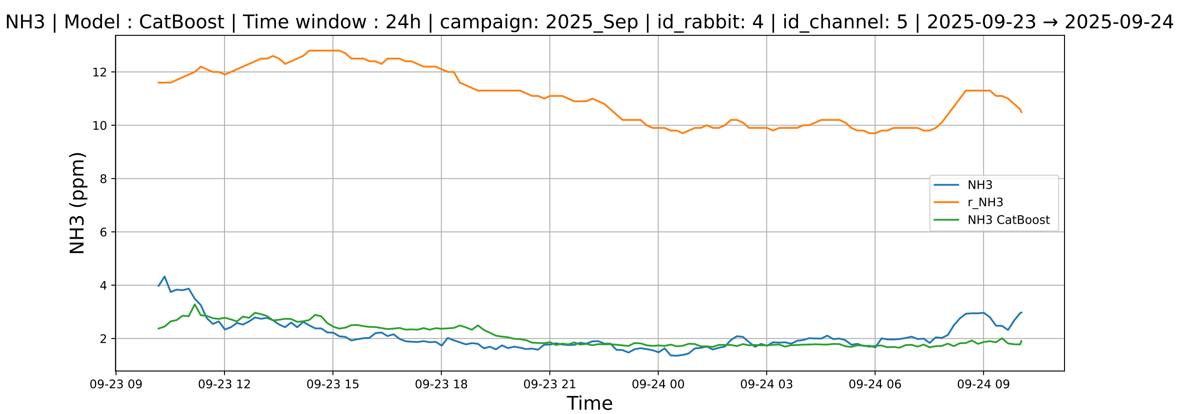


4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

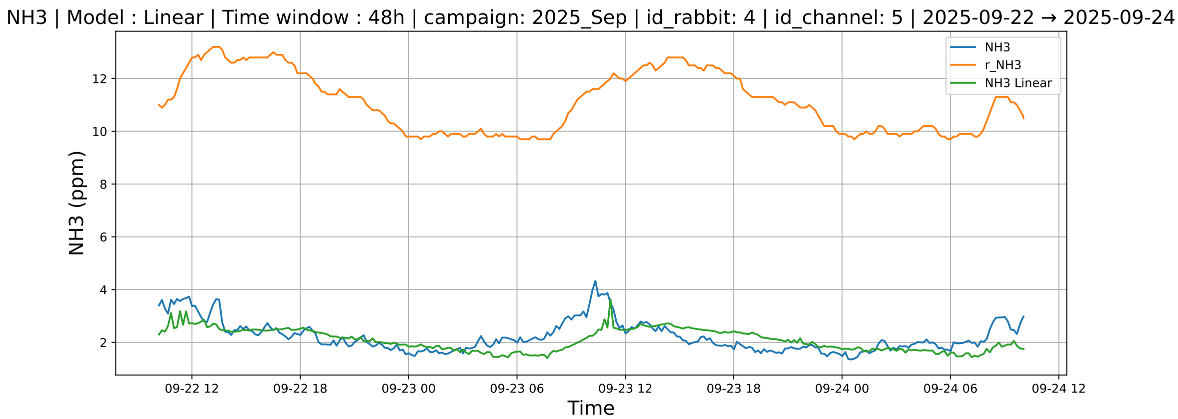
4.1.7.1 Time window : 24

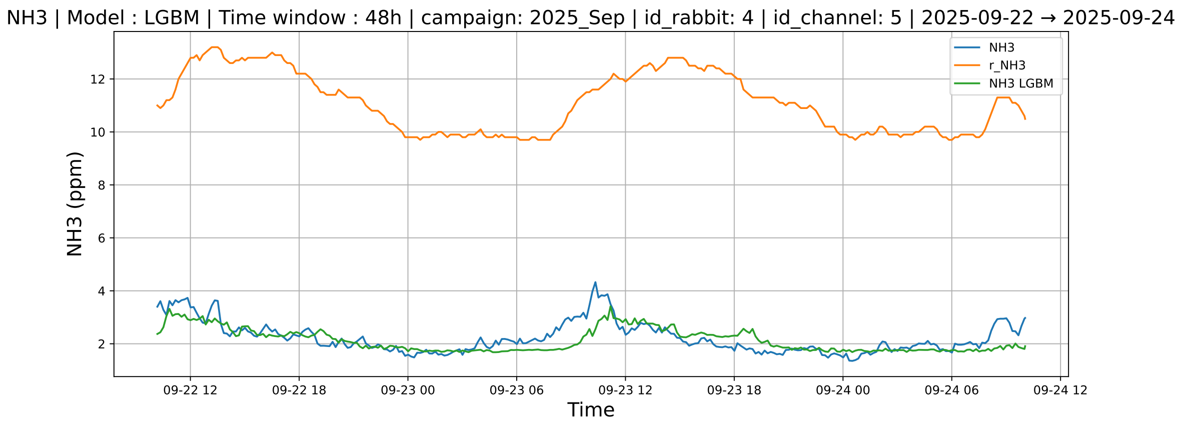


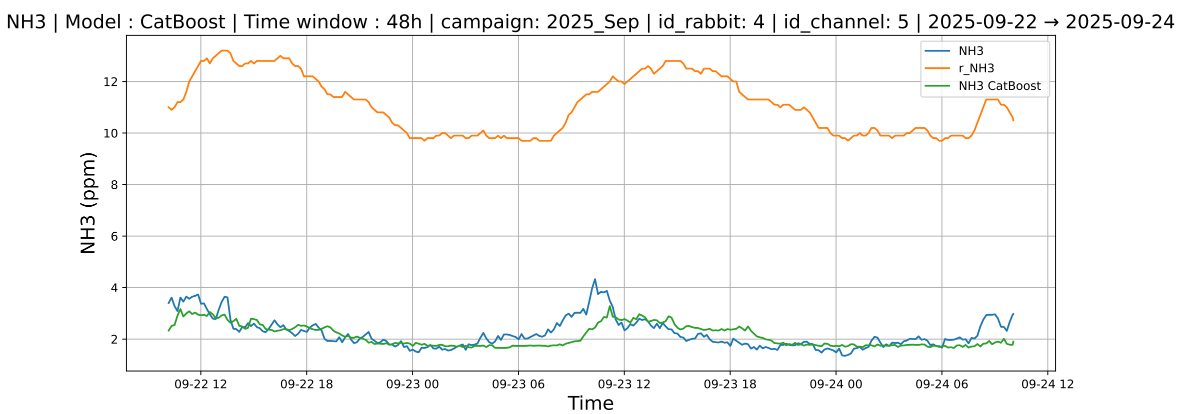




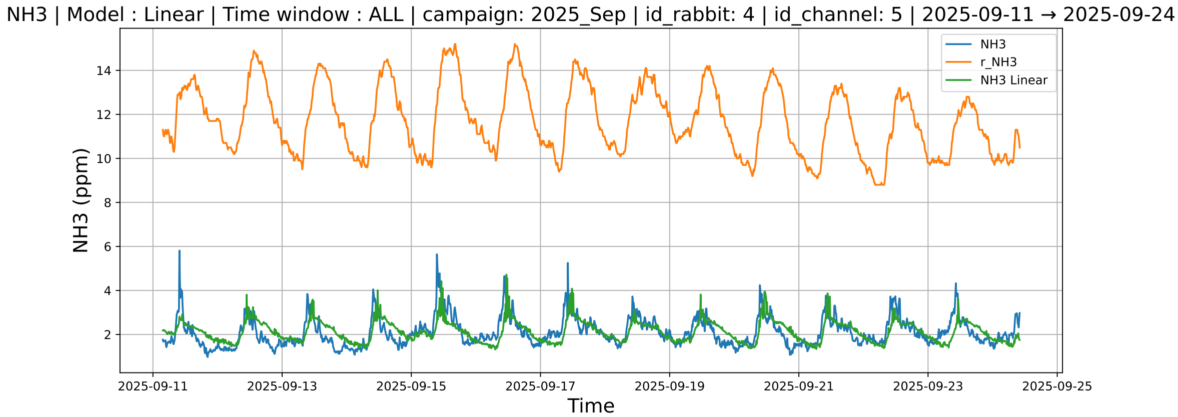
4.1.7.2 Time window : 48

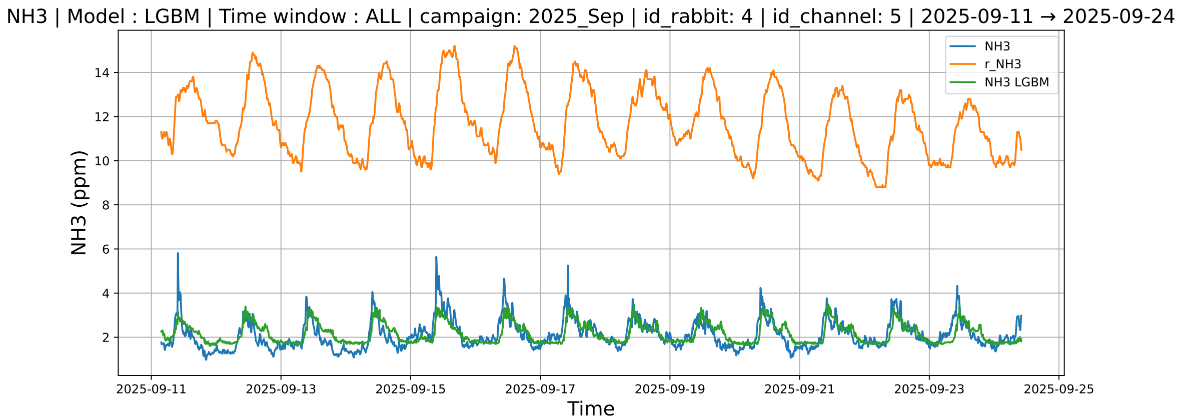


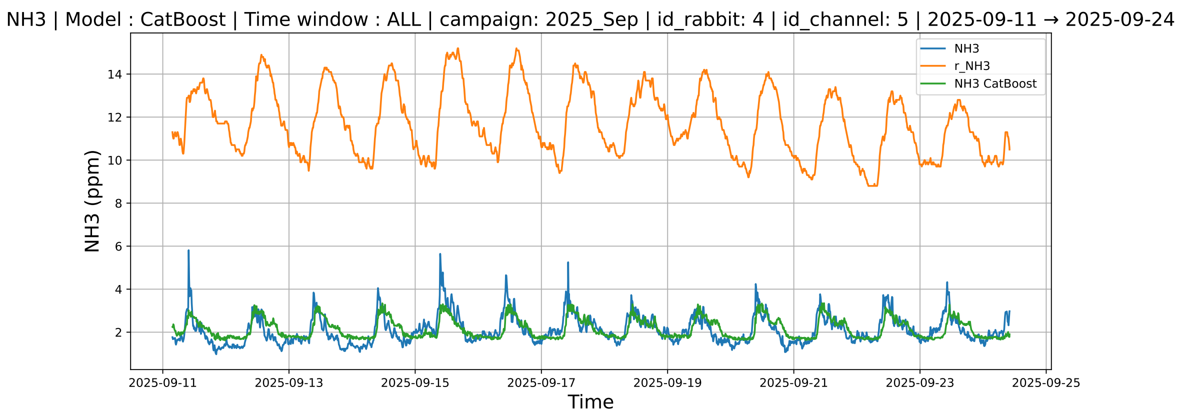




4.1.7.3 Time window : ALL



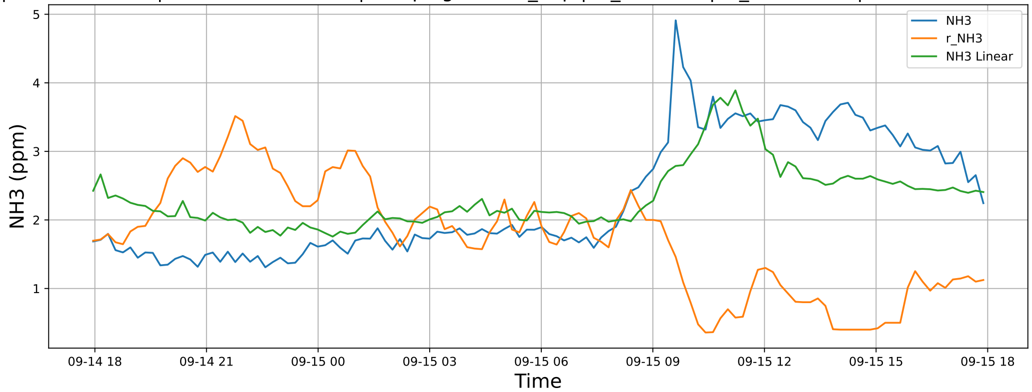




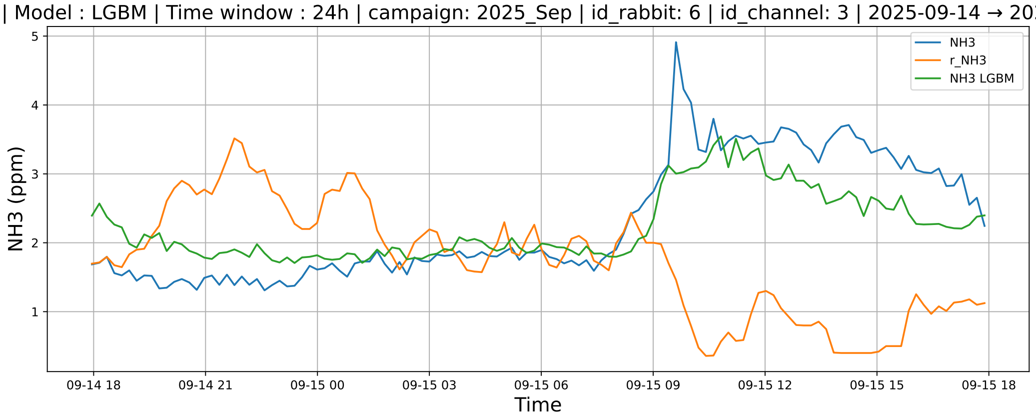
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

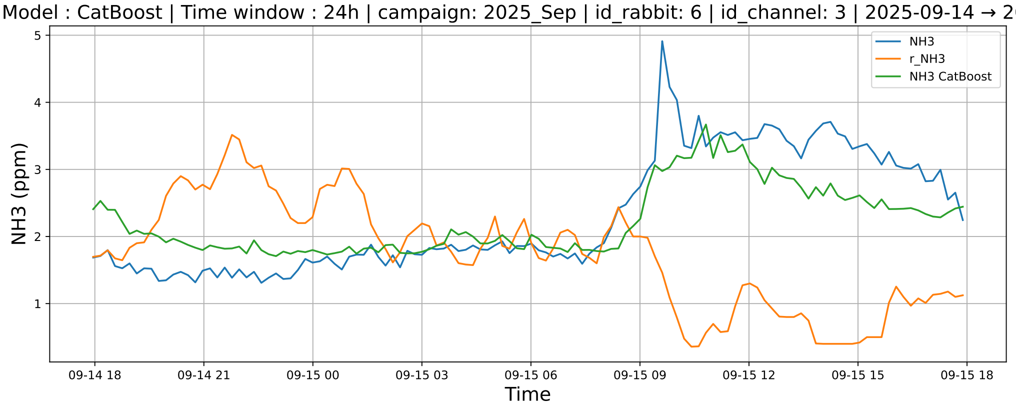
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

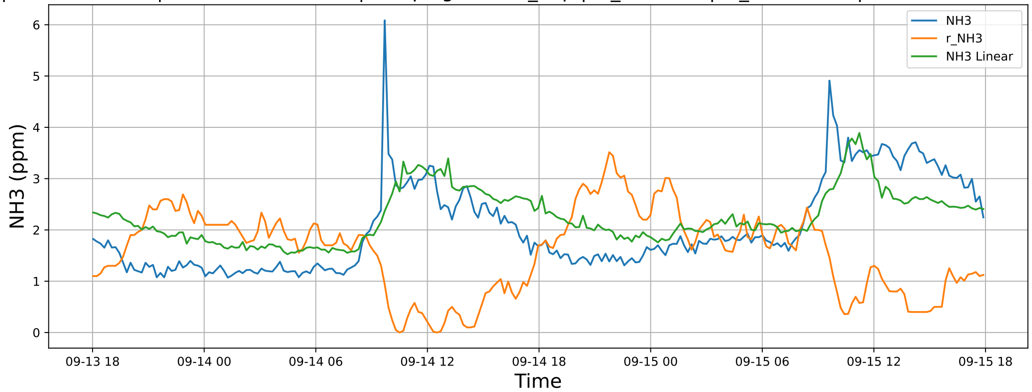


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

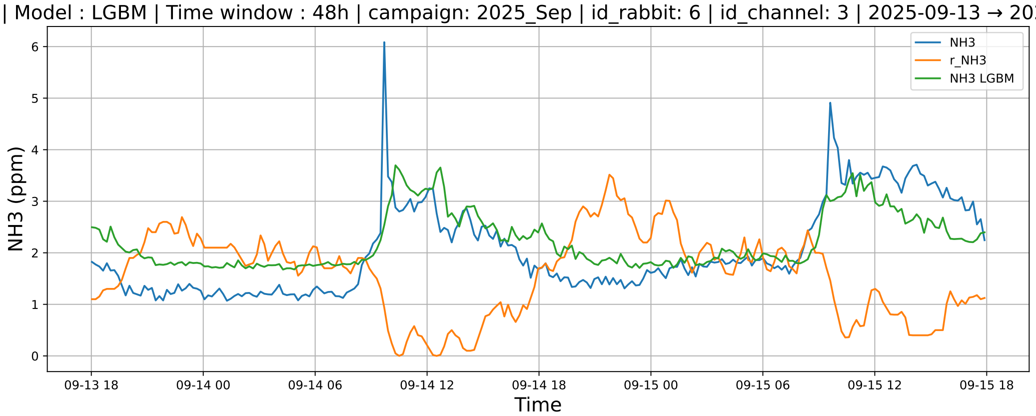


4.1.8.2 Time window : 48

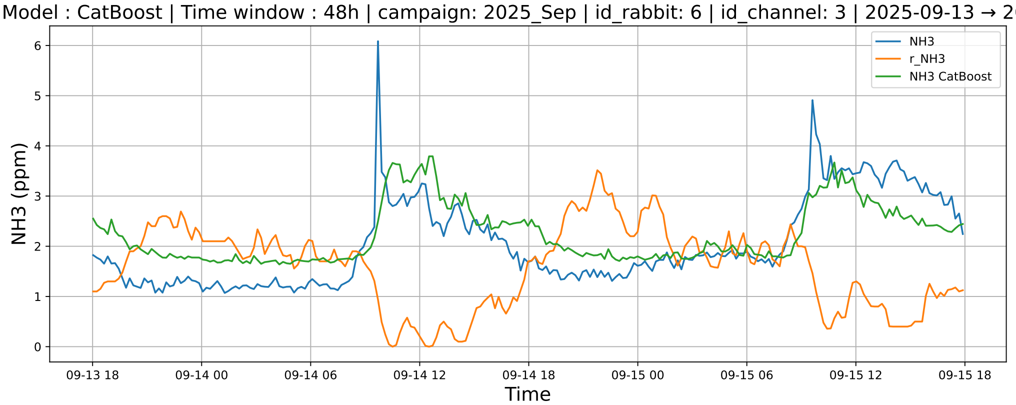
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



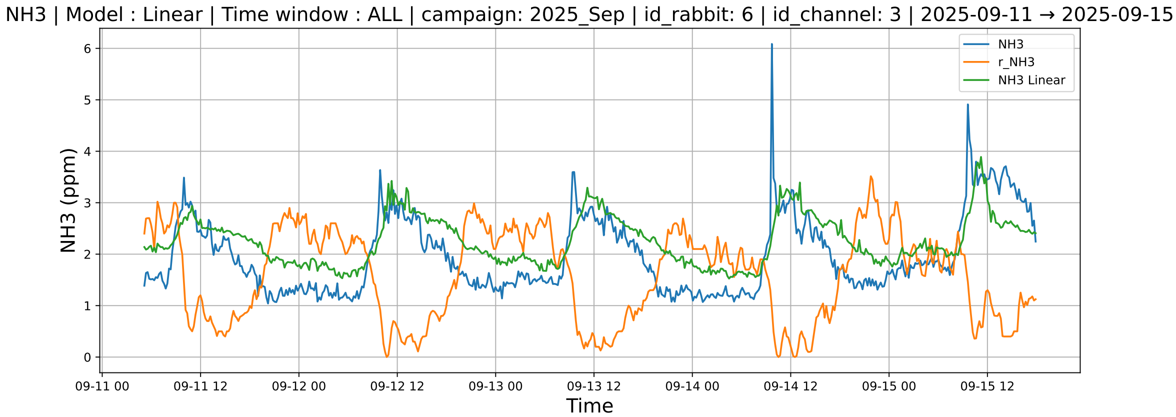
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

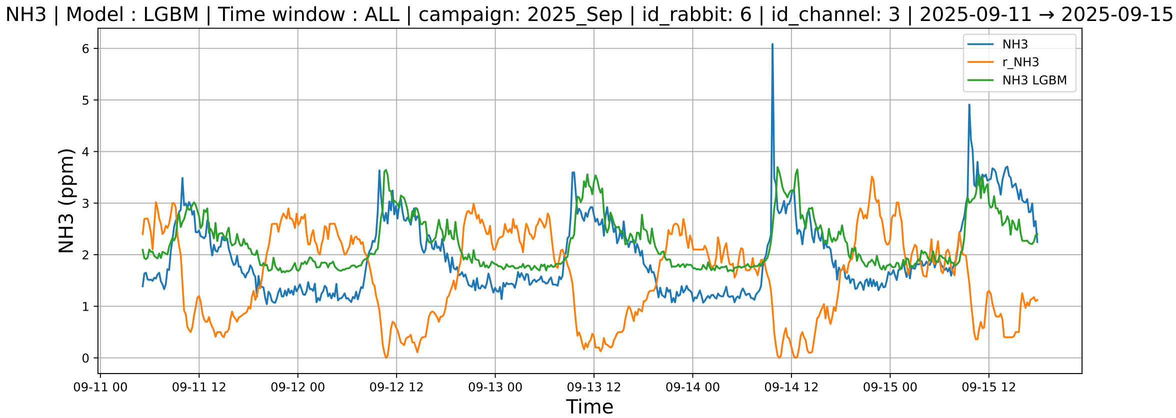


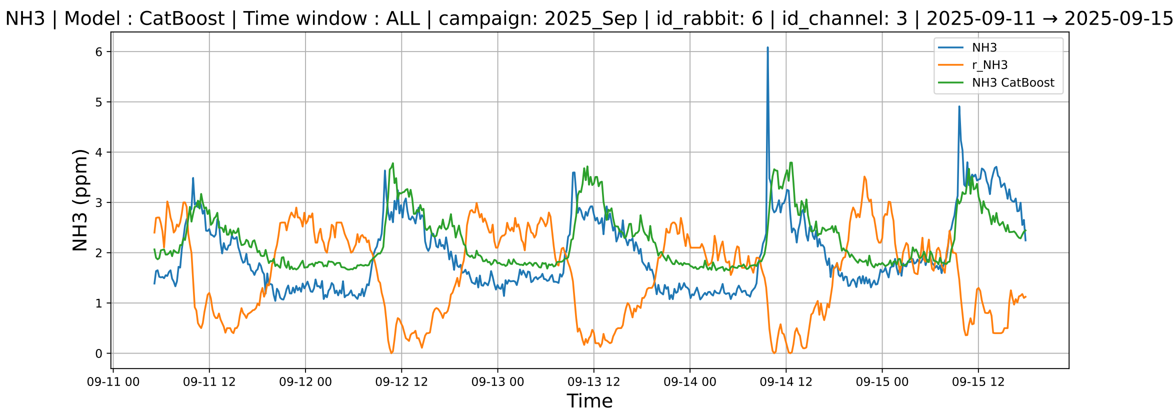
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.1.8.3 Time window : ALL



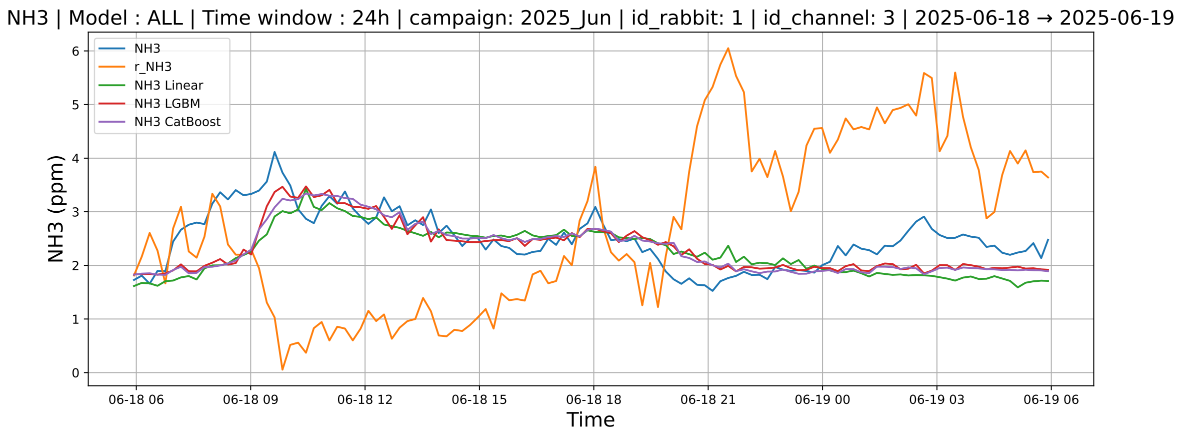




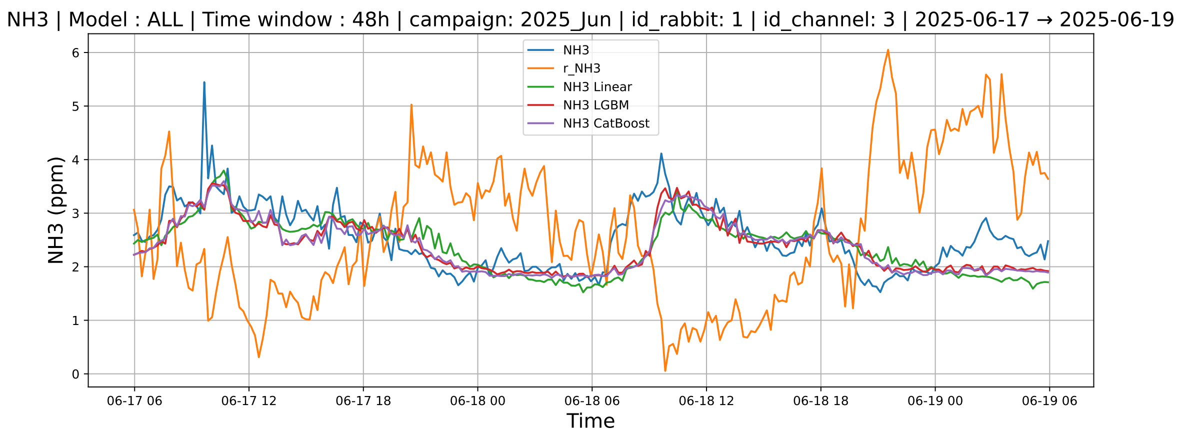
4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

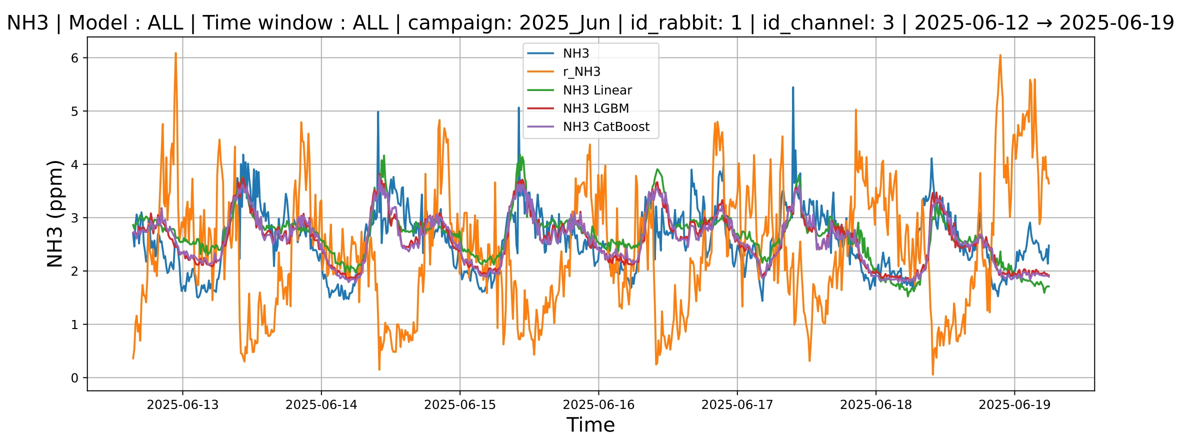
4.2.1.1 Time window : 24



4.2.1.2 Time window : 48



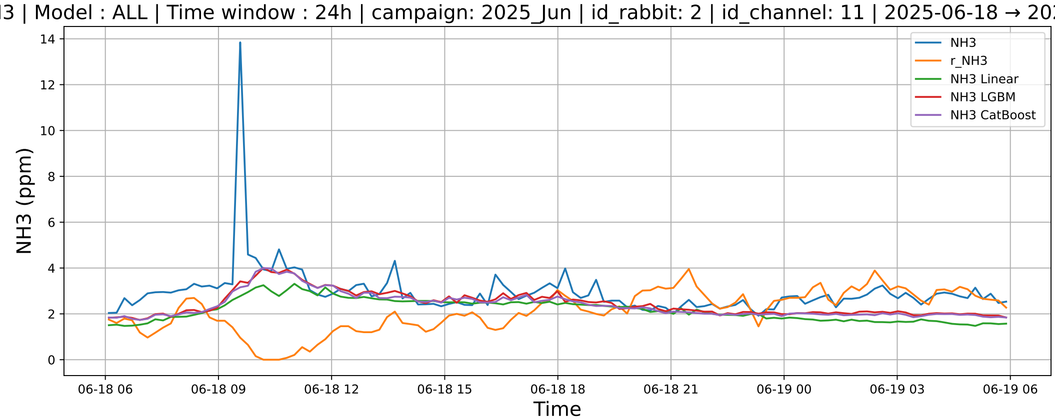
4.2.1.3 Time window : ALL



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

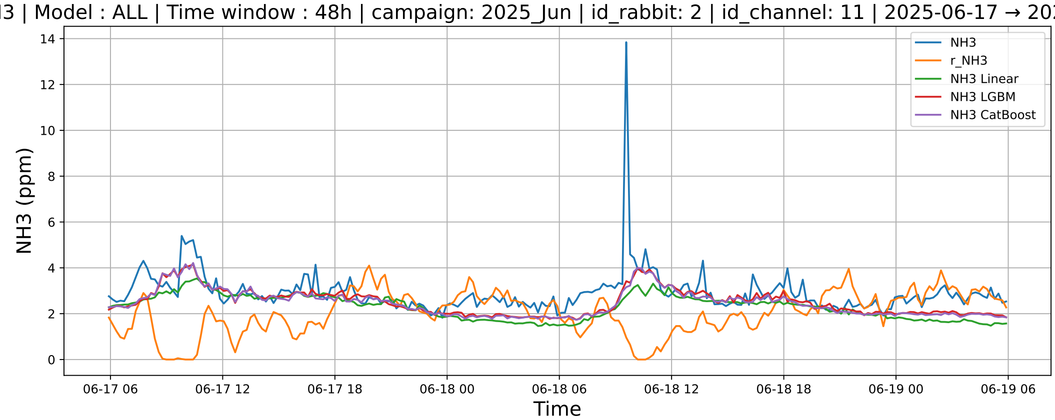
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



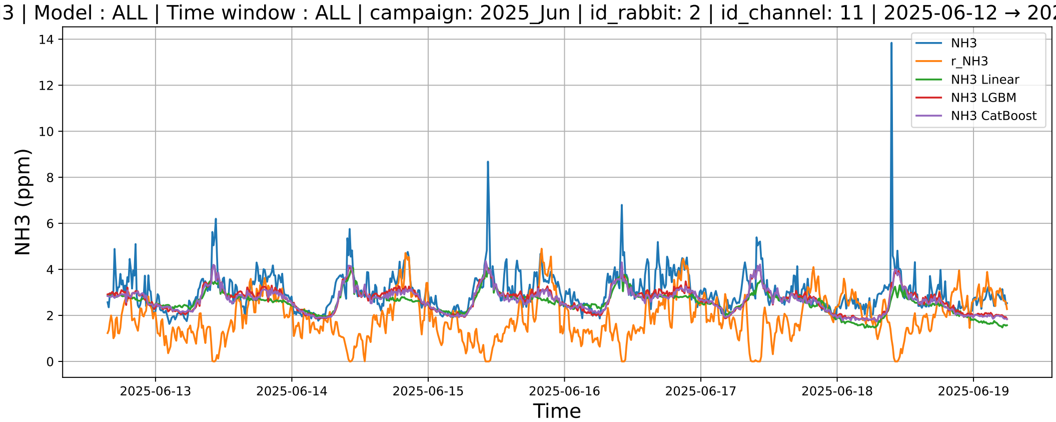
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

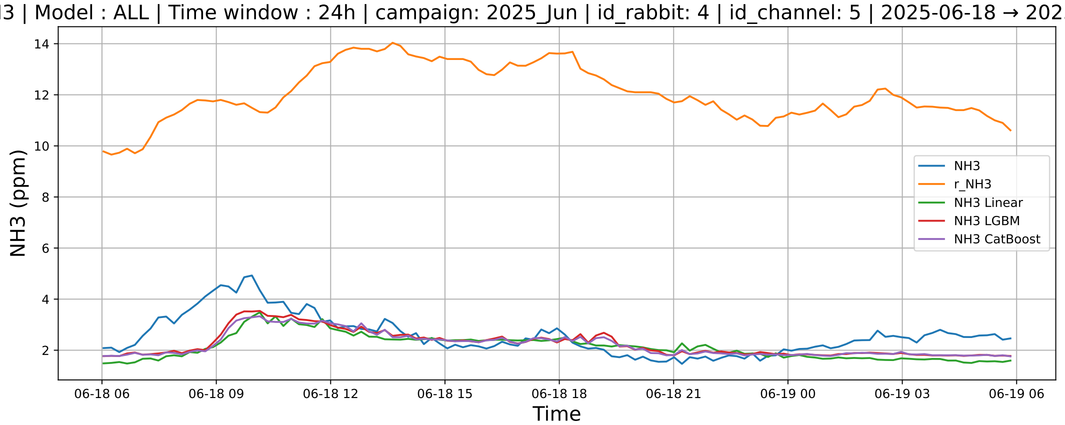
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



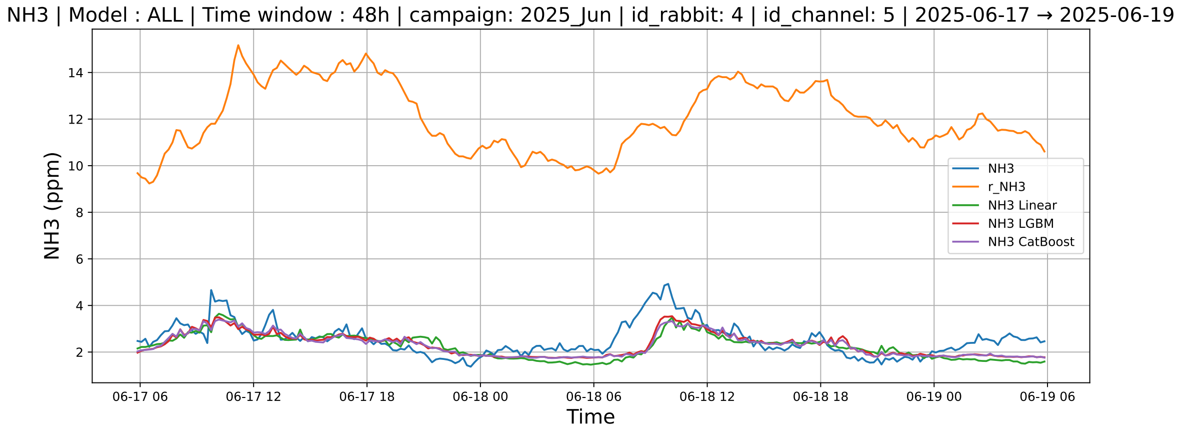
4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.2.3.1 Time window : 24

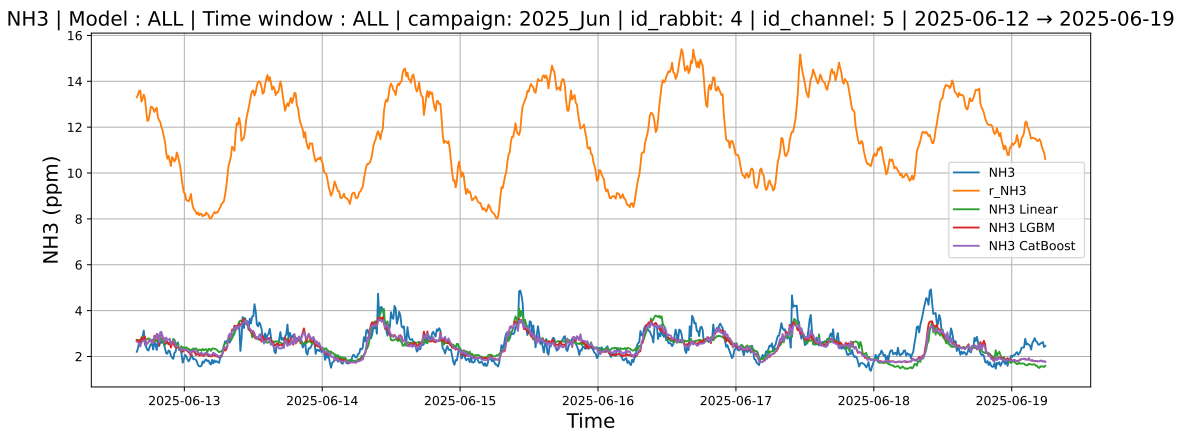
NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



4.2.3.2 Time window : 48



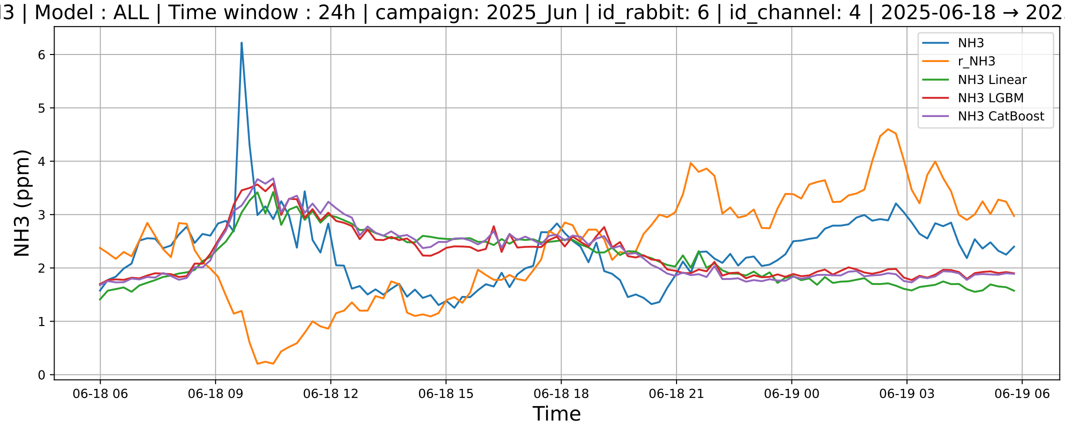
4.2.3.3 Time window : ALL



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

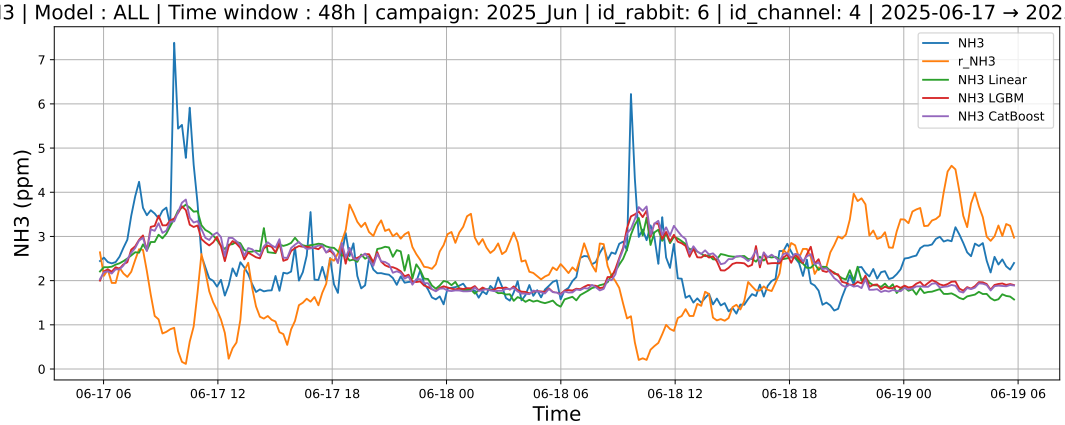
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



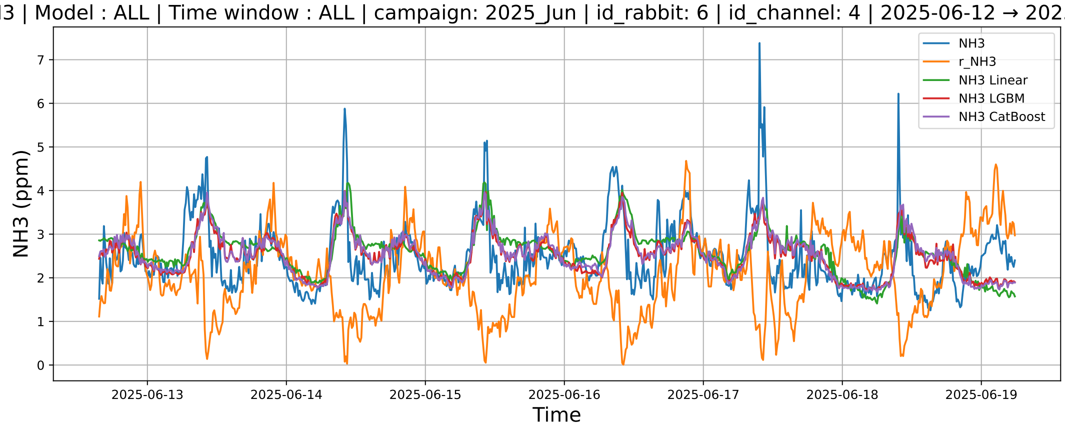
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

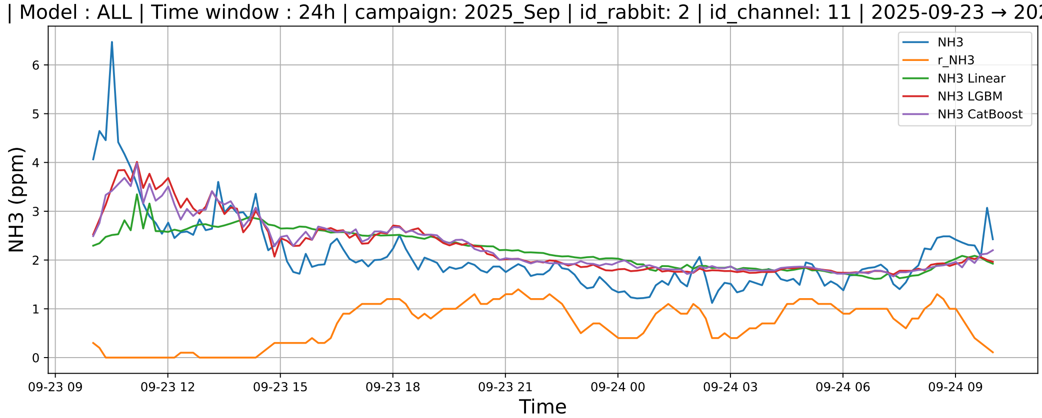
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

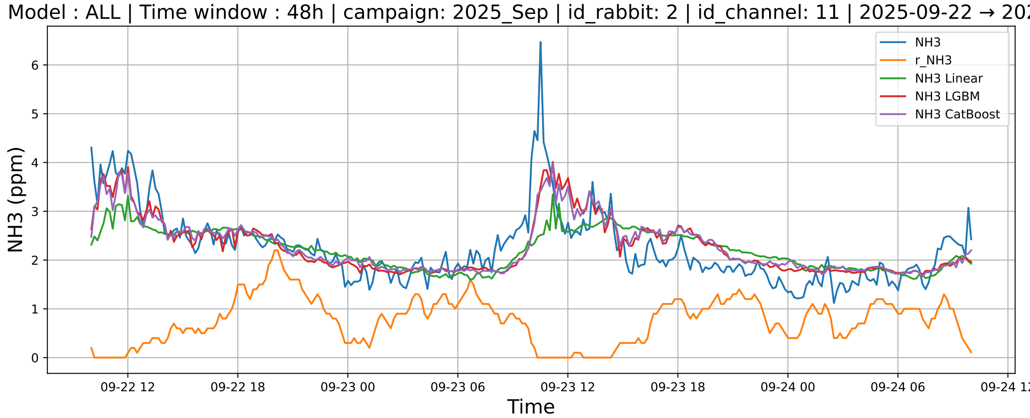
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

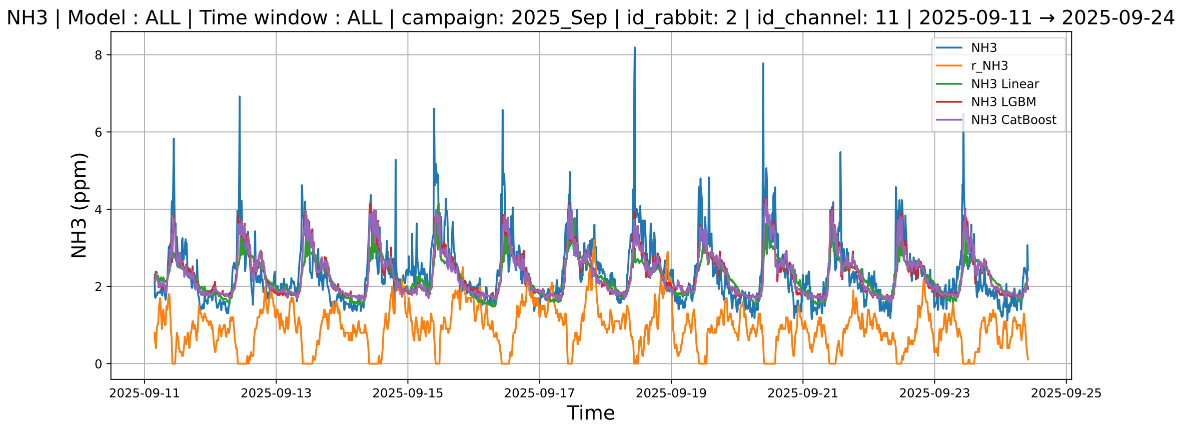


4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

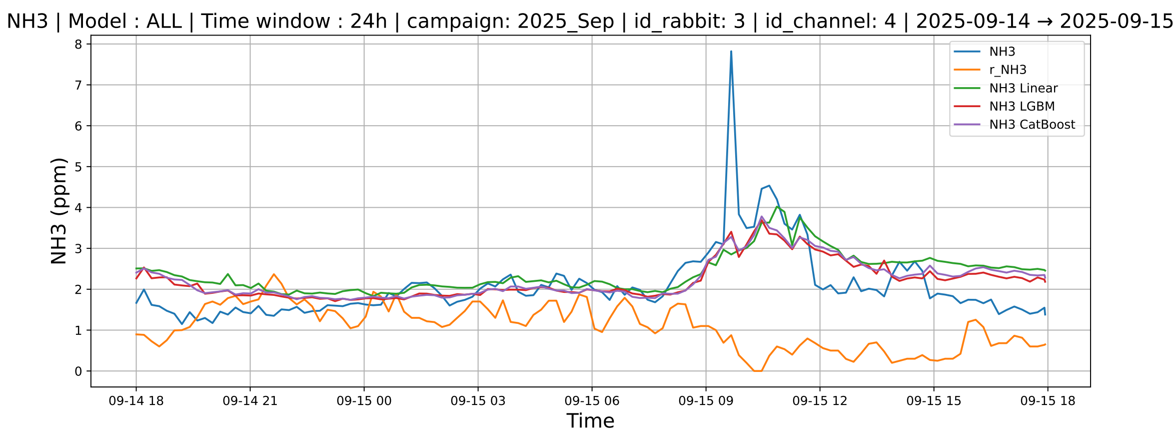


4.2.5.3 Time window : ALL

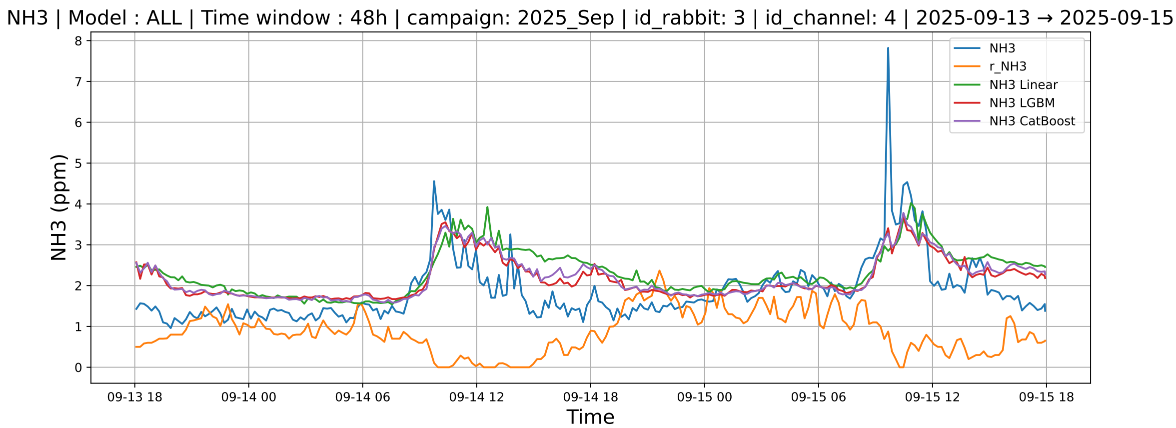


4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

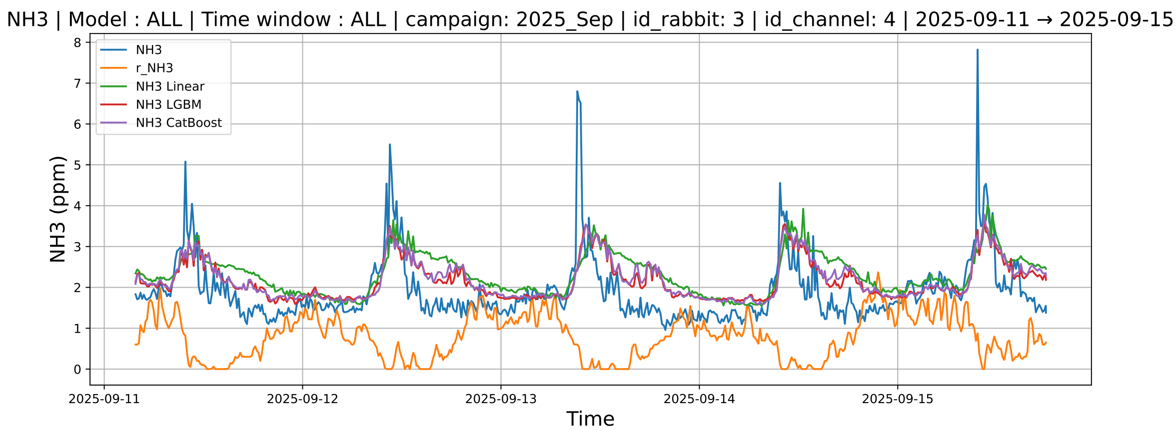
4.2.6.1 Time window : 24



4.2.6.2 Time window : 48

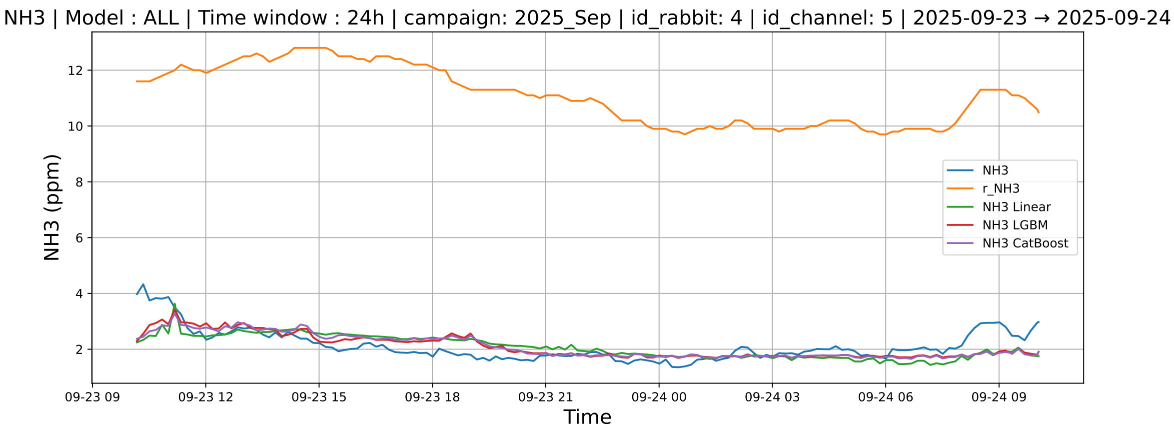


4.2.6.3 Time window : ALL

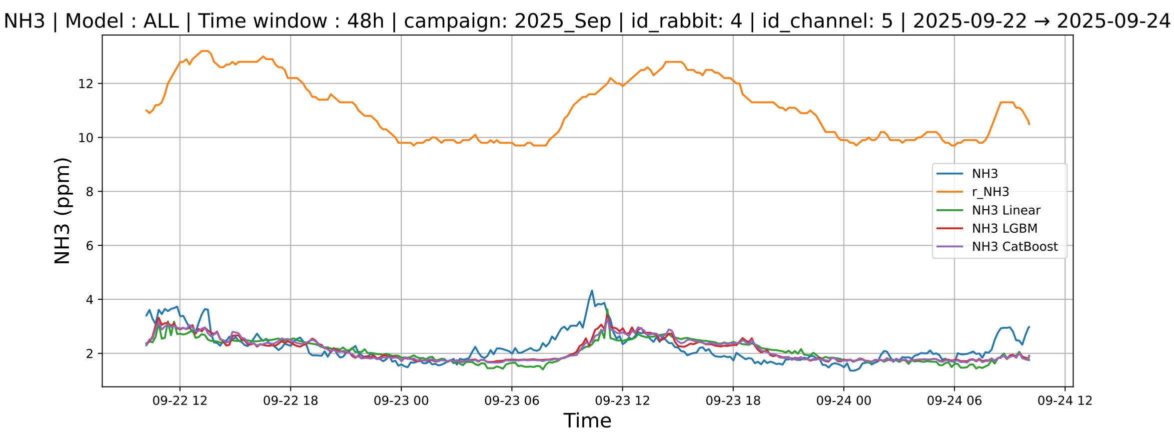


4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

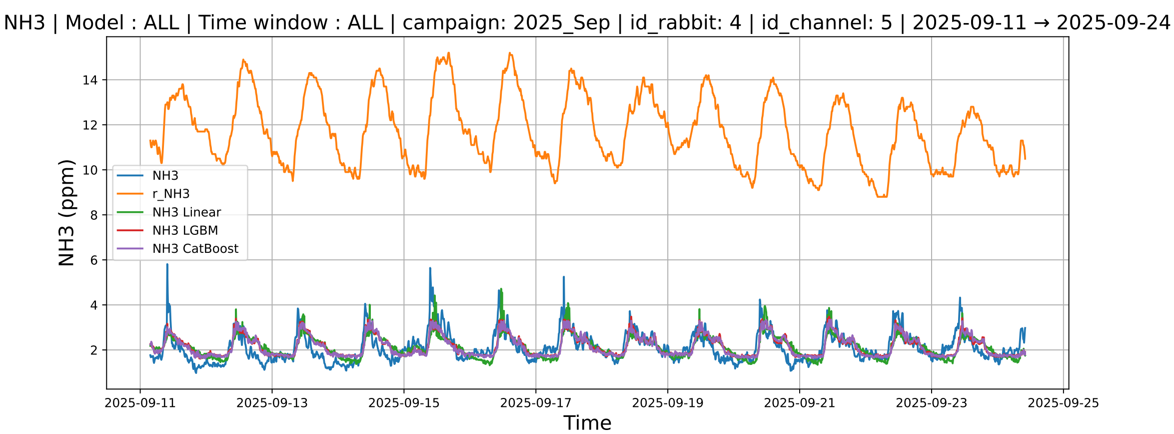
4.2.7.1 Time window : 24



4.2.7.2 Time window : 48



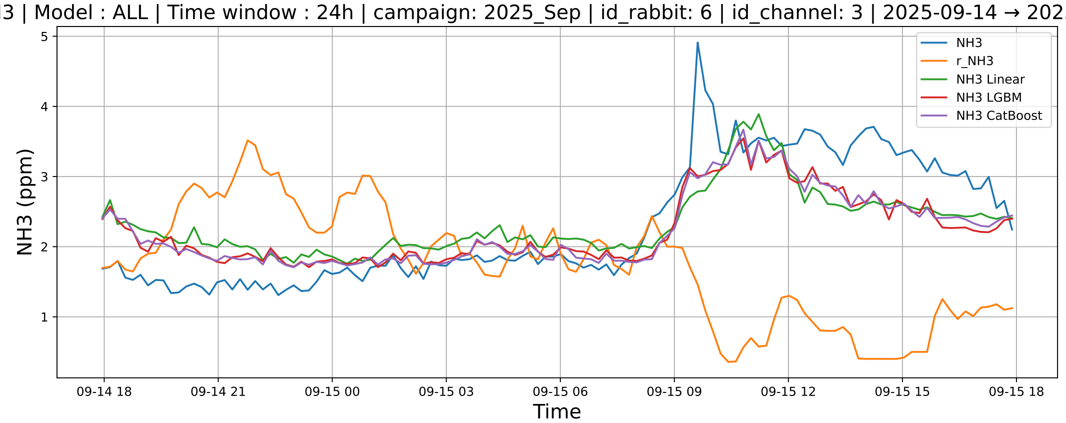
4.2.7.3 Time window : ALL



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

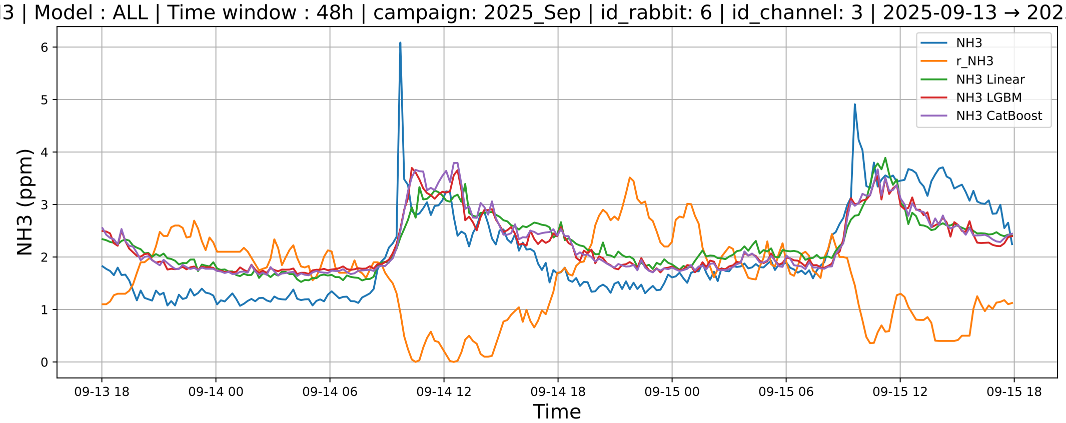
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

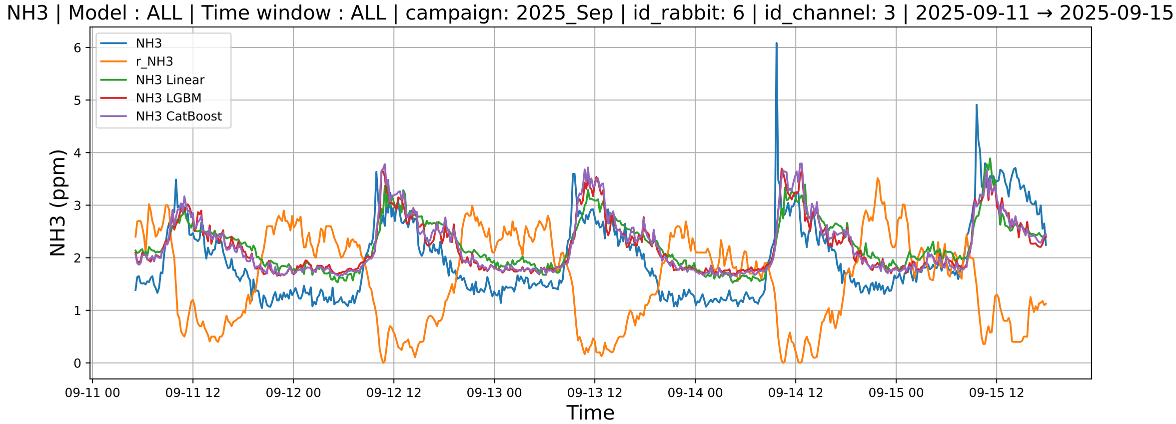


4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL


NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15
NH3
r_NH3
NH3 Linear
NH3 LGBM
NH3 CatBoost
NH3 (ppm)
Time
0
1
2
3
4
5
6
09-11 00
09-11 12
09-12 00
09-12 12
09-13 00
09-13 12
09-14 00
09-14 12
09-15 00
09-15 12